Control Systems

Detailed Solutions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

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Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Control Systems (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7^{th} question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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SOLUTIONS

1

Basics of Control Systems

Topics

1. Introduction to Control Systems
2. Laplace Transform and Concept of a System
3. Transfer Function
4. Concept of Feedback, OLTF and Sensitivity
5. Poles, Zeros and Final Value Theorem
6. Dominant Pole, Impulse and Step Response

Q1.15.1 MCQ | 2015 | 1M | Ans: B | ● ○ ○

The effects of negative feedback in a closed-loop control system are :

- **Reduction in Overall Gain:** The gain of a closed-loop system is $A_{CL} = \frac{G}{1+GH}$. Since the denominator $(1 + GH) > 1$, the overall gain is reduced.
- **Increase in Bandwidth:** For most systems, the **gain-bandwidth product remains constant**. Therefore, a decrease in gain results in an increase in bandwidth.
- **Improvement in Disturbance Rejection:** Negative feedback reduces the effect of external disturbances and noise signals entering the system thus **improving the stability**.
- **Reduction in Sensitivity:** ($S_G^T = \frac{1}{1+GH}$) It makes the system less sensitive to variations in internal plant parameters G.

$$\text{Gain} \downarrow \implies \text{Bandwidth} \uparrow$$

$$\boxed{\text{(B) reduce bandwidth}}$$

Q1.05.1 MCQ | 2005 | 1M | Ans: A | ● ○ ○

The physical components (such as actuators, sensors, and amplifiers) are not perfectly linear. They exhibit **non-linear characteristics** such as saturation, backlash, dead zones, and friction. These non-linearities can lead to phenomena like limit cycles or even total system instability.

$$\boxed{\text{(A) components used have non linearities}}$$

Mistakes to be avoided

Avoid confusing mathematical approximations with the physical cause of instability; while approximations might lead to design errors, the instability itself arises from the physical nature of the components.

Q1.03.1 MCQ | 2003 | 2M | Ans: B | ● ○ ○

From the given pole-zero plot: Zero: $s = -3$, $\text{Pole}_1 = -1 + j$ $\text{Pole}_2 = -1 - j$
The general expression for the driving point impedance $Z(s)$ is:

$$Z(s) = \frac{K(s+3)}{(s+1+j)(s+1-j)}$$

$$Z(s) = \frac{K(s+3)}{(s+1)^2 - j^2} = \frac{K(s+3)}{(s+1)^2 + 1}$$

From the problem statement, the value of $Z(0)$ at $\omega = 0$ is 3:

$$Z(0) = \frac{K(0+3)}{(0+1)^2 + 1} = \frac{3K}{2} = 3 \implies K = 2$$

Substituting $K = 2$ back into the expression for $Z(s)$:

$$Z(s) = \frac{2(s+3)}{s^2 + 2s + 1 + 1} = \frac{2(s+3)}{s^2 + 2s + 2}$$

$$\boxed{(B) \quad \frac{2(s+3)}{s^2 + 2s + 2}}$$

Q1.01.1 MCQ | 2001 | 2M | Ans: B | ● ○ ○

The closed-loop transfer function (CLTF) for a unity negative feedback system is given by:

$$\text{CLTF} = \frac{G(s)}{1 + G(s)H(s)} = \frac{s+4}{s^2+7s+13}$$

For unity feedback, $H(s) = 1$. Rearranging to find the open-loop transfer function $G(s)$:

$$\frac{1+G(s)}{G(s)} = \frac{s^2+7s+13}{s+4}$$

$$\frac{1}{G(s)} + 1 = \frac{s^2+7s+13}{s+4}$$

$$G(s) = \frac{s+4}{s^2+6s+9}$$

For DC gain, substitute $s = 0$:

$$\text{DC gain} = G(0) = \frac{0+4}{0^2+6(0)+9} = \frac{4}{9}$$

$$\boxed{\text{(B)}\ \frac{4}{9}}$$

Key Points

- For a unity feedback system ($H(s) = 1$), $G(s) = \dfrac{\text{CLTF}}{1-\text{CLTF}}$.
- DC gain is the value of the transfer function at steady state, calculated by setting $s = 0$.

Q1.98.1 MCQ | 1998 | 1M | Ans: A | ● ○ ○

A tachometer is an electromechanical device that converts mechanical energy (rad/s) into electrical energy (V).

The output voltage $e(t)$ produced by a tachometer is directly proportional to the angular velocity $\omega(t)$ of the shaft. Since $\omega(t)$ is the derivative of angular position $\theta(t)$, the relationship is:

$$e(t) = K_t \frac{d\theta(t)}{dt}$$

where K_t is the tachometer constant (sensitivity).
To find the transfer function, we take the Laplace Transform of both sides, assuming zero initial conditions:

$$E(s) = K_t s\theta(s)$$

$$\text{T.F.} = \frac{E(s)}{\theta(s)} = K_t s$$

Thus, a tachometer acts as a differentiator in the system.

$$\boxed{\text{(A)}\ Ks}$$

Q1.95.1 MCQ | 1995 | 1M | Ans: C | ● ○ ○

The transfer function is the ratio of the Laplace transform of the output to the Laplace transform of the input, under the strict assumption that all initial conditions are zero.

$$\text{T.F.} = \left.\frac{\mathcal{L}[\text{Output}]}{\mathcal{L}[\text{Input}]}\right|_{\text{initial conditions}=0}$$

$\boxed{\text{(C)}\quad \text{Ratio of the Laplace transform of the output and that of the input with all initial conditions zeroes}}$

Q1.94.1 MCQ | 1994 | 1M | Ans: True | ● ○ ○

Tachometer feedback is a form of derivative feedback. The inclusion of this feedback **adds a zero at the origin** in the feedback path, which effectively:

- Increases damping: Reduces oscillations and overshoot.
- Improves relative stability: Enhances the system's stability margins.

Hence, the statement that tachometer feedback enhances stability is $\boxed{\text{True}}$

DEBUG INFO

Chapter I

Basics of Control Systems

Total Questions : 7

MCQ : 7

MSQ : 0

NAT : 0

SOLUTIONS

II

Block Diagram and Signal Flow Graphs

Topics

1. Block Diagram Reduction Techniques
2. Signal Flow Graphs and Mason's Gain Formula
3. Limitations of Mason's Gain Formula

Q2.26.1 MCQ | 2026 | 1M | Ans: B | ● ○ ○

The given block diagram consists of a forward path and two feedback loops.

Method 1

Mason's Gain Formula: $\frac{C(s)}{R(s)} = \frac{\sum P_k \Delta_k}{\Delta}$

- Forward Path (P_1):

$$P_1 = \frac{1}{s+4}$$

- Individual Loops (L_m):

$$L_1 = \frac{1}{s+4} \times (-1) \times (-1) = \frac{1}{s+4} \quad ; \quad L_2 = \frac{1}{s+4} \times (+1) \times (-1) = -\frac{1}{s+4}$$

- Determinant (Δ):

$$\Delta = 1 - (L_1 + L_2) = 1 - \left(\frac{1}{s+4} - \frac{1}{s+4}\right) = 1 - 0 = 1$$

- Overall Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{P_1\Delta_1}{\Delta} = \frac{\left(\frac{1}{s+4}\right) \cdot 1}{1} = \frac{1}{s+4}$$

Method 2

Block Diagram Reduction: Consider the error signal E(s)

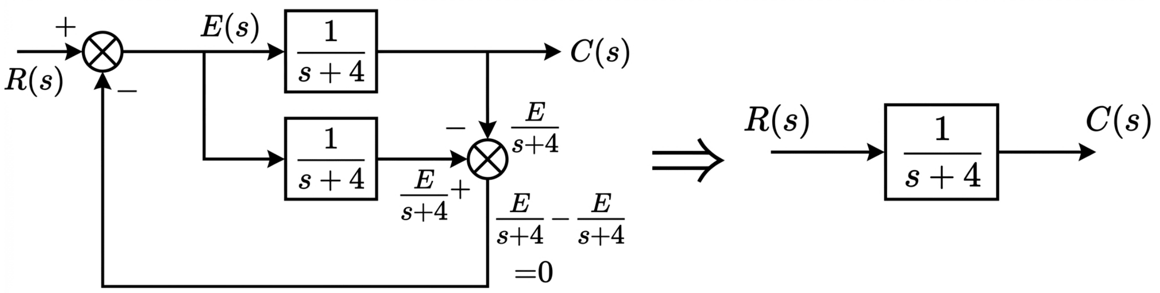


Fig. Solution Q2.26.1

$$\boxed{\text{(B)} \ \frac{1}{s+4}}$$

Key Points

- Mason's Gain Formula: $\frac{C(s)}{R(s)} = \frac{\sum P_k \Delta_k}{\Delta}$
  - P_k: Gain of the k^{th} forward path
  - $\Delta = 1 -$ (sum of individual loop gains) + (sum of gain products of two non-touching loops) $- \ldots$
  - Δ_k: The cofactor of the k^{th} forward path, calculated by removing the loops touching the k^{th} path.

Q2.25.1 MSQ 2025 2M Ans: B ● ○ ○

Method 1

Summing points and pick-off points in the block diagram are represented as nodes in SFG.

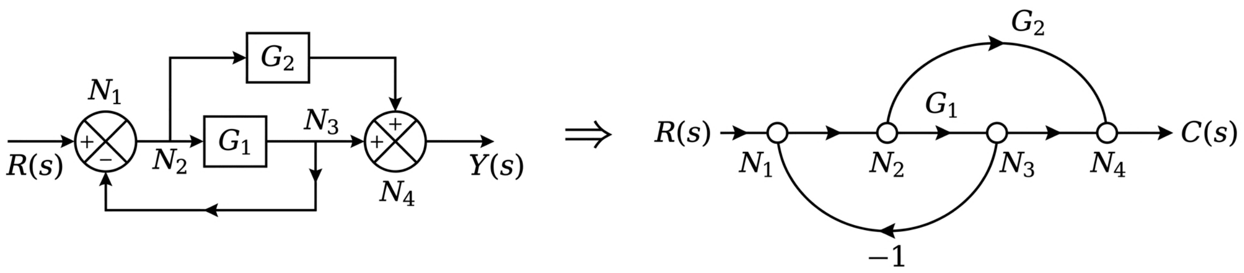


Fig. Solution Q2.25.1

Comparing this topology to the options, **Option (B)** is the only graph that maintains these exact connections and node relationships.

Method 2

Alternatively, we can use Mason's Gain Formula to compare the transfer functions. For the block diagram, the overall transfer function is:

$$\frac{Y(s)}{R(s)} = \frac{G_1 + G_2}{1 + G_1}$$

Evaluating the options:

- Option (A): There are 2 negative feedback loops ($-G_1$ and $-G_2$). The transfer function is $\frac{G_1 + G_2}{1 + G_1 + G_2}$, which is **wrong**.
- Option (B): There is only a single negative feedback loop. The transfer function is $\frac{G_1 + G_2}{1 + G_1}$, which is **correct**.
- Option (C): Represents 2 positive feedback loops : $\frac{G_1 + G_2}{1 - G_1 - G_2}$, which is **wrong**.
- Option (D): The SFG includes 2 negative feedback loops: $\frac{G_1 + G_2}{1 + G_1 + G_2}$, which is **wrong**.

Mistakes to be avoided

Ensure the feedback loop in the SFG starts and ends at the correct nodes corresponding to the block diagram pick-off and summing points.

Q2.23.1 MCQ 2023 2M Ans: A ● ● ○

The output is $Y(s) = G_1(s)R(s) + G_2(s)D(s)$. We use the **principle of superposition**.

Case 1: Output $Y_1(s)$ due to $R(s)$: $(D(s) = 0)$

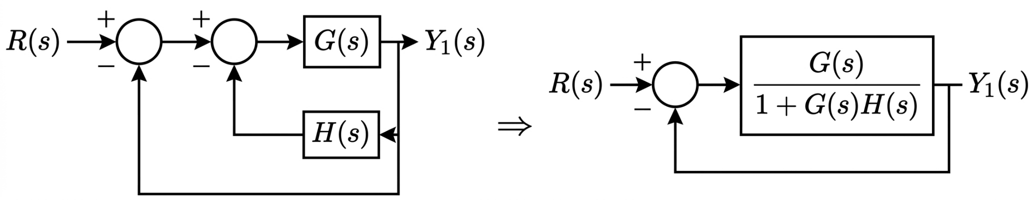


Fig. Solution Q2.23.1

$$\frac{Y_1(s)}{R(s)} = \frac{\frac{G(s)}{1+G(s)H(s)}}{1+\frac{G(s)}{1+G(s)H(s)}} = \frac{G(s)}{1+G(s)H(s)+G(s)} \implies G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$$

Case 2: Output $Y_2(s)$ due to $D(s)$: $(R(s) = 0)$

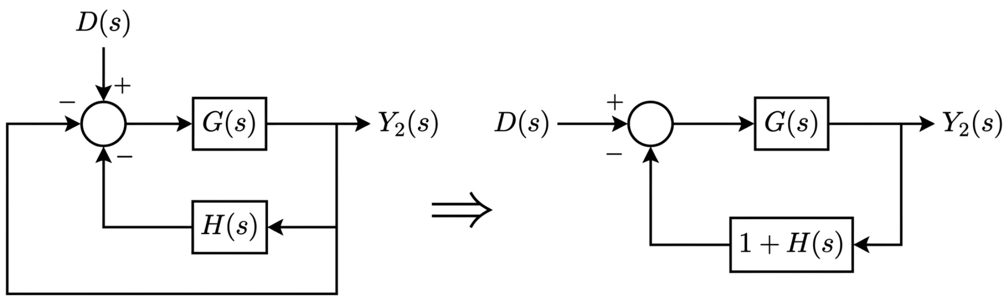


Fig. Solution Q2.23.1

$$\frac{Y_2(s)}{D(s)} = \frac{G(s)}{1+G(s)[1+H(s)]} = \frac{G(s)}{1+G(s)+G(s)H(s)} \implies G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}$$

$$\boxed{\text{(A)}\ G_1(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}, G_2(s) = \frac{G(s)}{1+G(s)+G(s)H(s)}}$$

Q2.21.1 MCQ | 2021 | 1M | Ans: C | ● ○ ○

The transfer function of the system is evaluated using Mason's gain formula.
Number of forward paths and their associate path factor :

$$P_1 = G_1, \quad \Delta_1 = 1 \quad \text{and} \quad P_2 = G_2, \quad \Delta_2 = 1$$

Individual loops : $L_1 = -G_1H$
Determinant of system : $\Delta = 1 - (L_1) = 1 - (-G_1H) = 1 + G_1H$

$$\frac{Y(s)}{X(s)} = \frac{P_1\Delta_1 + P_2\Delta_2}{\Delta} = \frac{G_1+G_2}{1+G_1H}$$

$$\boxed{\text{(C)}\ \frac{G_1+G_2}{1+G_1H}}$$

Q2.19.1 | MCQ | 2019 | 2M | Ans: B | ● ○ ○

Method 1

Using block diagram reduction:

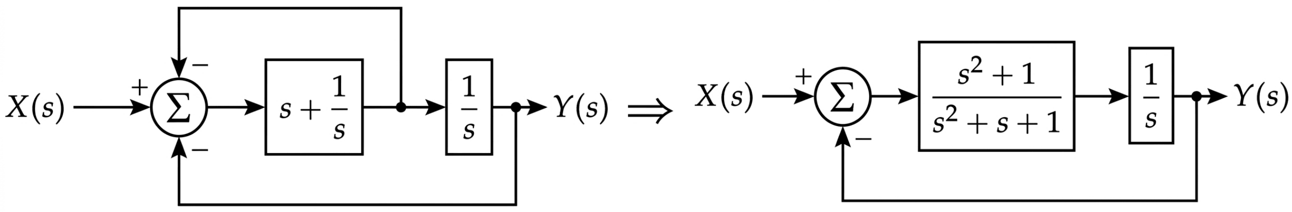


Fig. Solution Q2.19.1

The overall transfer function $H(s)$ with the final feedback path:

$$H(s) = \frac{\frac{s^2+1}{s(s^2+s+1)}}{1 + \frac{s^2+1}{s(s^2+s+1)}} = \frac{s^2+1}{s^3+2s^2+s+1}$$

Method 2

Using Mason's Gain Formula:

$$T.F = \frac{\sum P_k \Delta_k}{\Delta} = \frac{(s+\frac{1}{s})(\frac{1}{s})}{1-[-(s+\frac{1}{s})-(s+\frac{1}{s})(\frac{1}{s})]} = \frac{\frac{s^2+1}{s^2}}{1+\frac{s^2+1}{s}+\frac{s^2+1}{s^2}} = \frac{s^2+1}{s^3+2s^2+s+1}$$

$$\boxed{(B)\ \ H(s) = \frac{s^2+1}{s^3+2s^2+s+1}}$$

Q2.17.1 | NAT | 2017 | 1M | Ans: 0.95 to 1.05 | ● ○ ○

The given block diagram has the following parameters:
Number of forward paths = 2 (i.e, 1 and G(s)) ; Number of loops = 1 (i.e, -G(s))
By applying Mason's gain formula:

$$\frac{Y(s)}{X(s)} = \frac{1+G(s)}{1+G(s)} = \frac{1+2}{1+2} = \frac{3}{3} = 1$$

$$\boxed{1}$$

Q2.16.1 | MCQ | 2016 | 1M | Ans: B | ● ○ ○

The given block diagram of the feedback control system can be simplified as

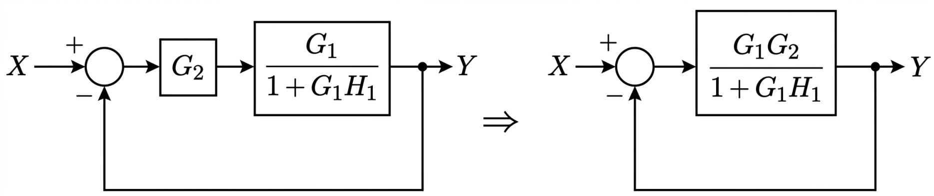


Fig. Solution Q2.16.1

$$\text{The overall close loop gain, } G = \frac{\frac{G_1G_2}{1+G_1H_1}}{1+\frac{G_1G_2}{1+G_1H_1}} = \frac{G_1G_2}{1+G_1G_2+G_1H_1}$$

$$\boxed{(B)\;\; G = \frac{G_1G_2}{1+G_1G_2+G_1H_1}}$$

Q2.15.1 MCQ 2015 1M Ans: B ● ○ ○

From the given signal flow graph, we can identify the following components:

- Forward path: $P_1 = G_1G_2G_3G_4$
- Individual loops: $L_1 = -G_1G_2H_1, \quad L_2 = -G_3G_4H_2, \quad L_3 = -G_2G_3H_3$
- Non-touching loops (two at a time): $L_1L_2 = (-G_1G_2H_1)(-G_3G_4H_2) = G_1G_2H_1G_3G_4H_2$

By applying Mason's Gain Formula:

$$\text{TF} = \frac{P_1\Delta_1}{\Delta} = \frac{G_1G_2G_3G_4}{1-(L_1+L_2+L_3)+(L_1L_2)} = \frac{G_1G_2G_3G_4}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2H_1G_3G_4H_2}$$

$$\boxed{(B)\;\; \frac{G_1G_2G_3G_4}{1+G_1G_2H_1+G_3G_4H_2+G_2G_3H_3+G_1G_2H_1G_3G_4H_2}}$$

Mistakes to be avoided

Touching vs. Non-touching: Ensure that loops designated as non-touching do not share even a single node.

Q2.15.2 MCQ 2015 1M Ans: B ● ○ ○

Given blocks have transfer functions $G_1(s)$ and $G_2(s)$.

(a) $G_1(s)G_2(s) \Rightarrow$ Realization is possible through a cascade connection.

(b) $\dfrac{G_1(s)}{G_2(s)} \Rightarrow$ Realization is **not possible** because division cannot be achieved through cascading, summing, or differencing.

(c) $G_1(s)\left(\dfrac{1}{G_1(s)} + G_2(s)\right) = 1 + G_1(s)G_2(s) \Rightarrow$ Realization is possible by summing a unity gain path (identity) with a cascade of $G_1(s)$ and $G_2(s)$.

(d) $G_1(s)\left(\dfrac{1}{G_1(s)} - G_2(s)\right) = 1 - G_1(s)G_2(s) \Rightarrow$ Realization is possible by differencing a unity gain path and a cascade of $G_1(s)$ and $G_2(s)$.

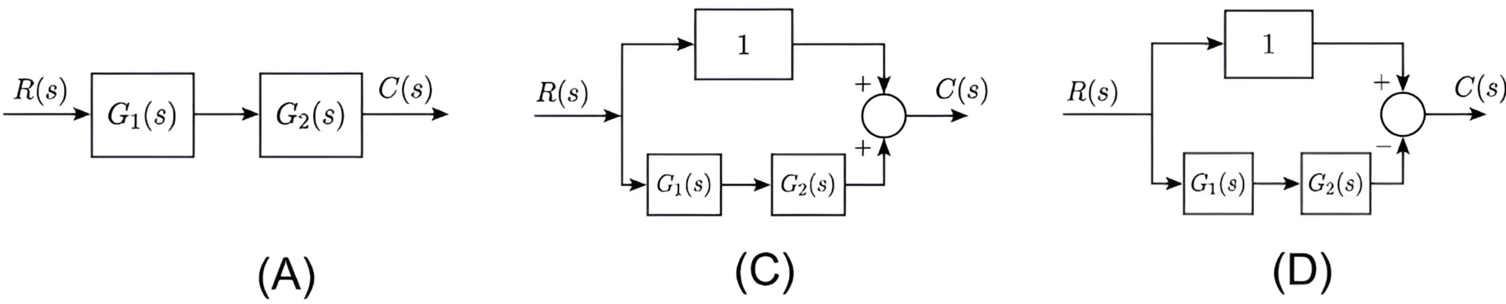


Fig. Solution Q2.15.2

$$\boxed{(B)\ \frac{G_1(s)}{G_2(s)}}$$

Key Points

Algebraic Constraint: Division of transfer functions $(\frac{G_1}{G_2})$ requires a feedback configuration or an explicit inverter block, which is not available through simple cascading or summing of G_1 and G_2.

Q2.15.3 | NAT | 2015 | 2M | Ans: -0.51 to -0.49 | ● ● ○

To find the steady-state position response due to the disturbance torque $T_d(s)$, we first determine the transfer function $\frac{\theta(s)}{T_d(s)}$ from the given block diagram.
Applying the closed-loop transfer function formula:

$$\frac{\theta(s)}{T_d(s)} = \frac{-\frac{1}{s}\left(\frac{1}{Js+B}\right)}{1-\left[\frac{K_T}{(R_a+L_as)}\frac{1}{(Js+B)}K_b(-1)\right]} = \frac{-\left(\frac{1}{s(Js+B)}\right)}{\frac{(R_a+L_as)(Js+B)+K_TK_b}{(R_a+L_as)(Js+B)}}$$

$$\frac{\theta(s)}{T_d(s)} = -\frac{(R_a+L_as)}{s[(R_a+L_as)(Js+B)+K_TK_b]}$$

For a unit impulse disturbance torque, $T_d(s) = 1$. The output position response is:

$$\theta(s) = -\frac{(R_a+L_as)}{s[(R_a+L_as)(Js+B)+K_TK_b]}$$

Using the Final Value Theorem to find the steady-state position response $\theta(\infty)$:

$$\theta(\infty) = \lim_{s\to 0} s\theta(s)$$

$$\theta(\infty) = \lim_{s\to 0} s\left[\frac{-(R_a+L_as)}{s[(R_a+L_as)(Js+B)+K_TK_b]}\right] = \frac{-R_a}{(R_aB)+K_TK_b}$$

Substituting the given parameter values: $K_T = 1$, $R_a = 1$, $B = 1$, and $K_b = 1$:

$$\theta(\infty) = \frac{-1}{(1)(1)+(1)(1)} = \frac{-1}{2} = -0.5$$

$$\boxed{\text{Steady State Position response= -0.5}}$$

Key Points

Transfer Function with Disturbance: When finding response to disturbance, treat other inputs (like $V_a(s)$) as zero.

Q2.14.1 MCQ 2014 1M Ans: D

Redrawing the block diagram with $X_1(s) = 0$:

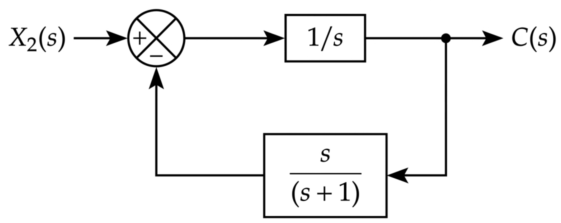


Fig. Solution Q2.14.1

Since it is a negative feedback system, the closed-loop transfer function is given by:

$$\frac{Y(s)}{X_2(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{1/s}{1+\frac{1}{s}\times\frac{s}{s+1}} = \frac{1/s}{\frac{(s+1)+1}{s+1}} = \frac{s+1}{s(s+2)}$$

$$\boxed{(D)\ \frac{s+1}{s(s+2)}}$$

Q2.14.2 MCQ 2014 1M Ans: C

Based on the provided block diagram, we can determine the transfer function using Mason's Gain Formula.

- Forward Paths: There are three forward paths from $R(s)$ to $C(s)$:

$$P_1 = G_1 \cdot G_2 \quad ; \quad P_2 = 1 \cdot G_2 = G_2 \quad ; \quad P_3 = 1 \cdot 1 = 1$$

- Loops: There are no feedback loops in the given diagram. $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 1$.

By using Mason's Gain Formula:

$$\text{Transfer function, } \frac{C(s)}{R(s)} = \frac{\sum P_i \Delta_i}{\Delta} = \frac{G_1G_2 + G_2 + 1}{1} = G_1G_2 + G_2 + 1$$

$$\boxed{(C)\ G_1G_2 + G_2 + 1}$$

Q2.13.1 MCQ 2013 2M Ans: A

We apply Mason's Gain Formula:

- Forward Paths:

$$P_1 = 1 \cdot s^{-1} \cdot s^{-1} \cdot 1 = \frac{1}{s^2} \qquad P_2 = 1 \cdot s^{-1} \cdot 1 \cdot 1 = \frac{1}{s}$$

- Individual Loops:

$$L_1 = \frac{-4}{s} \qquad L_2 = \frac{-2}{s^2} \qquad L_3 = \frac{-2}{s} \qquad L_4 = -4$$

- Path Factors (Δ_i):

$$\Delta_1 = 1 - 0 = 1 \qquad \Delta_2 = 1 - 0 = 1$$

- Determinant (Δ):

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) = 1 + \left(\frac{4}{s} + \frac{2}{s^2} + \frac{2}{s} + 4\right) = \frac{5s^2 + 6s + 2}{s^2}$$

$$\therefore \frac{Y(s)}{U(s)} = \frac{\sum P_i \Delta_i}{\Delta} = \frac{\frac{1}{s^2} \cdot 1 + \frac{1}{s} \cdot 1}{\left(\frac{5s^2 + 6s + 2}{s^2}\right)} = \frac{s+1}{5s^2 + 6s + 2}$$

$$\boxed{(A) \quad \frac{s+1}{5s^2 + 6s + 2}}$$

Q2.11.1 MCQ 2011 2M Ans: D ● ○ ○

The plant transfer function is given as:

$$H(s) = \frac{100}{s(s+10)^2} = \frac{100}{s(s^2 + 20s + 100)}$$

- From Option (A):

$$\text{TF} = \frac{\frac{1}{s} \cdot \frac{1}{s} \cdot \frac{1}{s} \cdot 100}{1 - \left(-\frac{10}{s} - \frac{10}{s}\right) + \left(-\frac{10}{s}\right)\left(-\frac{10}{s}\right)} = \frac{100}{s(s^2 + 20s + 100)}$$

- From Option (B):

$$\text{TF} = \frac{\frac{1}{s} \cdot \frac{1}{s} \cdot \frac{1}{s} \cdot 100}{1 - \left(-\frac{20}{s} - \frac{100}{s^2}\right)} = \frac{100}{s(s^2 + 20s + 100)}$$

- From Option (C):

$$\text{TF} = \frac{\frac{1}{s} \cdot \frac{1}{s} \cdot \frac{1}{s} \cdot 100}{1 - \left(-\frac{20}{s} - \frac{100}{s^2}\right)} = \frac{100}{s(s^2 + 20s + 100)}$$

- From Option (D):

$$\text{TF} = \frac{\frac{1}{s} \cdot \frac{1}{s} \cdot \frac{1}{s} \cdot 100}{1 - \left(-\frac{100}{s^2}\right)} = \frac{100}{s(s^2 + 100)}$$

Thus, Option (D) does not model the plant transfer function.

$$\boxed{D}$$

Q2.04.1 MCQ 2004 2M Ans: C ● ○ ○

To find the gain $\frac{x_5}{x_1}$ of the given signal flow graph, we apply Mason's Gain Formula:

- Forward Path: There is only one forward path from x_1 to x_5:

$$P_1 = a \cdot b \cdot c \cdot d = abcd$$

- Individual Loops: There are three individual feedback loops:

$$L_1 = b \cdot e = be \qquad L_2 = c \cdot f = cf \qquad L_3 = d \cdot g = dg$$

- Non-touching Loops:

$$\text{Product of non-touching loops} = L_1 \cdot L_3 = (be)(dg) = bedg$$

- Determinant (Δ):

$$\Delta = 1 - (L_1 + L_2 + L_3) + (L_1 L_3) = 1 - (be + cf + dg) + bedg$$

- Path Factor (Δ_1): Since the forward path touches all loops in the graph:

$$\Delta_1 = 1 - 0 = 1$$

$$\frac{x_5}{x_1} = \frac{P_1 \Delta_1}{\Delta} = \frac{abcd \cdot 1}{1 - (be + cf + dg) + bedg}$$

$$\boxed{(C) \quad \frac{abcd \cdot 1}{1 - (be + cf + dg) + bedg}}$$

Q2.03.1 MCQ 2003 2M Ans: D ● ○ ○

From the signal flow graph:

- Forward path: $P_1 = 1$; $\Delta_1 = 1 + \frac{3}{s} + \frac{24}{s} = \frac{s+27}{s}$
- Individual loops: $L_1 = \frac{-3}{s}, L_2 = \frac{-24}{s}, L_3 = \frac{-2}{s}$
- Non-touching loops: L_1 and L_3 are non-touching.

Substituting the values:

$$G(s) = \frac{\frac{(s+27)}{s}}{1 - \left(\frac{-3}{s} - \frac{24}{s} - \frac{2}{s}\right) + \left(\frac{-2}{s} \times \frac{-3}{s}\right)} = \frac{\frac{(s+27)}{s}}{1 + \frac{29}{s} + \frac{6}{s^2}} = \frac{s(s+27)}{s^2 + 29s + 6}$$

$$\boxed{(D) \quad \frac{s(s+27)}{s^2 + 29s + 6}}$$

Q2.01.1 MCQ 2001 1M Ans: D ● ○ ○

Moving the takeoff point from before a block to after a block requires dividing the branch gain by that block's transfer function.

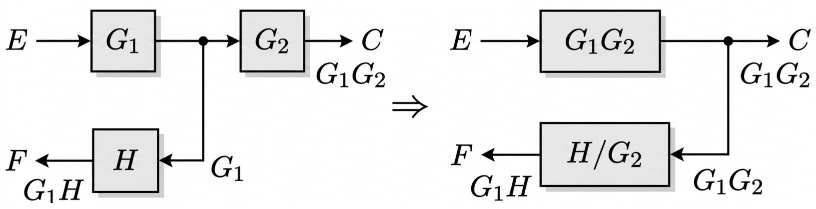


Fig. Solution Q2.01.1

$$\boxed{(D)}$$

Q2.01.2 MCQ 2001 2M Ans: C ● ● ○

To determine the transmittance values $G_2(s)$ and $H(s)$, we apply Kirchhoff's Voltage Law (KVL) to the circuit loops and compare them with the Signal Flow Graph (SFG) nodal equations.

Loop 1 Analysis:

$$V_i(s) = I_1(s)[Z_1(s) + Z_3(s)] - I_2(s)Z_3(s)$$

$$I_1(s) = \frac{V_i(s)}{Z_1(s) + Z_3(s)} + \frac{I_2(s)Z_3(s)}{Z_1(s) + Z_3(s)} \quad \ldots(1)$$

From the SFG:

$$I_1(s) = V_i(s)G_1(s) + I_2(s)H(s) \quad \ldots(3)$$

Comparing (1) and (3):

$$\therefore\ H(s) = \frac{Z_3(s)}{Z_1(s) + Z_3(s)}$$

Loop 2 Analysis:

$$[I_2(s) - I_1(s)]Z_3(s) + I_2(s)Z_2(s) + I_2(s)Z_4(s) = 0$$

$$I_2(s)[Z_2(s) + Z_3(s) + Z_4(s)] = I_1(s)Z_3(s)$$

$$I_2(s) = \frac{Z_3(s)}{Z_2(s) + Z_3(s) + Z_4(s)} I_1(s) \quad \ldots(2)$$

From the SFG:

$$I_2(s) = G_2(s)I_1(s) \quad \ldots(4)$$

Comparing (2) and (4):

$$G_2(s) = \frac{Z_3(s)}{Z_2(s) + Z_3(s) + Z_4(s)}$$

$$\boxed{(C)\ \frac{Z_3(s)}{Z_1(s) + Z_3(s)};\ \frac{Z_3(s)}{Z_2(s) + Z_3(s) + Z_4(s)}}$$

Q2.01.3 NAT 2001 5M Ans: * ● ○ ○

(a) At every summing point and take-off or pick-off point, consider the nodes as shown in the signal flow graph:

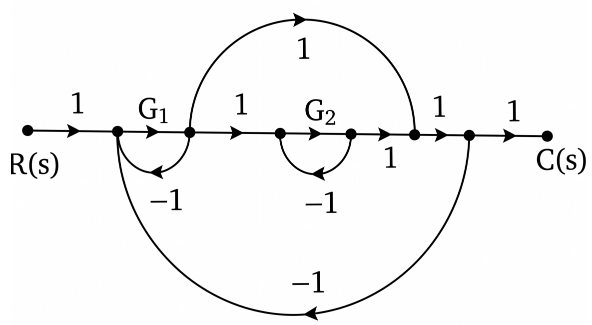


Fig. Solution Q2.01.3

(b) Individual Loops:

$$L_1 = -G_1, \quad L_2 = -G_2, \quad L_3 = -G_1, \quad L_4 = -G_1G_2$$

(c) Two Non-Touching Loops:

$$L_1L_2 = G_1G_2$$
$$L_2L_3 = G_1G_2$$

(d) Forward Paths: Number of forward paths = 2

$$P_1 = G_1G_2, \quad \Delta_1 = 1$$
$$P_2 = G_1, \quad \Delta_2 = 1 - (-G_2) = 1 + G_2$$

Applying Mason's Gain Formula, the overall transfer function is:

$$\text{TF} = \frac{C(s)}{R(s)} = \frac{G_1G_2 + G_1(1+G_2)}{1-(-G_1-G_2-G_1-G_1G_2)+G_1G_2+G_1G_2}$$
$$\frac{C(s)}{R(s)} = \frac{G_1G_2 + G_1(1+G_2)}{1+2G_1+G_2+3G_1G_2}$$

Q2.98.1 NAT 1998 5M Ans: * ● ○ ○

The signal flow graph for the given set of algebraic equations is drawn below:

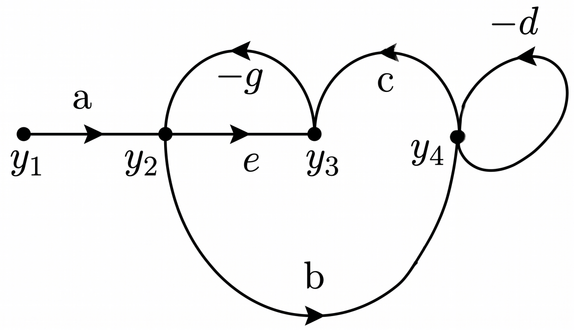


Fig. Solution Q2.98.1

From the constructed signal flow graph:

- Individual Loops (3): $L_1 = -eg$, $L_2 = -d$, $L_3 = -bcg$
- Two Non-Touching Loops (1 pair): $L_1L_2 = (-eg)(-d) = edg$
- Graph Determinant: $\Delta = 1 - (L_1 + L_2 + L_3) + L_1L_2 = 1 + eg + d + bcg + edg$

Case 1: Finding the gain $\frac{y_2}{y_1}$

- Forward paths from y_1 to y_2 is $P_1 = a$.
- Loop non-touching to P_1 is $L_2 = -d \implies \Delta_1 = 1 - (-d) = 1 + d$.

$$\boxed{\frac{y_2}{y_1} = \frac{a(1+d)}{1+d+eg+bcg+edg}}$$

Case 2: Finding the gain $\frac{y_3}{y_1}$

- Forward paths from y_1 to y_3 are:
  - $P_1 = ae$ (Touches all loops except $L_2 = -d$) $\implies \Delta_1 = 1 + d$
  - $P_2 = abc$ (Touches all loops) $\implies \Delta_2 = 1$

$$\boxed{\frac{y_3}{y_1} = \frac{ae(1+d)+abc}{1+d+eg+bcg+edg}}$$

Q2.97.1 MCQ 1997 2M Ans: C ● ○ ○

By applying Mason's Gain Formula, the transfer function is determined as follows:
Forward Path:

$$P_1 = 5 \times 2 \times 1 = 10$$

Loop Gain:

$$L_1 = 2 \times (-2) = -4$$

Determinant:

$$\Delta = 1 - L_1 = 1 - (-4) = 5$$

Transfer Function:

$$\frac{Y}{X} = \frac{P_1 \Delta_1}{\Delta} = \frac{10 \times 1}{5} = 2$$

$$\boxed{(C)\ \ 2}$$

Q2.95.1 MCQ 1995 1M Ans: C ● ○ ○

The signal flow graph (SFG) is a graphical representation of a set of linear algebraic equations. The main application of a signal flow graph is to determine the the **transfer function**.

$$\boxed{\text{(C)\ \ transfer function of the system}}$$

Mistakes to be avoided

Stability vs. Transfer Function: While the transfer function can be used to analyze stability (by looking at the poles), the SFG itself is a tool for finding the transfer function, not a direct method for determining stability.

Q2.94.1 NAT 1994 5M Ans: * ● ● ●

By eliminating the node e_5, the signal flow graph is transformed as shown below:

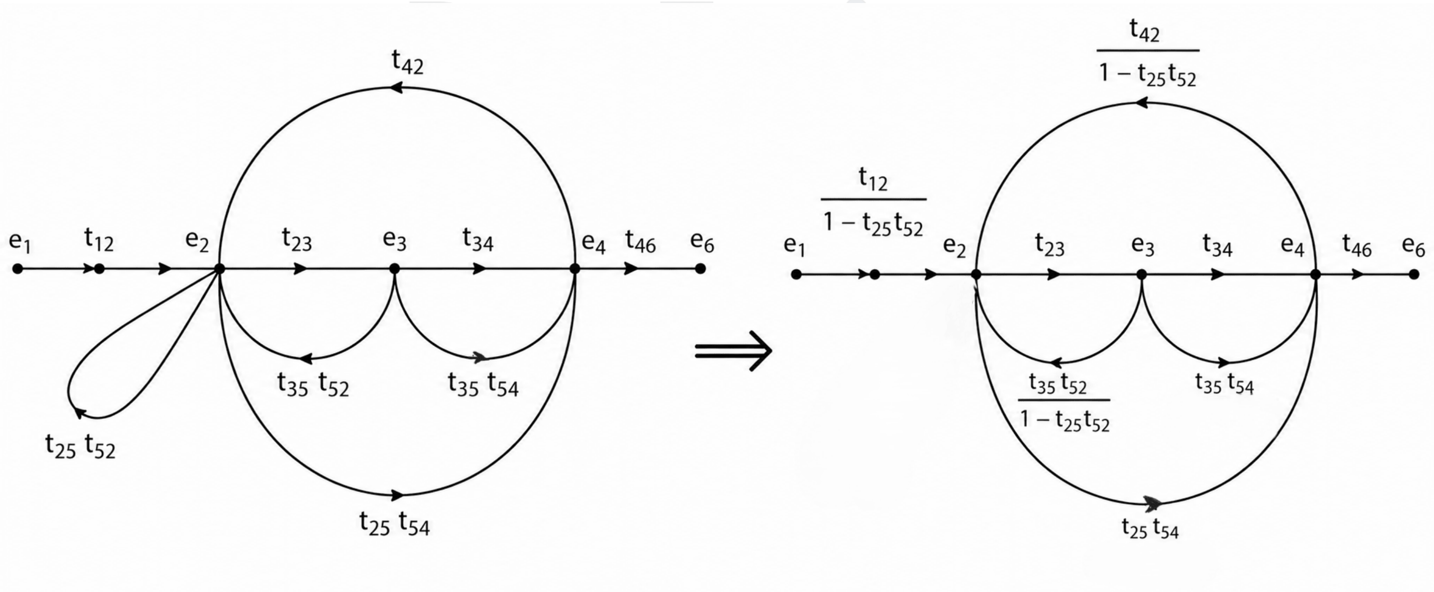


Fig. Solution Q2.94.1

The Node equation for e_2 is written as:

$$e_2 = e_1 t_{12} + e_2 t_{25} t_{52} + e_3 t_{35} t_{52} + e_4 t_{42}$$

$$e_2(1 - t_{25}t_{52}) = e_1t_{12} + e_3t_{35}t_{52} + e_4t_{42}$$
$$e_2 = e_1\frac{t_{12}}{1 - t_{25}t_{52}} + e_3\frac{t_{35}t_{52}}{1 - t_{25}t_{52}} + e_4\frac{t_{42}}{1 - t_{25}t_{52}}$$

Eliminating the self-loop at node e_2, the redrawn SFG is shown in the 2nd image

Mistakes to be avoided

When eliminating a node, ensure all possible combinations of incoming and outgoing paths passing through that node are included.

Q2.91.1

MCQ | 1991 | 2M | Ans: D | ● ○ ○

By applying Mason's Gain Formula to the signal flow graph:

$$\frac{C}{R} = \frac{\sum P_k\Delta_k}{\Delta} = \frac{(2\times3\times4\times1) + (5\times1)(3+1)}{1-(-2-3-4-5)+(-2)\times(-4)} = \frac{24+20}{1+14+8} = \frac{44}{23}$$

$$\boxed{(D)\ \frac{44}{23}}$$

Q2.89.1

MCQ | 1989 | 2M | Ans: C | ● ○ ○

The transfer function calculation for the given system using Mason's Gain Formula is shown below:

$$\frac{C}{R} = \frac{G_1G_2G_3G_4}{1-[G_1-G_2-G_3-G_4]+[G_1G_3+G_1G_4+G_2G_3+G_2G_4]}$$
$$\frac{C}{R} = \frac{G_1G_2G_3G_4}{1+G_1+G_2+G_3+G_4+G_1G_3+G_1G_4+G_2G_3+G_2G_4}$$

Factoring the denominator terms to obtain a product form:

$$\boxed{(C)\ \frac{G_1G_2G_3G_4}{(1+G_1+G_2)(1+G_3+G_4)}}$$

Q2.87.1

MCQ | 1987 | 2M | Ans: B | ● ○ ○

Method 1

Using Mason's Gain Formula:

$$\frac{C(s)}{R(s)} = \frac{\frac{10}{s(s+1)}}{1-\left(-\frac{10}{s(s+1)}-\frac{10s}{s(s+1)}\right)} = \frac{10}{s(s+1)+10+10s} = \frac{10}{s^2+11s+10}$$

Method 2

Using Block Diagram Reduction: The inner loop with feedback gain s has a transfer function $G_{inner}(s)$:

$$G_{inner}(s) = \frac{\frac{10}{s(s+1)}}{1+\frac{10}{s(s+1)}\times s} = \frac{10}{s(s+1)+10s}$$

$$\frac{C(s)}{R(s)} = \frac{G_{inner}(s)}{1 + G_{inner}(s) \times 1} = \frac{\frac{10}{s^2 + 11s}}{1 + \frac{10}{s^2 + 11s}} = \frac{10}{s^2 + 11s + 10}$$

$$\boxed{(B) \quad \frac{10}{s^2 + 11s + 10}}$$

Q2.87.2 MCQ 1987 2M Ans: D ● ○ ○

Applying Mason's Gain Formula to the signal flow graph: (1 forward path (5) and 1 loop (0.5))

$$T = \frac{X_2}{X_1} = \frac{5}{1 - 0.5} = \frac{5}{0.5} = 10$$

$$\Rightarrow X_2 = 10X_1$$

$$\text{T} = 10$$

$$\boxed{(D) \;\; 10}$$

DEBUG INFO

Chapter II

Block Diagram and Signal Flow Graphs

Total Questions : 27

MCQ : 21

MSQ : 1

NAT : 5

SOLUTIONS

III

Time Response Analysis

Topics

1. Impulse, Step, Ramp and Parabolic Response
2. Concept of Damping in 2nd Order Systems
3. Complex Frequency and Damping in RLC Circuits
4. Step Response of 2nd Order Systems
5. Time Domain Performance Parameters and their variation
6. Steady State Error - Unity and Non-Unity Feedback
7. Error Constants and System Types

Q3.24.1 MCQ 2024 1M Ans: D ● ● ○

For a unity feedback system, the characteristic equation is given by:

$$1 + G(s) = 0$$

$$1 + \frac{6}{s(s+1)(s+2)} = 0$$

$$s^3 + 3s^2 + 2s + 6 = 0$$

To check the stability of the system, we compare the product of the internal coefficients and external coefficients for this third-order characteristic equation $as^3 + bs^2 + cs + d = 0$:

$$\text{Internal product} = b \times c = 3 \times 2 = 6$$

$$\text{External product} = a \times d = 1 \times 6 = 6$$

Since the internal product is equal to the external product ($bc = ad$), the system is marginally stable.
For a marginally stable system, the roots lie on the $j\omega$ axis, which results in **sustained oscillations** in the time domain. Therefore, for a unit step input $r(t)$, the error signal $e(t)$ will be oscillatory, and the limit $\lim_{t\to\infty} e(t)$ **does not exist**.

$$\boxed{(D)\ \lim_{t\to\infty} e(t) \text{does not exist, } e(t) \text{is oscillatory}}$$

Mistakes to be avoided

Do not blindly apply the Final Value Theorem to find steady-state error without first verifying the stability of the closed-loop system. The Final Value Theorem ($\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$) is only applicable if all poles of $sF(s)$ lie in the left half of the s-plane.

Q3.23.1 MCQ 2023 2M Ans: A ● ○ ○

From the given block diagram, the open-loop transfer function is:

$$G(s)H(s) = \frac{k}{s(s+2)}$$

The steady-state error e_{ss} for a ramp input in a unity feedback system is defined as:

$$e_{ss} = \frac{\alpha}{K_v}$$

where α is the magnitude of the ramp input and K_v is the static velocity error constant.

$$K_v = \lim_{s\to 0} [sG(s)H(s)]$$

$$K_v = \lim_{s\to 0} \left[\frac{s \times k}{s(s+2)}\right] = \frac{k}{2}$$

$$e_{ss} = \frac{\alpha}{k/2} = \frac{2\alpha}{k}$$

$$\boxed{(A)\ \frac{2\alpha}{k}}$$

Mistakes to be avoided

Ensure the system is Type-1 before using the K_v formula for a **finite error**; Type-0 systems result in **infinite error** and Type-2 systems result in **zero error** for ramp inputs.

Q3.22.1 MSQ 2022 2M Ans: A ● ○ ○

For system 1:

$$G_1(s) = \frac{10}{s^2+s+1}$$

Characteristic equation: $s^2+s+1=0$
Comparing with standard eqn: $s^2+2\xi\omega_n s+\omega_n^2=0$

$$\omega_n = 1, \quad \xi = 0.5$$

$$\omega_d = 1\sqrt{1-(0.5)^2} = 0.866$$

Settling time (2% criterion):

$$t_{s1} = \frac{4}{\xi\omega_n} = \frac{4}{0.5\times 1} = 8\,\text{sec}$$

Steady-state value:

$$y_1(\infty) = \lim_{s\to 0}\frac{10}{s^2+s+1} = 10$$

For system 2:

$$G_2(s) = \frac{10}{s^2+\sqrt{10}s+10}$$

Characteristic equation: $s^2+\sqrt{10}s+10=0$
Comparing with standard eqn: $s^2+2\xi\omega_n s+\omega_n^2=0$

$$\omega_n = \sqrt{10}, \quad \xi = 0.5$$

$$\omega_d = \sqrt{10}\sqrt{1-(0.5)^2} = 2.739$$

Settling time (2% criterion):

$$t_{s2} = \frac{4}{0.5\sqrt{10}} = 2.535\,\text{sec}$$

Steady-state value:

$$y_2(\infty) = \lim_{s\to 0}\frac{10}{s^2+\sqrt{10}s+10} = 1$$

Since the damping ratio ξ is the same (0.5) for both systems, the percentage peak overshoot ($M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}\times 100\%$) is identical.

$$M_{p1} = M_{p2}$$

$$\boxed{(A)\quad y_1(t) \text{ and } y_2(t) \text{ have the same percentage peak overshoot}}$$

Key Points

Steady-state value of a unit step response is given by the DC gain of the system, $\lim_{s\to 0} G(s)$.

Q3.22.2 NAT 2022 2M Ans: -0.11 to -0.09 ● ● ○

$R(s)=0$, So the given above block diagram can be reduced as below,

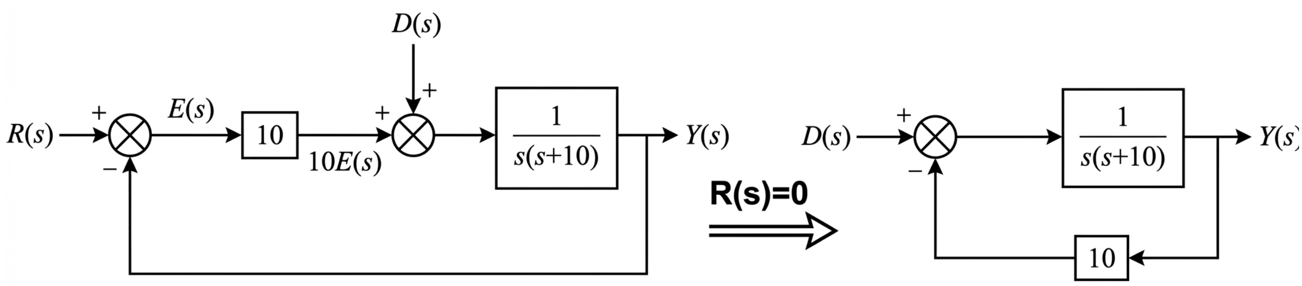


Fig. Solution Q3.22.2

Thus, the transfer function of the reduced block diagram is,

$$\frac{Y(s)}{D(s)} = \frac{\frac{1}{s(s+10)}}{1+\frac{10}{s(s+10)}} = \frac{1}{s(s+10)+10} \quad \ldots \text{(i)}$$

According to the question, the error signal is defined as $e(t) = r(t) - y(t)$. Since $r(t) = 0$:

$$e(t) = -y(t) \implies E(s) = -Y(s)$$

Substituting this into equation (i):

$$-E(s) = \frac{D(s)}{s(s+10)+10}$$

Steady-state error $e(\infty)$ under a unit step disturbance $\left[D(s) = \frac{1}{s}\right]$ is calculated using the **Final Value Theorem**:

$$e(\infty) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\left[\frac{-D(s)}{s(s+10)+10}\right] = \lim_{s\to 0} s\left[\frac{-\frac{1}{s}}{s(s+10)+10}\right] = -\frac{1}{10} = -0.1$$

$$\boxed{e(\infty) = -0.1}$$

Q3.20.1 NAT 2020 2M Ans: 30 ● ○ ○

Open loop transfer function is given by,

$$\text{OLTF} = G(s)C(s)$$

$$G'(s) = \frac{1}{s(s+1)}\frac{K(s+1)}{(s+3)} = \frac{K}{s(s+3)}$$

Given: For ramp input, $e_{ss} = 0.1$
K_v(Velocity error constant) :

$$K_v = \lim_{s\to 0} s[\text{OLTF}]$$

$$K_v = \lim_{s\to 0} sG'(s) = \lim_{s\to 0} s\frac{K}{s(s+3)} = \frac{K}{3}$$

Steady state error is given by,

$$e_{ss} = \frac{1}{K_v} \implies e_{ss} = \frac{1}{K/3} = \frac{3}{K}$$

So, $\frac{3}{K} = 0.1 \implies K = 30$

$$\boxed{K = 30}$$

Key Points

Shortcut to find K_v:

$$K_v = \frac{\text{Numerator Constant of OLTF}}{\text{Denominator Constant of OLTF}}$$

Q3.19.1 MCQ | 2019 | 2M | Ans: B | ● ● ○

$$G(s) = \frac{2}{s^2 + 2s + 1} = \frac{1}{(s+1)^2}$$

$$C(s) = G(s)R(s) = \frac{1}{s(s+1)^2}$$

Using partial fraction expansion, we get,

$$C(s) = \frac{A}{s} + \frac{B}{(s+1)} + \frac{C}{(s+1)^2}$$

$$\therefore\ C(s) = \frac{1}{s} - \frac{1}{(s+1)} - \frac{1}{(s+1)^2}$$

$$c(t) = (1 - e^{-t} - te^{-t})u(t)$$

$$\lim_{t\to\infty} c(t) = 1$$

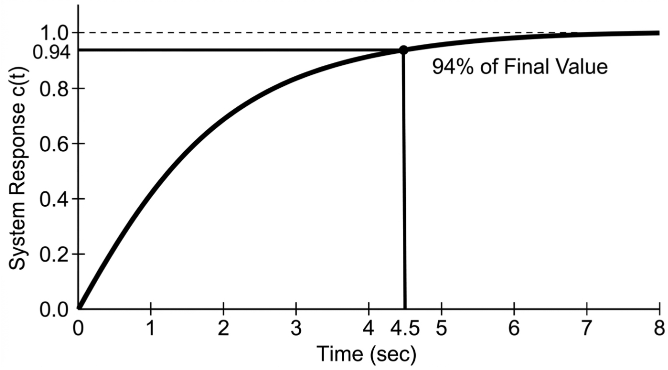


Fig. Solution Q3.19.1

In order to reach 94% of its steady-state value : $(1 - e^{-t} - te^{-t}) = 0.94$
By trial and error, we get,

$$\boxed{(B)\ \ t \approx 4.50\,\text{sec}}$$

Q3.17.1 NAT | 2017 | 1M | Ans: 0.99 to 1.01 | ● ○ ○

Given: Steady-state error for unit step input = 0. Steady-state error for unit ramp input = 6 (finite).
Analysis: Based on the steady-state error relationship with system type and input:

Condition	e_{ss}
Type = Input	Constant
Type > Input	0
Type < Input	∞

- **Type-0 & Step:** $e_{ss} = \dfrac{A}{1 + K_p}$
- **Type-1 & Ramp:** $e_{ss} = \dfrac{A}{K_v}$
- **Type-2 & Parabolic:** $e_{ss} = \dfrac{A}{K_a}$

The given system must be of Type-1.
In the transfer function $G(s) = \dfrac{(s+1)}{s^p(s+2)(s+3)}$, the parameter p defines the system type.

$$\boxed{p = 1}$$

Mistakes to be avoided

Order vs Type: Do not confuse system order (total poles) with system type (poles at origin).

Q3.16.1 NAT | 2016 | 1M | Ans: 0.96 to 1.04 | ● ● ○

Given the transfer function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{s-2}{(s+1)(s+3)}$$

For a unit step input, $U(s) = \frac{1}{s}$. Thus, the response in s-domain is:

$$Y(s) = \frac{s-2}{s(s+1)(s+3)}$$

Method 1

Using the **Initial Value Theorem** for the derivative:

$$\mathcal{L}\left[\frac{dy}{dt}\right] = sY(s) - y(0^-)$$

Assuming zero initial conditions:

$$\mathcal{L}\left[\frac{dy}{dt}\right] = sY(s) = \frac{s-2}{(s+1)(s+3)}$$

By Initial Value Theorem:

$$\left.\frac{dy}{dt}\right|_{t=0^+} = \lim_{s\to\infty} s[sY(s)]$$

Evaluating the limit:

$$\left.\frac{dy}{dt}\right|_{t=0^+} = \lim_{s\to\infty} \frac{s(s-2)}{(s+1)(s+3)} = 1$$

Method 2

Using **Partial Fraction Expansion**:

$$Y(s) = \frac{-2}{3s} + \frac{3}{2(s+1)} - \frac{5}{6(s+3)}$$

Taking the inverse Laplace transform:

$$y(t) = \left[-\frac{2}{3} + \frac{3}{2}e^{-t} - \frac{5}{6}e^{-3t}\right]u(t)$$

Differentiating with respect to t:

$$\frac{dy}{dt} = -\frac{3}{2}e^{-t} + \frac{5}{2}e^{-3t}$$

At $t = 0^+$:

$$\left.\frac{dy}{dt}\right|_{t=0^+} = -\frac{3}{2} + \frac{5}{2} = 1$$

$$\boxed{1}$$

Mistakes to be avoided

Theorem Application: Ensure you are applying the limit to the Laplace transform of the derivative, not just the original $Y(s)$.

Q3.16.2 NAT 2016 2M Ans: 2.7 to 3.0 ● ○ ○

Given: Peak overshoot $(M_p) = 10\% = 0.1$; Open-loop transfer function: $G(s) = \dfrac{K}{s(s+2)}$

Analysis: The peak overshoot is related to the damping ratio (ξ) as:

$$M_p = e^{-\pi\xi/\sqrt{1-\xi^2}}$$

$$0.1 = e^{-\pi\xi/\sqrt{1-\xi^2}} \implies \ln(0.1) = \frac{-\pi\xi}{\sqrt{1-\xi^2}}$$

Solving for ξ, we get $\xi \approx 0.59$.

The characteristic equation (CE) for unity feedback is:

$$1 + G(s) = 0 \implies 1 + \frac{K}{s(s+2)} = 0 \implies s^2 + 2s + K = 0$$

Comparing this with the standard second-order equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$2\xi\omega_n = 2 \implies \xi\omega_n = 1$$

Substituting $\xi = 0.59$:

$$0.59 \times \omega_n = 1 \implies \omega_n = \frac{1}{0.59} \approx 1.69\,\text{rad/sec}$$

$$K = \omega_n^2 = (1.69)^2 \approx 2.87$$

$$\boxed{K = 2.87}$$

Q3.16.3 NAT 2016 2M Ans: 0.95 to 1.05 ● ● ○

Given the open-loop transfer function:

$$G(s) = \frac{1}{s^2 + 2s}$$

The characteristic equation for the unity feedback system shown in the figure is:

$$1 + KG(s) = 0 \implies 1 + \frac{K}{s^2 + 2s} = 0 \implies s^2 + 2s + K = 0$$

Comparing this with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$2\xi\omega_n = 2 \implies \xi\omega_n = 1$$

$$\omega_n^2 = K \implies \omega_n = \sqrt{K}$$

For the step response to have **no overshoot** and **minimum settling time**, the system must be **critically damped** : $\xi = 1$

$$1 \cdot \omega_n = 1 \implies \omega_n = 1\,\text{rad/s}$$

$$K = \omega_n^2 = (1)^2 = 1$$

$$\boxed{K = 1}$$

Key Points

- **No Overshoot:** Occurs when the damping ratio $\xi \geq 1$.
- **Minimum Settling Time (without overshoot):** Achieved exactly at the critically damped condition where $\xi = 1$.

Q3.15.1 NAT 2015 1M Ans: 400 ● ○ ○

The characteristic equation for the unity feedback system is:

$$1 + G(s) = 0 \implies 1 + \frac{K}{s(s+10)} = 0$$

$$s^2 + 10s + K = 0$$

Comparing with the standard second-order equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = K \implies \omega_n = \sqrt{K}$$

$$2\xi\omega_n = 10 \implies \xi\omega_n = 5$$

Given the damping ratio $\xi = 0.25$:

$$0.25 \times \sqrt{K} = 5$$

$$\sqrt{K} = \frac{5}{0.25} = 20$$

$$\boxed{K = 400}$$

Q3.14.1 | NAT | 2014 | 1M | Ans: 38.13 to 38.19 | ● ○ ○

Given : $\omega_n = 40$ rad/sec; Damping ratio, $\xi = 0.3$

$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

$$\omega_d = 40\sqrt{1-(0.3)^2} = 40\sqrt{0.91} = 38.1576\,\text{rad/sec}$$

$$\boxed{\omega_d = 38.16\,\text{rad/sec}}$$

Key Points

- This formula specifically applies to underdamped systems ($0 < \xi < 1$).
- For a critically damped system ($\xi = 1$) or overdamped system ($\xi > 1$), the system does not oscillate, and ω_d is not defined in the same context.

Q3.14.2 | MCQ | 2014 | 1M | Ans: C | ● ○ ○

The characteristic equation for a closed-loop system is given by $1 + G(s)H(s) = 0$. For unity feedback ($H(s) = 1$):

$$1 + \frac{4}{s(s+4)} = 0$$

$$s^2 + 4s + 4 = 0$$

Comparing this with the standard form of the second-order characteristic equation: $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$\omega_n^2 = 4 \implies \omega_n = 2\,\text{rad/sec}$$

$$\boxed{(C)\ 2}$$

Q3.14.3 | NAT | 2014 | 2M | Ans: 0.49 to 0.51 | ● ● ○

The error signal $E(s)$ in a feedback system is the signal after the summing junction, defined as:

$$E(s) = R(s) - C(s)H(s)$$

From the block diagram, the output is $C(s) = E(s)G(s)$. Substituting this into the error equation:

$$R(s) - (E(s)G(s))H(s) = E(s)$$

$$E(s)[1 + G(s)H(s)] = R(s) \implies \frac{E(s)}{R(s)} = \frac{1}{1+G(s)H(s)}$$

For a unit step input, $R(s) = \frac{1}{s}$. The steady-state error e_{ss} is found using the Final Value Theorem:

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\left[R(s)\frac{1}{1+G(s)H(s)}\right]$$

Given $G(s) = \frac{4}{s+2}$ and $H(s) = \frac{2}{s+4}$:

$$e_{ss} = \lim_{s\to 0} s\cdot\frac{1}{s}\left[\frac{1}{1+\left(\frac{4}{s+2}\right)\left(\frac{2}{s+4}\right)}\right] = \frac{1}{1+\left(\frac{4}{2}\right)\left(\frac{2}{4}\right)} = \frac{1}{1+1} = 0.5$$

$$\boxed{e_{ss} = 0.5}$$

Key Points

Actuating error for non-unity feedback: $E(s) = \frac{R(s)}{1+G(s)H(s)}$.

Mistakes to be avoided

- $e_{ss} = \lim_{s\to 0} s[R(s) - C(s)]$ is strictly for unity feedback systems ($H(s) = 1$). Do not use it for non-unity feedback systems.
- **Conversion to Unity Feedback:** Converting a non-unity system to an equivalent unity feedback system results in a wrong answer for error because the "Error Signal" is redefined.
  - The Core Conflict: While the conversion preserves the overall Transfer Function $\frac{C(s)}{R(s)}$, it changes the physical location and mathematical definition of the error signal.
  - Hence,for steady-state error analysis, always solve the system in its original non-unity form.

Q3.13.1 MCQ 2013 2M Ans: C ● ○ ○

Given the open-loop transfer function:

$$G(s) = \frac{10K_a}{1+10s}$$

The time constant of the open-loop system is : $T_{OLTF} = 10\,\text{sec}$.

$$\text{CLTF} = \frac{G(s)}{1+G(s)} = \frac{\frac{10K_a}{1+10s}}{1+\frac{10K_a}{1+10s}} = \frac{10K_a}{10s + (1+10K_a)}$$

The time constant of the closed-loop system is : $T_{CLTF} = \frac{10}{1+10K_a}$.

According to the problem, the closed-loop time constant should be $\frac{1}{100}$ of the open-loop time constant:

$$T_{CLTF} = \frac{1}{100} T_{OLTF} \implies \frac{10}{1+10K_a} = \frac{10}{100}$$

$$1 + 10K_a = 100 \implies K_a = 9.9$$

$$\boxed{(C)\ \ 10}$$

Key Points

- For a first-order system $G(s) = \frac{K}{\tau s+1}$, τ is the time constant.
- Feedback reduces the time constant by a factor of $(1 + GH)$, thereby increasing the bandwidth and speed of response.

Q3.11.1 MCQ 2011 1M Ans: A ● ○ ○

Given the differential equation for the system:

$$100\frac{d^2y}{dt^2} - 20\frac{dy}{dt} + y = x(t)$$

Taking the Laplace transform on both sides (assuming zero initial conditions):

$$100s^2Y(s) - 20sY(s) + Y(s) = X(s)$$

$$G(s) = \frac{Y(s)}{X(s)} = \frac{1}{100s^2 - 20s + 1} = \frac{1}{(10s-1)^2}$$

For a unit step input $x(t) = u(t)$, $X(s) = \frac{1}{s}$. The output $Y(s)$ is:

$$Y(s) = \frac{1}{s(10s-1)^2}$$

The poles of the system are located at:

$$s = 0, \frac{1}{10}, \frac{1}{10}$$

Since the closed-loop poles $s = 0.1$ lie on the RHS of the s-plane, the system is unstable. An unstable system's step response will grow without bound over time.
Among the given options in the reference figure, only **option (A)** represents an unstable, unbounded waveform.

Q3.09.1 MCQ 2009 2M Ans: B ● ○ ○

Given that the steady-state value of the unit step response is −2 and the system is an underdamped second-order system. All options satisfy the steady-state value.

Check for stability and the underdamped condition $(0 < \xi < 1)$:

- Options (c) and (d) have a negative coefficient for the s term in the denominator ($s^2 - 2.59s + 1.12$ and $s^2 - 1.91s + 1.91$), which implies the poles are in the right-half plane, making the systems **unstable**.
- For option (a): $\omega_n^2 = 1.12 \Rightarrow \omega_n = 1.058$.

$$2\xi\omega_n = 2.59 \Rightarrow \xi = \frac{2.59}{2 \times 1.058} = 1.22$$

  Since $\xi > 1$, option (a) is **overdamped**.
- For option (b): $\omega_n^2 = 1.91 \Rightarrow \omega_n = \sqrt{1.91} \approx 1.38$.

$$2\xi\omega_n = 1.91 \Rightarrow \xi = \frac{1.91}{2 \times 1.38} \approx 0.69$$

  Since $0 < \xi < 1$, the system is **underdamped**.

$$\boxed{(B)\ \frac{-3.82}{s^2 + 1.91s + 1.91}}$$

Mistakes to be avoided

Do not forget to check the stability of the system first; a system with negative coefficients in the characteristic equation is unstable.

Q3.08.1 MCQ 2008 1M Ans: C ● ○ ○

The percentage peak overshoot ($\%M_p$) for a second-order underdamped system is defined by:

$$\%M_p = e^{-\left(\frac{\pi\xi}{\sqrt{1-\xi^2}}\right)} \times 100\%$$

$$\text{where, } \xi = \cos\theta$$

Given that the percentage overshoot is the same for all three systems, the damping ratio ξ must be the same for all three. This implies that the **angle** θ **must remain constant** for all the poles.

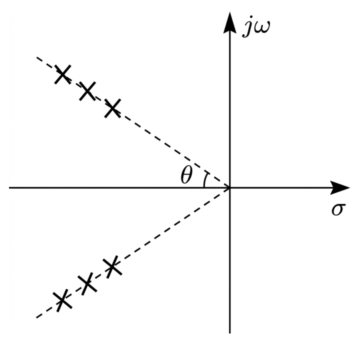


Fig. Solution Q3.08.1

Q3.08.2 MCQ 2008 2M Ans: D ● ○ ○

Compare Group I transfer functions with the standard second-order characteristic equation:

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

- For P: $s^2 + 25 = 0$. Here $\omega_n = 5$ and $2\xi\omega_n = 0 \Rightarrow \xi = 0$. The system is **undamped**, resulting in continuous oscillations. Matches with waveform (3).
- For Q: $s^2 + 20s + 36 = 0$. Here $\omega_n = 6$ and $2\xi\omega_n = 20 \Rightarrow \xi = \frac{20}{12} = 1.66$. Since $\xi > 1$, the system is **overdamped** resulting in sluggish response without overshoot. Matches with waveform (4).
- For R: $s^2 + 12s + 36 = 0$. Here $\omega_n = 6$ and $2\xi\omega_n = 12 \Rightarrow \xi = \frac{12}{12} = 1$. Since $\xi = 1$, the system is **critically damped** resulting in fastest response without overshoot. Matches with waveform (1).
- For S: $s^2 + 7s + 49 = 0$. Here $\omega_n = 7$ and $2\xi\omega_n = 7 \Rightarrow \xi = \frac{7}{14} = 0.5$. Since $0 < \xi < 1$, the system is **underdamped** resulting in decaying oscillations. Matches with waveform (2).

$$\boxed{(D)\ \ P-3, Q-4, R-1, S-2}$$

Q3.08.3 MCQ 2008 2M Ans: C ● ○ ○

Based on the given poles and zero, the transfer function TF is formulated as:

$$TF = \frac{K(s+1)}{(s+2)(s+4)}$$

Given the input is a unit step, $R(s) = \frac{1}{s}$. The output in the s-domain is:

$$C(s) = R(s) \cdot TF = \frac{1}{s} \cdot \frac{K(s+1)}{(s+2)(s+4)}$$

The steady-state output is given as 1 (Using Final Value Theorem):

$$\lim_{t\to\infty} c(t) = 1 = \lim_{s\to 0} sC(s)$$

$$\lim_{s\to 0} s \cdot \frac{K(s+1)}{s(s+2)(s+4)} = 1$$

$$\frac{K(1)}{(2)(4)} = 1 \implies \frac{K}{8} = 1 \implies K = 8$$

Now, substituting $K = 8$ into the transfer function and performing partial fraction expansion:

$$TF = \frac{8(s+1)}{(s+2)(s+4)} = \frac{-4}{s+2} + \frac{12}{s+4}$$

$$IR = \mathcal{L}^{-1}[TF] = (-4e^{-2t} + 12e^{-4t})u(t)$$

$$\boxed{(C)\ \ (-4e^{-2t} + 12e^{-4t})u(t)}$$

Q3.07.1 MCQ | 2007 | 2M | Ans: D

The given transfer function is:

$$T(s) = \frac{5}{(s+5)(s^2+s+1)}$$

The poles of the system are located at $s = -5$ and $s = -1/2 \pm j\sqrt{3}/2$.

The pole at $s = -5$ is a fast pole, whose time constant $(1/5)$ sec is '5' times less than the time constant of the dominant pole $(1/0.5)$ sec $= 2$ sec.
Hence, we neglect the pole at $s = -5$ while ensuring the steady-state gain remains unchanged.

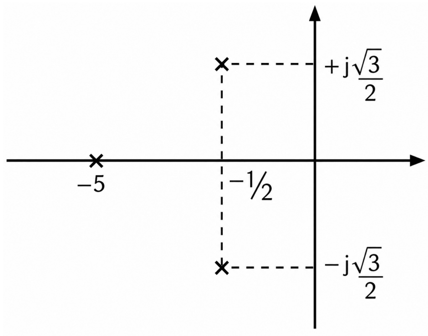


Fig. Solution Q3.07.1

To keep the DC gain constant, we represent the term in time-constant form $(1 + s\tau)$:

$$T(s) = \frac{5}{5\left(1+\frac{s}{5}\right)(s^2+s+1)}$$

Approximating by neglecting the fast pole ($\frac{s}{5} \to 0$):

$$\boxed{(D)\quad \frac{1}{s^2+s+1}}$$

ity

Key Points

- **Dominant Poles:** Poles closer to the imaginary axis dominate the transient response as they have larger time constants.
- **Insignificant Poles:** Poles located far to the left (typically > 5 times the real part of dominant poles) can be neglected.

Mistakes to be avoided

Do not simply remove the $(s+5)$ term without normalizing the numerator; failing to account for the gain will result in an incorrect steady-state value.

Q3.05.1 MCQ | 2005 | 2M | Ans: A

Steady state error to a Ramp input is

$$e_{ss} = \frac{A}{K_v}$$

Given steady state error is 5%:

$$e_{ss} = 0.05, A = 1, (\text{Type} = 1)$$

$$0.05 = \frac{1}{K_v}$$

$$K_v = \frac{1}{0.05} = 20$$

$$\boxed{(A)\quad 1 \text{ and } 20}$$

Q3.05.2 MCQ | 2005 | 2M | Ans: C | ● ● ○

The standard expression for the peak percent overshoot (M_p) of a second-order system is given by:

$$M_p = \exp\left(\frac{-\pi\xi}{\sqrt{1-\xi^2}}\right) \times 100\%$$

In the derivation of this specific formula, the following conditions are required:

- The system must be Linear and Time-Invariant (LTI).
- The system is assumed to have zero initial conditions to utilize transfer function analysis.
- The transfer function is a standard second-order system with a pair of complex conjugate poles and no zeros.

The presence or absence of a transportation delay does not change the peak overshoot value itself, only the time at which that peak occurs (t_p) changes.

$$\boxed{\text{(C) There is no transportation delay in the system}}$$

Q3.04.1 MCQ | 2004 | 2M | Ans: C | ● ○ ○

Given $G(s) = \frac{1}{s+2}$ and $r(t) = 10u(t)$.
Output in s-domain:

$$C(s) = G(s)R(s) = \frac{10}{s(s+2)}$$

Using partial fractions:

$$\frac{10}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \implies A = 5, B = -5$$

$$C(s) = \frac{5}{s} - \frac{5}{s+2}$$

Applying Inverse Laplace Transform:

$$c(t) = 5(1 - e^{-2t})u(t)$$

As $t \to \infty$, steady state value = 5.
For 99% of steady state value:

$$5(1 - e^{-2t}) = 5 \times 0.99$$
$$1 - e^{-2t} = 0.99$$
$$e^{-2t} = 0.01$$
$$-2t = \ln(0.01) \approx -4.60$$
$$t = 2.3 \text{ sec}$$

$$\boxed{(C)\ \ 2.3 \text{ sec}}$$

Key Points

- **Step Response:** For a first order system $\frac{1}{s\tau + 1}$, the response is $(1 - e^{-t/\tau})$.
- **Settling Time:** The time taken to reach within a specified percentage (99% or 95%) of the final value.

Q3.03.1 MCQ | 2003 | 2M | Ans: B | ● ○ ○

Given the Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 4s + 4}$$

Comparing the denominator with $s^2 + 2\xi\omega_n s + \omega_n^2$:

$$\omega_n^2 = 4 \implies \omega_n = 2 \text{ rad/s} \quad ; \quad 2\xi\omega_n = 4 \implies \xi = \frac{4}{2 \times 2} = 1$$

Since $\xi = 1$, the system is **critically damped**. For a critically damped system, the step response is an exponential rise without oscillations.
The settling time (t_s) for a 2% tolerance band is:

$$t_s = 4\tau = \frac{4}{\xi\omega_n} = \frac{4}{1 \times 2} = 2\,\text{sec}$$

The waveform in **option (B)** represents a critically damped response with $t_s \approx 2$ sec.

$$\boxed{(B)}$$

Q3.02.1 | MCQ | 2002 | 1M | Ans: C | ● ○ ○

Given transfer function is:

$$G(s) = \frac{s+6}{Ks^2+s+6} = \frac{s+6}{K\left[s^2+\frac{s}{K}+\frac{6}{K}\right]}$$

Comparing the denominator with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = \frac{6}{K} \implies \omega_n = \sqrt{\frac{6}{K}}$$

$$2\xi\omega_n = \frac{1}{K}$$

Given damping ratio $\xi = 0.5$:

$$2 \times 0.5 \times \sqrt{\frac{6}{K}} = \frac{1}{K}$$

Squaring both sides:

$$\frac{6}{K} = \frac{1}{K^2} \implies 6 = \frac{1}{K} \implies K = \frac{1}{6}$$

$$\boxed{(C)\ \frac{1}{6}}$$

Mistakes to be avoided

Ensure the coefficient of s^2 is unity before comparing coefficients.

Q3.02.2 | MCQ | 2002 | 2M | Ans: B | ● ● ○

The given transfer function is:

$$G(s) = \frac{100}{(s+1)(s+100)}$$

The poles of the system are located at $s = -1$ and $s = -100$. Since these poles are real and distinct, the system is **overdamped**, and the standard second-order formulas for oscillatory systems are not applicable.

In such cases, the transient response is determined by the dominant pole, which is the pole closest to the imaginary axis. Here, the dominant pole is at $s = -1$. The corresponding time constant (τ) is:

$$\tau = \frac{1}{|\text{pole}|} = \frac{1}{1} = 1\,\text{sec}$$

For the 2% settling time criterion, the settling time (t_s) is approximately 4τ:

$$t_s = 4\tau = 4 \times 1 = 4\,\text{sec}$$

$$\boxed{(B)\ 4\,\text{sec}}$$

Q3.01.1 MCQ | 2001 | 1M | Ans: C | ● ○ ○

The given characteristic equation is: $s^2 + 2s + 2 = 0$ Comparing this with the standard second-order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

- $\omega_n^2 = 2 \implies \omega_n = \sqrt{2} \approx 1.414\,\text{rad/s}$
- $2\xi\omega_n = 2 \implies \xi = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.707$

Since the damping ratio lies in the range $0 < \xi < 1$, the system is classified as **underdamped**.

$$\boxed{\text{(C) underdamped}}$$

Q3.00.1 NAT | 2000 | 5M | Ans: * | ● ○ ○

$$G_{\text{cl}}(s) = \frac{G(s)}{1 + G(s)H(s)} = \frac{\frac{G}{s(s+3)}}{1 + \frac{G}{s(s+3)}} = \boxed{\frac{G}{s^2 + 3s + G}}$$

Q3.00.2 NAT | 2000 | 5M | Ans: * | ● ○ ○

The system's characteristic equation is:

$$s^2 + 3s + G = 0$$

Comparing with the standard second-order system equation $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \sqrt{G}, \quad 2\xi\omega_n = 3 \implies \xi = \frac{3}{2\sqrt{G}}$$

The peak overshoot from the step response graph is $M_p = 0.1$ (10%):

$$M_p = \exp\left(\frac{-\pi\xi}{\sqrt{1-\xi^2}}\right) = 0.1 \implies \xi \approx 0.5912 \approx 0.6$$

$$\sqrt{G} = \frac{3}{2 \times 0.6} = 2.5 \implies G = 2.5^2 = \boxed{6.25}$$

Q3.00.3 NAT | 2000 | 5M | Ans: * | ● ● ○

For $G' = 2G = 2 \times 6.25 = 12.5$:

$$\omega_n = \sqrt{12.5} \approx 3.535\,\text{rad/s}$$

Since $2\xi\omega_n = 3$, the new damping factor becomes:

$$\xi = \frac{3}{2 \times 3.535} \approx 0.424$$

The damped frequency of oscillation ω_d is calculated as:

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 3.535\sqrt{1 - 0.424^2} \approx 3.20\,\text{rad/s}$$

The time period T corresponding to the oscillation cycles is:

$$T = \frac{2\pi}{\omega_d} = \frac{2\pi}{3.20} \approx \boxed{1.96\,\text{sec}}$$

Key Points

The time interval between successive peak overshoots or successive undershoots is given by $T = \frac{2\pi}{\omega_d}$.

Q3.00.4

NAT | 2000 | 8M | Ans: *

Each integrator block has a transfer function of $\frac{1}{s}$. From the block diagram:

- Forward path gain, $P_1 = \frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$
- Inner loop gain, $L_1 = -\frac{4}{s}$
- Outer loop gain, $L_2 = -\frac{1}{s} \cdot \frac{1}{s} \cdot 3 = -\frac{3}{s^2}$

Applying Mason's Gain Formula:

$$\frac{Y(s)}{X(s)} = \frac{P_1}{1-(L_1+L_2)} = \frac{\frac{1}{s^2}}{1-\left(-\frac{4}{s}-\frac{3}{s^2}\right)}$$

$$\boxed{\frac{Y(s)}{X(s)} = \frac{1}{s^2+4s+3}}$$

Q3.00.5

NAT | 2000 | 8M | Ans: *

Given $x(t) = u(t) \implies X(s) = \frac{1}{s}$.

$$Y(s) = \frac{1}{s^2+4s+3} \cdot \frac{1}{s} = \frac{1}{s(s+1)(s+3)}$$

Using partial fraction expansion:

$$Y(s) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3}$$

$$A = \left.\frac{1}{(s+1)(s+3)}\right|_{s=0} = \frac{1}{3}; \quad B = \left.\frac{1}{s(s+3)}\right|_{s=-1} = -\frac{1}{2}; \quad C = \left.\frac{1}{s(s+1)}\right|_{s=-3} = \frac{1}{6}$$

$$Y(s) = \frac{1/3}{s} - \frac{1/2}{s+1} + \frac{1/6}{s+3}$$

Taking the inverse Laplace transform:

$$\boxed{y(t) = \left(\frac{1}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t}\right)u(t)}$$

Q3.00.6

NAT | 2000 | 8M | Ans: *

The pulse input can be represented as:

$$x(t) = u(t-1) - u(t-2)$$

Taking the Laplace transform:

$$X(s) = \frac{1}{s}\left(e^{-s} - e^{-2s}\right)$$

The output response in Laplace domain is:

$$Y(s) = \frac{1}{s(s^2+4s+3)}\left(e^{-s} - e^{-2s}\right)$$

Let $y_{step}(t) = \left(\frac{1}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t}\right)u(t)$ be the response from part (b). By linearity and time shifting:

$$y(t) = y_{step}(t-1) - y_{step}(t-2)$$

Expanding the terms explicitly:

$$y(t) = \left[\frac{1}{3} - \frac{1}{2}e^{-(t-1)} + \frac{1}{6}e^{-3(t-1)}\right]u(t-1)$$
$$- \left[\frac{1}{3} - \frac{1}{2}e^{-(t-2)} + \frac{1}{6}e^{-3(t-2)}\right]u(t-2)$$

Key Points

Time-shifting property: $\mathcal{L}\{f(t-t_0)u(t-t_0)\} = e^{-st_0}F(s)$.

Q3.99.1 MCQ | 1999 | 1M | Ans: B | ● ○ ○

Comparing the denominator of the given transfer function $T(s) = \frac{9}{s^2+4s+9}$ with the standard equation:

$$2\xi\omega_n = 4 \implies \xi\omega_n = 2$$

For a ±2% tolerance band, the settling time t_s is defined as:

$$t_s = 4\tau = \frac{4}{\xi\omega_n} = \frac{4}{2} = 2\,\text{sec}$$

$$\boxed{(B)\ \ 2.0}$$

Q3.99.2 MCQ | 1999 | 2M | Ans: D | ● ● ○

For a unity negative feedback system ($H(s) = 1$), the open-loop transfer function $G(s)$ is related to the closed-loop transfer function (CLTF) as:

$$G(s) = \frac{\text{CLTF}}{1 - \text{CLTF}}$$

Given:

$$\text{CLTF} = \frac{a_{n-1}s + a_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n}$$

Calculating $G(s)$:

$$G(s) = \frac{a_{n-1}s + a_n}{(s^n + a_1 s^{n-1} + \cdots + a_{n-1}s + a_n) - (a_{n-1}s + a_n)}$$

$$G(s) = \frac{a_{n-1}s + a_n}{s^n + a_1 s^{n-1} + \cdots + a_{n-2}s^2}$$

Factoring s^2 from the denominator shows that the system is Type-2:

$$G(s) = \frac{a_{n-1}s + a_n}{s^2(s^{n-2} + a_1 s^{n-3} + \cdots + a_{n-2})}$$

The steady-state error e_{ss} for a unit ramp input is $e_{ss} = \frac{1}{K_v}$, where K_v is the velocity error constant:

$$K_v = \lim_{s\to 0} sG(s) = \lim_{s\to 0} \frac{s(a_{n-1}s + a_n)}{s^2(s^{n-2} + \cdots + a_{n-2})} = \infty$$

$$e_{ss} = \frac{1}{K_v} = \frac{1}{\infty} = 0$$

$$\boxed{\text{(D)}\ \ \text{zero}}$$

Mistakes to be avoided

System Type Confusion: Do not determine the system type directly from the closed-loop transfer function; always find $G(s)H(s)$ first.

Q3.98.1 MCQ | 1998 | 1M | Ans: B | ● ○ ○

Given loop transfer function:

$$G(s)H(s) = \frac{K(1 + 0.5s)}{s(1 + s)(1 + 2s)}$$

The type of a control system is determined by the number of poles of the open-loop transfer function located at the origin ($s = 0$). Since there is one pole at the origin in $G(s)H(s)$, it is a **type one** system.

$$\boxed{(B)\ \ 1}$$

Q3.98.2 | MCQ | 1998 | 1M | Ans: A | ● ○ ○

For the given open-loop transfer function $G(s) = \frac{K}{s(s+1)}$ and unity feedback ($H(s) = 1$), the positional error constant K_p is calculated as:

$$K_p = \lim_{s\to 0} G(s)H(s) = \lim_{s\to 0} \frac{K}{s(s+1)} = \infty$$

$$e_{ss} = \frac{A}{1+K_p} = \frac{1}{1+\infty} = 0$$

$$\boxed{(A)\ \ zero}$$

Q3.98.3 | NAT | 1998 | 10M | Ans: 0.2, 0.605, 0.163 | ● ○ ○

Determining Parameter 'a':

Given $G(s) = \dfrac{16}{s(s+0.8)}$ and $H(s) = 1 + as$.

$$\text{CLTF} = \frac{G(s)}{1+G(s)H(s)}$$

$$\text{CLTF} = \frac{\dfrac{16}{s(s+0.8)}}{1+\dfrac{16(1+as)}{s(s+0.8)}}$$

$$\text{CLTF} = \frac{16}{s^2 + s(0.8+16a) + 16}$$

Comparing with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n^2 = 16 \implies \omega_n = 4\,\text{rad/sec}$$

Given damping ratio $\xi = 0.5$:

$$2\xi\omega_n = 0.8 + 16a$$

$$2\times 0.5\times 4 = 0.8 + 16a$$

$$4 = 0.8 + 16a \implies a = 0.2$$

Calculating Transient Performance Parameters:

The damped natural frequency ω_d is:

$$\omega_d = \omega_n\sqrt{1-\xi^2} = 4\sqrt{1-0.5^2} = 3.464\,\text{rad/sec}$$

For $\xi = 0.5$:

$$\theta = \tan^{-1}\left(\frac{\sqrt{1-0.5^2}}{0.5}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \approx 1.047\,\text{rad}$$

The rise time t_r is evaluated as:

$$t_r = \frac{\pi-\theta}{\omega_d} = \frac{\pi - \dfrac{\pi}{3}}{3.464} = \frac{2\pi}{3\times 3.464} \approx 0.605\,\text{sec}$$

The maximum overshoot M_p is given by:

$$M_p = e^{-\frac{\pi\times 0.5}{\sqrt{1-0.5^2}}} = e^{-\frac{\pi}{\sqrt{3}}} \approx e^{-1.814} \approx 0.163$$

$$\boxed{a = 0.2,\ t_r = 0.605,\ M_p = 0.163}$$

Q3.97.1 | NAT | 1997 | 5M | Ans: * | ● ○ ○

Given the impulse responses:

$$h_A(t) = e^{-t}u(t) \implies H_A(s) = \frac{1}{s+1} \quad ; \quad h_B(t) = e^{-2t}u(t) \implies H_B(s) = \frac{1}{s+2}$$

The open-loop forward path transfer function is:

$$G(s) = H_A(s)\cdot H_B(s) = \frac{1}{(s+1)(s+2)} = \frac{1}{s^2+3s+2}$$

For $k = 1$, the closed-loop transfer function is:

$$\text{T.F.} = \frac{G(s)}{1+k\cdot G(s)} = \frac{\dfrac{1}{(s+1)(s+2)}}{1+1\cdot\dfrac{1}{(s+1)(s+2)}} = \boxed{\frac{1}{s^2+3s+3}}$$

Q3.97.2 NAT 1997 5M Ans: * ● ○ ○

For $k = 0$, the feedback loop is disconnected. The overall transfer function equals the open-loop transfer function:

$$H(s) = \frac{1}{(s+1)(s+2)}$$

Using partial fraction expansion:

$$H(s) = \frac{A}{s+1} + \frac{B}{s+2} \quad where, \quad A = \left.\frac{1}{s+2}\right|_{s=-1} = 1, \quad B = \left.\frac{1}{s+1}\right|_{s=-2} = -1$$

$$H(s) = \frac{1}{s+1} - \frac{1}{s+2}$$

Taking the inverse Laplace transform to obtain the impulse response $h(t)$:

$$\boxed{h(t) = (e^{-t} - e^{-2t})u(t)}$$

Q3.97.3 NAT 1997 5M Ans: * ● ○ ○

$$T(s) = \frac{G(s)}{1 + k \cdot G(s)} = \frac{1}{s^2 + 3s + 2 + k}$$

The characteristic equation is: $s^2 + 3s + (2 + k) = 0$ RH Table :

$$\begin{array}{c|cc} s^2 & 1 & 2+k \\ s^1 & 3 & 0 \\ s^0 & 2+k & \end{array}$$

For the system to be unstable, there must be a sign change in the first column:

$$2 + k < 0 \implies \boxed{k < -2}$$

Q3.95.1 MCQ 1995 1M Ans: A ● ○ ○

Method 1

The positional error constant K_p is defined as:

$$K_p = \lim_{s\to 0} G(s)H(s)$$

$$K_p = \lim_{s\to 0} \frac{1}{(s+6)(s+1)}$$

$$K_p = \frac{1}{(0+6)(0+1)} = \frac{1}{6}$$

Method 2

Shortcut Method:

$$K_p = \frac{\text{Numerator constant}}{\text{Denominator constant}}$$

Numerator constant = 1; Denominator constant = 6

$$K_p = \frac{1}{6}$$

$$\boxed{(A)\ \frac{1}{6}}$$

Q3.95.2 MCQ 1995 1M Ans: C ● ○ ○

The characteristic polynomial for a standard second order system is given by:

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

When the damping ratio (ξ) is in the range $0 < \xi < 1$, the system is underdamped. The roots (poles) of the characteristic equation are complex conjugates:

$$s_{1,2} = -\xi\omega_n \pm j\omega_n\sqrt{1-\xi^2}$$

Since the roots have both real and imaginary parts, they appear as a pair of **complex conjugates** in the s-plane.

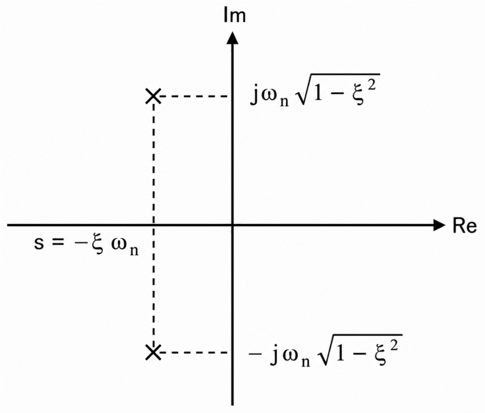


Fig. Solution Q3.95.2

Q3.94.1 MCQ 1994 2M Ans: a-1, b-5, c-4 ● ● ○

Based on the characteristics of control systems and components:

- **(a) → (1):** A system that exhibits a very low response at very high frequencies naturally attenuates high-frequency signals, which is the primary characteristic of a **Low Pass system**.
- **(b) → (5):** Maximum overshoot (M_p) in a second-order system is mathematically defined as $M_p = e^{-\left(\frac{\xi\pi}{\sqrt{1-\xi^2}}\right)}$. This clearly shows that overshoot depends directly on the **Damping ratio** (ξ).
- **(c) → (4):** A synchro-control transformer acts as an error detector. Its output is an AC voltage whose magnitude depends on the angular displacement and whose phase depends on the direction of displacement, representing **Phase-sensitive modulation**.

$$\boxed{a-1, b-5, c-4}$$

Q3.94.2 MCQ 1994 2M Ans: a-1, b-5, c-2 ● ● ○

The damping behavior of a second-order system, such as an LCR circuit, is determined by the specific location of the roots (poles) of its characteristic polynomial in the s-plane.

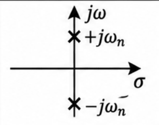
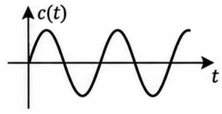
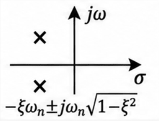
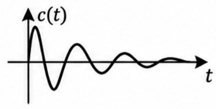
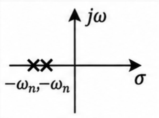
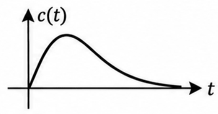
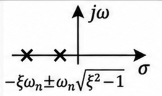
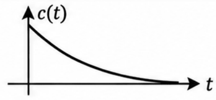
ξ	Damping	s-plane	Impulse Response	Poles	Stability
$\xi = 0$	Undamped			Imaginary Conjugate	Marginally stable
$0 < \xi < 1$	Underdamped			Complex conjugate roots on the LHS of s-plane	Stable
$\xi = 1$	Critically damped			Negative Equal Real Roots	Stable
$\xi > 1$	Overdamped			Negative Unequal Real Roots	Stable

Fig. Solution Q3.94.2

By analyzing the transfer function characteristics:

- **Overdamped:** When $\xi > 1$, the poles are real, negative, and distinct, lying on the **negative real axis**.
- **Critically damped:** When $\xi = 1$, the system has **multiple (repeated) poles on the negative real axis**.
- **Oscillatory/Undamped:** When $\xi = 0$, the poles are purely **imaginary conjugates** ($\pm j\omega_n$), lying on the imaginary axis.

$$\boxed{a - 1, b - 5, c - 2}$$

Q3.92.1 NAT 1992 8M Ans: 0.316, 0.2 ● ○ ○

Part 1: Damping Ratio (ξ) Given $K_t = 0$, $K_A = 5$, and $H(s) = 1$:

$$G(s) = \frac{5}{s(0.5s+1)} = \frac{10}{s(s+2)}$$

The closed-loop characteristic equation is:

$$1 + G(s)H(s) = 0 \implies s^2 + 2s + 10 = 0$$

Comparing with standard form:

$$\omega_n^2 = 10 \implies \omega_n = \sqrt{10}\,\text{rad/sec}$$

$$2\xi\omega_n = 2 \implies \xi = \frac{1}{\sqrt{10}} \approx \boxed{0.316}$$

Part 2: Steady-State Error (e_{ss})
The velocity error constant K_v is:

$$K_v = \lim_{s\to 0} s \cdot \frac{10}{s(s+2)} = \frac{10}{2} = 5$$

Thus, the steady-state error is:

$$e_{ss} = \frac{1}{K_v} = \frac{1}{5} = \boxed{0.2}$$

Q3.92.2 NAT 1992 8M Ans: 25, 4 ● ● ○

Formulating the Loop Relations:

With both K_A and K_t active, the forward path transfer function with the inner loop simplified is:

$$G(s) = \frac{2K_A}{s[s + (2 + 2K_t)]}$$

The velocity error coefficient K_v becomes:

$$K_v = \lim_{s \to 0} sG(s) = \frac{2K_A}{2 + 2K_t}$$

$$K_v = \frac{K_A}{1 + K_t}$$

To keep the steady-state error unchanged ($e_{ss} = 0.2 \implies K_v = 5$):

$$\frac{K_A}{1 + K_t} = 5 \implies 1 + K_t = 0.2K_A \quad \cdots (1)$$

The characteristic equation of the system is:

$$s^2 + (2 + 2K_t)s + 2K_A = 0$$

From this, $\omega_n = \sqrt{2K_A}$ and:

$$2\xi\omega_n = 2 + 2K_t \implies \xi = \frac{1}{\sqrt{2}} \frac{(1 + K_t)}{\sqrt{K_A}}$$

Substitute $\xi = 0.7$ and equation (1):

$$0.7 = \frac{0.2K_A}{\sqrt{2K_A}} \implies 0.7\sqrt{2} = 0.2\sqrt{K_A}$$

$$\sqrt{K_A} = \frac{0.7\sqrt{2}}{0.2} = 3.5\sqrt{2} = \sqrt{24.5} \approx 4.95$$

$$\boxed{K_A = 24.5 \approx 25}$$

$$1 + K_t = 0.2(25) = 5 \implies \boxed{K_t = 4}$$

Key Points

To maintain an unchanged steady-state error performance (e_{ss}), the forward path open-loop gain must be compensated appropriately when changing loop parameters.

Q3.91.1 MCQ 1991 2M Ans: A ● ○ ○

Given the open-loop transfer function:

$$G(s) = \frac{4(1 + 2s)}{s^2(s + 2)}$$

For a unity-feedback system ($H(s) = 1$), the steady-state error e_{ss} for a unit ramp input ($A = 1$) is given by:

$$e_{ss} = \frac{A}{K_v}$$

$$K_v = \lim_{s \to 0} s \cdot G(s) = \lim_{s \to 0} s \cdot \frac{4(1 + 2s)}{s^2(s + 2)} = \lim_{s \to 0} \frac{4(1 + 2s)}{s(s + 2)} = \frac{4}{0} = \infty$$

The steady-state error is:

$$e_{ss} = \frac{1}{\infty} = 0$$

$$\boxed{(A)\ 0}$$

Q3.91.2 MCQ 1991 2M Ans: A ● ○ ○

Compare $s^2 + 8s + 25$ with the standard 2^{nd} order characteristic equation $s^2 + 2\xi\omega_n s + \omega_n^2$:

$$\omega_n^2 = 25 \implies \omega_n = 5\,\text{rad/sec}$$

$$2\xi\omega_n = 8 \implies \xi = \frac{8}{2 \times 5} = 0.8$$

The damped frequency of oscillation ω_d is:

$$\omega_d = \omega_n\sqrt{1 - \xi^2} = 5\sqrt{1 - (0.8)^2} = 3\,\text{rad/sec}$$

The time for peak overshoot is given by: $t_p = \dfrac{n\pi}{\omega_d} = \dfrac{3\pi}{\omega_d} = \dfrac{3\pi}{3} = \pi$ sec

$$\boxed{(A)\ \pi\ sec}$$

Mistakes to be avoided

- Peak time formula: $t_p = \dfrac{n\pi}{\omega_d}$, where $n = 1, 3, 5\ldots$ are maxima and $n = 2, 4, 6\ldots$ are minima.
- The question asks for the second peak in the response, which refers to the second maximum (at $n = 3$).

Q3.90.1 MCQ 1990 2M Ans: B ● ○ ○

The steady-state error e_{ss} for a unit step function $R(s) = \frac{1}{s}$ is given by:

$$e_{ss} = \lim_{s\to 0} \frac{s \cdot R(s)}{1 + G(s)H(s)}$$

For unity feedback, $H(s) = 1$. For a unit step input:

$$e_{ss} = \frac{1}{1 + \lim_{s\to 0} G(s)}$$

The term $\lim_{s\to 0} G(s)$ is defined as the positional error coefficient K_p. For a Type-0 system, K_p is a finite constant. Thus:

$$\boxed{(B)\ \ e_{ss} = \frac{1}{1 + K_p}}$$

Q3.87.1 MCQ 1987 2M Ans: A ● ○ ○

For Unit Step Input:
The steady-state error (e_{ss}) is given by:

$$e_{ss} = \frac{A}{1 + K_p}$$

Where K_p is the position error constant:

$$K_p = \lim_{s\to 0} G(s)H(s)$$

Given the system forward path transfer function implicitly contains a pole at the origin, i.e., $G(s)H(s) = \dfrac{K}{s(s+10)}$:

$$K_p = \lim_{s\to 0} \frac{K}{s(s+10)} = \infty$$

$$e_{ss} = \frac{1}{1+\infty} = 0$$

Hence, the system has zero steady-state error for a step input.

For Unit Ramp Input:
The steady-state error (e_{ss}) is given by:

$$e_{ss} = \frac{A}{K_v}$$

Where K_v is the velocity error constant:

$$K_v = \lim_{s\to 0} s \cdot G(s)H(s)$$

$$K_v = \lim_{s\to 0} s \cdot \frac{K}{s(s+10)} = \frac{K}{10}$$

Substituting K_v back into the error equation for a unit ramp input ($A = 1$):

$$e_{ss} = \frac{1}{K/10} = \frac{10}{K}$$

Thus, the steady-state error for a ramp input is $\dfrac{10}{K}$.

$$\boxed{\text{(A) Zero steady state position error}}$$

DEBUG INFO

Chapter III

Time Response Analysis

Total Questions : 52

MCQ : 30

MSQ : 1

NAT : 21

SOLUTIONS

IV

Routh’s Stability Criterion

Topics

1. Hurwitz Polynomial and Routh Array
2. Special Cases in Routh Stability
3. Mapping Theorem in Routh-Hurwitz

Q4.26.1 NAT | 2026 | 2M | Ans: 1386 | ● ○ ○

For the system to be marginally stable, we analyze the characteristic equation:

$$1 + G(s) = 0$$

$$1 + \frac{K}{s(s+7)(s+11)} = 0$$

$$s^3 + 18s^2 + 77s + K = 0$$

Applying Routh-Hurwitz stability criterion for a third-order system $as^3 + bs^2 + cs + d = 0$, the condition for marginal stability is:

$$bc = ad$$

$$18 \times 77 = 1 \times K_{\text{mar}}$$

$$\boxed{K_{\text{mar}} = 1386}$$

Key Points

Marginal stability occurs when a row in the Routh array becomes zero, leading to symmetric roots on the $j\omega$ axis.

Q4.25.1 MCQ | 2025 | 2M | Ans: A | ● ○ ○

Given the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. We construct the Routh-Hurwitz (RH) table to determine the root distribution.

$$\begin{array}{c|ccc}
s^5 & 1 & 3 & 2 \\
s^4 & 7 & -33 & -40 \\
s^3 & \frac{54}{7} & \frac{54}{7} & 0 \\
s^2 & -40 & -40 & 0 \\
s^1 & 0(-80) & 0 & 0 \\
s^0 & -40 & &
\end{array}$$

A row of zeros occurs at s^1. We form the Auxiliary Equation [A.E.] from the coefficients of the s^2 row:

$$-40s^2 - 40 = 0$$

$$s^2 + 1 = 0 \implies s = \pm j1$$

Since the roots are $\pm j1$, the number of purely imaginary roots is $I = 2$.
To complete the RH table, we differentiate the A.E. with respect to s:

$$\frac{d}{ds}(-40s^2 - 40) = -80s$$

The coefficient -80 replaces the zero in the s^1 row.
There is exactly 1 sign change in the 1st column of the RH Table (from $\frac{54}{7}$ to -40). So the number of roots in the right-half of the s-plane is $R = 1$.
The number of roots in the left-half of the s-plane is (L = Total roots $- I - R$): $L = 5 - 2 - 1 = 2$

$$\boxed{(A)\quad L = 2, \quad I = 2, \quad R = 1}$$

Key Points

The Auxiliary Equation is always of even degree and its roots are a subset of the original polynomial's roots.

Q4.24.1 MCQ | 2024 | 2M | Ans: A | ● ● ○ | CLICK FOR VIDEO SOLUTION

The impulse response of a closed-loop system decays faster than e^{-t} if all closed-loop poles lie to the left of $s = -1$ in the s-plane. Given the open-loop transfer function: $G(s) = \frac{K}{(s+1)(s+2)(s+3)}$ The characteristic equation for the unity feedback system is:

$$1 + G(s) = 0 \implies (s+1)(s+2)(s+3) + K = 0$$

$$s^3 + 6s^2 + 11s + (6 + K) = 0$$

To ensure all poles are to the left of $s = -1$, we perform a variable shift by substituting $s = z - 1$:

$$(z - 1 + 1)(z - 1 + 2)(z - 1 + 3) + K = 0$$

$$z(z+1)(z+2) + K = 0$$

$$\implies z^3 + 3z^2 + 2z + K = 0$$

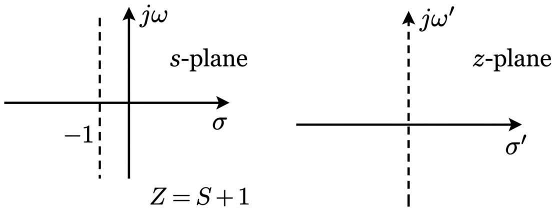


Fig. Solution Q4.24.1

Applying the Routh-Hurwitz stability criterion to the shifted equation:

$$\begin{array}{c|cc} z^3 & 1 & 2 \\ z^2 & 3 & K \\ z^1 & \frac{6-K}{3} & 0 \\ z^0 & K & 0 \end{array}$$

For stability in the z-plane, all coefficients in the first column must be greater than zero:

- From the z^0 row: $K > 0$
- From the z^1 row: $\frac{6-K}{3} > 0 \implies K < 6$

Combining these conditions, we find the range for K is $0 < K < 6$.

$$\boxed{(A)\ \ 1 \le K < 5}$$

Key Points

A system response e^{-at} requires all poles to satisfy $\mathrm{Re}(s) < -a$.

Q4.22.1 MCQ | 2022 | 2M | Ans: A | ● ○ ○

Given $p(s) = s^4 + 5s^2 + (4 + K)$.
It is mentioned that all 4 roots lie on the imaginary axis. Hence, the s^3 row is zero.
The auxiliary equation is $A(s) = s^4 + 5s^2 + (4 + K)$. The derivative $\frac{dA(s)}{ds} = 4s^3 + 10s$ provides the coefficients to replace the zero row.

$$\begin{array}{c|ccc} s^4 & 1 & 5 & (4+K) \\ s^3 & 0 & 0 & \text{...Zero Row} \\ s^3 & 4 & 10 & \text{...Replaced Row} \\ s^2 & 2.5 & (4+K) & \\ s^1 & \frac{25-4(4+K)}{2.5} & & \\ s^0 & 4+K & & \end{array}$$

For all roots to lie on the imaginary axis, the first column of the R-H table must be all positive:
1. $\dfrac{25 - 16 - 4K}{2.5} > 0 \implies 9 > 4K \implies K < \frac{9}{4}$
2. $4 + K > 0 \implies K > -4$

$$\boxed{(A)\ \ \text{-4 < K < 9/4}}$$

Q4.20.1 NAT 2020 1M Ans: 160 ● ○ ○

Characteristic equation $q(s)$ for the given open-loop system will be:

$$q(s) = s^3 + 10s^2 + (16+K)s + 11K = 0$$

Method 1

R-H Table:

$$\begin{array}{c|cc} s^3 & 1 & 16+K \\ s^2 & 10 & 11K \\ s^1 & \dfrac{10(16+K)-11K}{10} & 0 \\ s^0 & 11K & 0 \end{array}$$

For marginal stability, the s^1 row must be zero:

$$\frac{10(16+K)-11K}{10} = 0$$

$$160 + 10K - 11K = 0$$

$$\boxed{K = 160}$$

Method 2

Shortcut Method : At $K = K_{\text{marginal}}$,

$$\text{Inner product} = \text{Outer product}$$

$$1 \times 11K_{\text{marginal}} = 10(16 + K_{\text{marginal}})$$

$$11K_{\text{marginal}} - 10K_{\text{marginal}} = 160$$

$$\boxed{K_{\text{marginal}} = 160}$$

Key Points

This 'Inner Product = Outer Product' shortcut is a direct derivation from the Routh-Hurwitz s^1 row being zero.

Q4.17.1 NAT 2017 2M Ans: 0.74 to 0.76 ● ○ ○

For unity feedback, the characteristic equation (CE) is $1 + G(s) = 0$:

$$s^3 + ks^2 + 2s + 1 + 2s + 2 = 0 \implies s^3 + ks^2 + 4s + 3 = 0$$

Method 1

Constructing the Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 4 \\ s^2 & k & 3 \\ s^1 & \dfrac{4k-3}{k} & 0 \\ s^0 & 3 & \end{array}$$

For the system to oscillate, the s^1 row must be zero:

$$\frac{4k-3}{k} = 0 \implies 4k - 3 = 0$$

$$k = 0.75$$

Method 2

For a 3^{rd} order system $as^3+bs^2+cs+d = 0$ to be marginally stable:

$$\text{Inner Product} = \text{Outer Product}$$

$$bc = ad$$

Substituting $a = 1, b = k, c = 4, d = 3$:

$$(k)(4) = (1)(3)$$

$$k = \frac{3}{4} = 0.75$$

$$\boxed{0.75}$$

Key Points

The frequency of oscillation ω is derived from the auxiliary equation: $A(s) = ks^2 + 3 = 0 \implies \omega = \sqrt{3/k}$.

Q4.17.2 MCQ 2017 2M Ans: C ● ○ ○

The given characteristic equation is: $Q(s) = s^4 + s^2 + 1 = 0$

Routh-Hurwitz Table:

$$\begin{array}{c|ccc} s^4 & 1 & 1 & 1 \\ s^3 & 0\,(4) & 0\,(2) & 0 \\ s^2 & 0.5 & 1 & \\ s^1 & -6 & & \\ s^0 & 1 & & \end{array}$$

The s^3 row is a Row of Zeros (ROZ). We form the Auxiliary Equation (AE) from the s^4 row:

$$A(s) = s^4 + s^2 + 1 = 0$$

$$\frac{dA(s)}{ds} = 4s^3 + 2s$$

Looking at the first column of the Routh array, there are exactly **two sign changes**.
Number of RHP roots = 2.
Since the AE is of 4th order, all 4 roots are symmetric about the origin.
Thus, the remaining 2 roots must lie in the LHP to maintain symmetry.

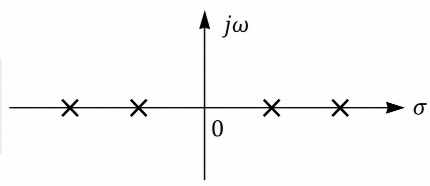


Fig. Solution Q4.17.2

$\boxed{\text{(C) Two RHP roots and two LHP roots}}$

Mistakes to be avoided

Do not assume an ROZ always means roots are purely on the imaginary axis. Check for sign changes below the auxiliary row to see if roots leave the $j\omega$-axis.

Q4.16.1 MCQ 2016 1M Ans: D ● ○ ○

Based on Routh's stability criterion for a characteristic equation:

- **Stable System ($X \to P$):** When all elements of the first column are positive, the system is stable.
- **Unstable System ($Y \to R$):** When there is a change in sign of coefficients in the first column, the system is unstable.
- **Test Breaks Down ($Z \to Q$):** When any one element in the first column is zero, the Routh array cannot be completed normally without special modifications.

First Column Visual Representation:

P	+ + +	Stable
Q	+ 0 +	Routh array breaks
R	+ − +	Unstable

$\boxed{(D)\ \ X \to P, \quad Y \to R, \quad Z \to Q}$

Q4.16.2 MCQ 2016 2M Ans: D ● ○ ○

Routh-Hurwitz Table:

$$\begin{array}{c|cc} s^3 & 1 & 2k+3 \\ s^2 & 2k & 4 \\ s^1 & \dfrac{2k(2k+3)-4}{2k} & 0 \\ s^0 & 4 & \end{array}$$

For stability, all elements in the first column must be greater than zero.

From row s^2: $2k > 0 \implies k > 0$

From row s^1: $4k^2 + 6k - 4 > 0$

$$(2k-1)(k+2) > 0$$

$$k > 0.5 \quad \text{or} \quad k < -2$$

Combining $k > 0$ and $k > 0.5$:

$$\boxed{(D)\ \ 0.5 < k < \infty}$$

Q4.16.3 NAT 2016 2M Ans: 3 ● ○ ○

Given the characteristic polynomial:

$$H(s) = 2s^4 - 5s^3 + 5s - 2$$

We construct the Routh array as follows:

$$\begin{array}{c|ccc} s^4 & 2 & 0 & -2 \\ s^3 & -5 & 5 & \\ s^2 & 2 & -2 & \\ s^1 & 0\,(2) & & \\ s^0 & -2 & & \end{array}$$

The total number of sign changes in the first column is 3. According to Routh's stability criterion, the number of sign changes equals the number of roots(zeros) lying in the right-half of the s-plane.

$$\boxed{3}$$

Q4.15.1 NAT 2015 2M Ans: 2 ● ● ○

Routh Table Construction:

$$\begin{array}{c|ccc} s^5 & 1 & 3 & -4 \\ s^4 & 2 & 6 & -8 \\ s^3 & 0\,(8) & 0\,(12) & 0 \\ s^2 & 3 & -8 & 0 \\ s^1 & \dfrac{100}{3} & 0 & 0 \\ s^0 & -8 & & \end{array}$$

Auxiliary equation from s^4 row:

$$A(s) = 2s^4 + 6s^2 - 8 = 0$$

Taking derivative with respect to s:

$$\frac{dA(s)}{ds} = 8s^3 + 12s = 0$$

There is exactly **1 sign change** in the first column, which occurs below the row of zeros (s^3).
Analyzing the auxiliary equation $A(s)$:

$$2s^4 + 6s^2 - 8 = 0 \implies s^4 + 3s^2 - 4 = 0$$

$$(s^2+4)(s^2-1) = 0$$

The roots of $A(s)$ are:

$$s = \pm 2j, \quad s = \pm 1$$

- Roots in RHS (RHP) = 1 (at $s = 1$)
- Roots on $j\omega$-axis = 2 (at $s = \pm 2j$)
- Total roots = 5

Roots in LHS (LHP) = $5 - 1 - 2 = \boxed{2}$.

Key Points

If there is a sign change above the row of zeros, the symmetric roots lie on the imaginary axis. However, if there is a sign change below the row of zeros, the root lies on the RHP and its symmetric root lies on the LHP.

Q4.15.2 MCQ 2015 2M Ans: A ● ○ ○

The characteristic equation of the closed-loop control system is given by:

$$1 + G(s) = 0$$

Substituting the given loop transfer function:

$$1 + \frac{K_p s + K_I}{s} \cdot \frac{1}{s(s+2)} = 0$$

$$s^2(s+2) + (K_p s + K_I) = 0$$

$$s^3 + 2s^2 + K_p s + K_I = 0$$

Method 1

Using Routh-Hurwitz Array:

$$\begin{array}{c|cc} s^3 & 1 & K_p \\ s^2 & 2 & K_I \\ s^1 & \dfrac{2K_p - K_I}{2} & 0 \\ s^0 & K_I & \end{array}$$

For stability, elements in the first column must be positive:

$$\frac{2K_p - K_I}{2} > 0 \implies K_p > \frac{K_I}{2}$$

and $K_I > 0$

Method 2

Shortcut ($bc > ad$):
For a standard cubic equation:

$$as^3 + bs^2 + cs + d = 0$$

where $a = 1$, $b = 2$, $c = K_p$, and $d = K_I$.
For absolute stability:

1. All coefficients $> 0 \implies K_I > 0, K_p > 0$
2. Inner product > Outer product:

$$bc > ad \implies 2K_p > 1 \cdot K_I$$

$$K_p > \frac{K_I}{2}$$

$$\boxed{(A)\ K_p > \frac{K_I}{2} > 0}$$

Q4.14.1 NAT 2014 1M Ans: 2.24 to 2.26 ● ○ ○

The characteristic equation is defined by $1 + G(s) = 0$:

$$1 + \frac{k}{s^2 + s - 2} = 0 \implies s^2 + s + (k-2) = 0$$

Method 1

Standard Comparison:
Comparing with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$:

$$\omega_n = \sqrt{k-2} \quad \text{— (1)}$$

For both poles at the same location, the system must be **critically damped** ($\xi = 1$):

$$2(1)\omega_n = 1 \implies \omega_n = 0.5 \quad \text{— (2)}$$

Equating (1) and (2):

$$\sqrt{k-2} = 0.5 \implies k = 2.25$$

Method 2

Discriminant Method: For a second-order polynomial equation to have both roots (poles) at identical locations, its discriminant (Δ) must be zero:

$$\Delta = b^2 - 4ac = 0$$

From $s^2 + s + (k-2) = 0$:

$$(1)^2 - 4(1)(k-2) = 0$$

$$1 - 4k + 8 = 0$$

$$4k = 9 \implies k = 2.25$$

$$\boxed{k = 2.25}$$

Q4.14.2 | NAT | 2014 | 2M | Ans: 1.9 to 2.1 | ● ○ ○

The given transfer function $G_p(s)$ is :

$$G_p(s) = \frac{ps^2 + 3ps - 2}{s^2 + (3+p)s + (2-p)}$$

The characteristic equation is:

$$s^2 + (3+p)s + (2-p) = 0$$

Method 1

Routh-Hurwitz Table:

$$\begin{array}{c|cc} s^2 & 1 & 2-p \\ s^1 & 3+p & 0 \\ s^0 & 2-p & \end{array}$$

For a stable system, all elements in the first column must be positive:

1. $3 + p > 0 \implies p > -3$
2. $2 - p > 0 \implies p < 2$

Given that p is a positive real parameter ($p > 0$), the valid range is $0 < p < 2$.

Method 2

Shortcut:
For a second-order characteristic equation:

$$as^2 + bs + c = 0$$

The system is stable if and only if all coefficients have the same sign and are non-zero. Since $a = 1 > 0$:

1. $b = 3 + p > 0 \implies p > -3$
2. $c = 2 - p > 0 \implies p < 2$

Therefore, for stability, p must satisfy the inequality condition:

$$p < 2$$

Q4.13.1 | MCQ | 2013 | 1M | Ans: D | ● ○ ○

The given polynomial is:

$$f(x) = a_4x^4 + a_3x^3 + a_2x^2 + a_1x - a_0$$

where all coefficients $a_4, a_3, a_2, a_1, a_0 > 0$.

$$\begin{array}{c|ccc} x^4 & a_4 & a_2 & -a_0 \\ x^3 & a_3 & a_1 & \\ x^2 & c_1 = \dfrac{a_3a_2 - a_1a_4}{a_3} & -a_0 & \\ x^1 & d_1 = \dfrac{c_1a_1 - a_3(-a_0)}{c_1} = \dfrac{c_1a_1 + a_3a_0}{c_1} & & \\ x^0 & -a_0 & & \end{array}$$

We analyze the signs in the first column (Row $x^4 \to$ Row x^0) to find the number of roots in the right-half plane:

1. If $c_1 > 0$, then $d_1 > 0$. The signs in the first column become $(+, +, +, +, -)$, resulting in exactly 1 sign change.
2. If $c_1 < 0$, then d_1 can be either positive or negative.

$$\boxed{\text{(D) at least one positive and one negative real root}}$$

Q4.12.1 MCQ | 2012 | 2M | Ans: A | ● ● ○

The characteristic equation for the given negative feedback loop is determined by $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s+1)}{s^3 + as^2 + 2s + 1} = 0$$

$$s^3 + as^2 + (K+2)s + (K+1) = 0$$

RH Table :

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & a & K+1 \\ s^1 & \dfrac{a(K+2)-(K+1)}{a} & 0 \\ s^0 & K+1 & \end{array}$$

For sustained oscillations, the elements of the s^1 row must equal zero:

$$\frac{a(K+2)-(K+1)}{a} = 0$$

$$a = \frac{K+1}{K+2} \quad \text{— (1)}$$

Forming the Auxiliary Equation (AE) from the s^2 row:

$$as^2 + (K+1) = 0 \quad \text{— (2)}$$

Substituting equation (1) into equation (2) :

$$\left[\frac{K+1}{K+2}\right] s^2 + (K+1) = 0$$

$$s^2 + K + 2 = 0 \implies s = \pm j\sqrt{K+2}$$

Given that the system oscillates at a frequency $\omega_n = 2$ rad/sec :

$$\sqrt{K+2} = 2 \implies K+2 = 4$$

$$K = 2$$

Substituting $K = 2$ back into equation (1) yields a:

$$a = \frac{2+1}{2+2} = \frac{3}{4} = 0.75$$

$$\boxed{(A)\ \ \text{K = 2 and a = 0.75}}$$

Key Points

Shortcut: The roots of the auxiliary equation immediately preceding the zero row determine the exact natural frequency of oscillation via $\omega = \sqrt{\dfrac{\text{independent term}}{\text{coefficient of } s^2}}$.

Q4.08.1 MCQ | 2008 | 2M | Ans: C | ● ○ ○

The poles of the transfer function $G(s)$ are given by the roots of its denominator polynomial:

$$s^5 + 2s^4 + 3s^3 + 6s^2 + 5s + 3 = 0$$

Since the first element of the s^3 row in the Routh table becomes zero, we substitute it with a small positive number $\varepsilon > 0$:

$$\begin{array}{c|ccc} s^5 & 1 & 3 & 5 \\ s^4 & 2 & 6 & 3 \\ s^3 & \varepsilon\ (+\text{ve}) & 1.5 & \\ s^2 & \dfrac{6\varepsilon - 3}{\varepsilon} \approx -\dfrac{3}{\varepsilon}\ (-\text{ve}) & 3 & \\ s^1 & \dfrac{1.5\left(\dfrac{6\varepsilon-3}{\varepsilon}\right) - 3\varepsilon}{\dfrac{6\varepsilon-3}{\varepsilon}} \approx 1.5\ (+\text{ve}) & & \\ s^0 & 3 & & \end{array}$$

As $\varepsilon \to 0^+$, the first column transitions through signs in the sequence: $(+), (+), (+), (-), (+), (+)$. Since there are exactly 2 sign changes, the system has 2 open right-half-plane poles.

$$\boxed{(C)\ 2}$$

Q4.08.2 MCQ | 2008 | 2M | Ans: C | ● ● ○

The characteristic equation is determined by setting $1 + G(s) = 0$:

$$1 + \frac{s+8}{s^2 + \alpha s - 4} = 0 \implies s^2 + (\alpha + 1)s + 4 = 0$$

RH Table:

$$\begin{array}{c|cc} s^2 & 1 & 4 \\ s^1 & \alpha + 1 & 0 \\ s^0 & 4 & \end{array}$$

For a second-order system to be stable, all coefficients must be strictly positive:

$$\alpha + 1 > 0 \implies \alpha > -1$$

$$\boxed{\text{(C) For all positive values of } \alpha \text{, the closed-loop system is stable}}$$

Q4.07.1 MCQ | 2007 | 1M | Ans: D | ● ○ ○

Given the closed-loop transfer function:

$$T(s) = \frac{s-5}{(s+2)(s+3)}$$

- Poles: $s = -2, -3$; Since all poles lie in the left-half of the s-plane, the system is stable.
- Zeros: $s = 5$; Since there is a zero located in the right-half of the s-plane, the system is a **non-minimum phase system**.

$$\boxed{(D)\ \text{a non-minimum phase system}}$$

Key Points

- **Minimum Phase System:** A system is minimum-phase if all its poles and zeros lie strictly in the left-half of the s-plane (excluding the $j\omega$-axis for poles).
- **Non-Minimum Phase System:** If at least one zero (or pole) lies in the right-half of the s-plane, it is a non-minimum phase system.

Q4.04.1 MCQ | 2004 | 2M | Ans: B | ● ○ ○

Given polynomial:

$$P(s) = s^5 + s^4 + 2s^3 + 2s^2 + 3s + 15$$

$$\begin{array}{c|ccc}
s^5 & 1 & 2 & 3 \\
s^4 & 1 & 2 & 15 \\
s^3 & \epsilon & -12 & \\
s^2 & \dfrac{2\epsilon+12}{\epsilon} \approx \dfrac{12}{\epsilon}\,(+\text{ve}) & 15 & \\
s^1 & \dfrac{\left[\frac{12}{\epsilon}(-12)\right]-15\epsilon}{\frac{12}{\epsilon}} \approx -12\,(-\text{ve}) & & \\
s^0 & 15 & &
\end{array}$$

Since a first-element row value becomes zero (s^3 row), it is replaced by a small positive number $\epsilon \to 0^+$.

As there are 2 sign changes in the first column, 2 roots lie in the right half of the s-plane.

$$\boxed{(B)\ \ 2}$$

Q4.03.1 | MCQ | 2003 | 2M | Ans: A | ● ○ ○

The open-loop transfer function is $G(s) = \dfrac{k}{s(s^2+s+2)(s+3)}$.

For a unity feedback system, the characteristic equation is $1 + G(s) = 0$:

$$1 + \frac{k}{s(s^2+s+2)(s+3)} = 0$$

$$(s^3+s^2+2s)(s+3) + k = 0$$

$$s^4 + 4s^3 + 5s^2 + 6s + k = 0$$

$$s^4 + 4s^3 + 5s^2 + 6s + k = 0$$

RH Table:

$$\begin{array}{c|ccc}
s^4 & 1 & 5 & k \\
s^3 & 4 & 6 & 0 \\
s^2 & \frac{7}{2} & k & 0 \\
s^1 & \frac{2(21-4k)}{7} & 0 & \\
s^0 & k & &
\end{array}$$

For stability, the first column must be positive:

- From s^0 row: $k > 0$
- From s^1 row: $21 - 4k > 0 \implies k < \frac{21}{4}$

Combining the bounds:

$$\boxed{(A)\ \ \frac{21}{4} > k > 0}$$

Q4.02.1 | MCQ | 2002 | 2M | Ans: C | ● ● ○

The characteristic equation is given by: $q(s) = 2s^5 + s^4 + 4s^3 + 2s^2 + 2s + 1 = 0$

RH Table:

s^5	2	4	2
s^4	1	2	1
s^3	0 (4)	0 (4)	
s^2	1	1	
s^1	0 (2)		
s^0	1		

$$A(s) = s^4 + 2s^2 + 1 = 0$$

Differentiating with respect to s:

$$\frac{dA(s)}{ds} = 4s^3 + 4s = 0$$

Roots of A.E.:

$$(s^2 + 1)^2 = 0 \implies s = \pm j1, \pm j1$$

Since there are no sign changes in the first column, no roots lie in the right-half of the s-plane. However, the auxiliary equation reveals **repeated roots on the imaginary axis** ($s = \pm j1$) as shown in the pole-zero configuration plot below. This condition makes the system **unstable**.

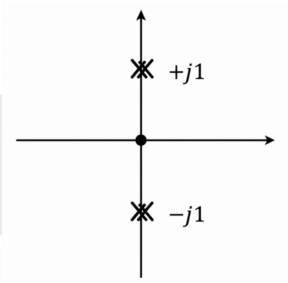


Fig. Solution Q4.02.1

$$\boxed{(C)\ unstable}$$

Mistakes to be avoided

Marginal vs. Unstable: Do not automatically classify a system as marginally stable just because a row of zeros occurs with no sign changes. Check for multiplicity of the imaginary roots.

Q4.02.2 MCQ 2002 2M Ans: D ● ○ ○

Applying Mason's gain formula to the given signal flow graph:

$$\text{TF} = \frac{Y(s)}{R(s)} = \frac{\frac{k}{s}}{1 - (\frac{3}{s} - \frac{k}{s})}$$

$$\text{TF} = \frac{k}{s + k - 3}$$

The closed-loop pole is located at:

$$s = 3 - k$$

For the system to remain stable, the pole must lie strictly in the left-half of the s-plane:

$$3 - k < 0 \implies k > 3$$

$$\boxed{(D)\ \ k > 3}$$

Q4.01.1 MCQ 2001 2M Ans: C ● ○ ○

Given : $G(s) = \dfrac{k(s-2)}{(s+2)^2}$ *and* $H(s) = s-2.$
The closed-loop characteristic equation is given by $1 + G(s)H(s) = 0$:

$$1 + k\frac{(s-2)}{(s+2)^2}(s-2) = 0 \implies (k+1)s^2 + (4-4k)s + (4+4k) = 0$$

RH Table:

s^2	$k+1$	$4+4k$
s^1	$4-4k$	0
s^0	$4+4k$	

For stability, all elements in the first column must be strictly positive:

- From s^1 row: $4-4k > 0 \implies k < 1$
- From s^0 row: $4+4k > 0 \implies k > -1$

Given the physical constraint $k \geq 0$, the valid range for stability is:

$$\boxed{(C)\ only\ if\ \ 0 \leq k < 1}$$

Q4.01.2 NAT 2001 5M Ans: * ● ○ ○

Closed-loop Transfer Function
First, simplify the inner feedback loop containing block 3:

$$G_{inner}(s) = \frac{\frac{1}{s}}{1+\frac{1}{s}(3)} = \frac{1}{s+3}$$

Now, the forward path cascade is:

$$G(s) = \left[\frac{K(s+1)}{s}\right] \cdot \left[\frac{1}{s+3}\right] = \frac{K(s+1)}{s(s+3)}$$

The overall feedback loop has $H(s) = \dfrac{1}{s+0.1}$.

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{\frac{K(s+1)}{s(s+3)}}{1+\frac{K(s+1)}{s(s+3)}\cdot\frac{1}{s+0.1}}$$

$$= \frac{K(s+1)(s+0.1)}{s(s+3)(s+0.1)+K(s+1)}$$

$$= \frac{K(s+1)(s+0.1)}{s^3+3.1s^2+(K+0.3)s+K}$$

The characteristic equation is the denominator:

$$\boxed{s^3+3.1s^2+(K+0.3)s+K=0}$$

Q4.01.3 NAT 2001 5M Ans: * ● ○ ○

From the characteristic equation, we construct the Routh array:

s^3	1	$K+0.3$
s^2	3.1	K
s^1	$\dfrac{3.1(K+0.3)-1(K)}{3.1}$	0
s^0	K	

For absolute stability, all components in the first column must be strictly positive (> 0):

1. From row s^0: $K > 0$
2. From row s^1: $\dfrac{2.1K+0.93}{3.1} > 0$

$$2.1K + 0.93 > 0 \implies K > -0.44$$

Taking the intersection of $K > 0$ and $K > -0.44$:

$$\boxed{K > 0}$$

Q4.00.1 MCQ | 2000 | 1M | Ans: B | ● ○ ○

The characteristic equation of the system is $s^3 + \alpha s^2 + ks + 3 = 0$.

RH Table:

$$\begin{array}{c|cc} s^3 & 1 & k \\ s^2 & \alpha & 3 \\ s^1 & \dfrac{\alpha k - 3}{\alpha} & 0 \\ s^0 & 3 & \end{array}$$

For a stable system, all elements in the first column must be strictly positive:

- From s^2 row: $\alpha > 0$
- From s^1 row: $\dfrac{\alpha k - 3}{\alpha} > 0$

Since $\alpha > 0$, the numerator must satisfy:

$$\alpha k - 3 > 0 \implies \alpha k > 3$$

$$\boxed{(B)\ \ \alpha > 0\ ,\ \ \alpha k > 3}$$

Q4.00.2 MCQ | 2000 | 1M | Ans: B | ● ○ ○

For the given data, the block diagram of the system is shown below:

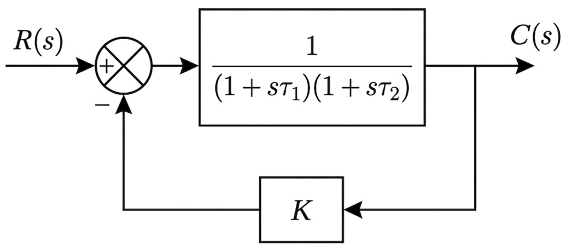


Fig. Solution Q4.00.2

With a resistive negative feedback network having a feedback factor K, the closed-loop transfer function is given by:

$$T(s) = \frac{G(s)}{1 + K \cdot G(s)} = \frac{A_0}{(1 + s\tau_1)(1 + s\tau_2) + KA_0}$$

Expanding the denominator gives the characteristic equation:

$$s^2\tau_1\tau_2 + s(\tau_1 + \tau_2) + 1 + KA_0 = 0$$

Since $\tau_1 > 0$, $\tau_2 > 0$, and $KA_0 > -1$, all coefficients of this second-order characteristic equation remain strictly positive for any positive feedback factor K. Consequently, both roots will always lie in the left half of the s-plane.

$$\boxed{\text{(B)\ \ will be stable for all frequency}}$$

Key Points

A second-order system with negative feedback never becomes unstable or goes into sustained oscillations because the maximum phase shift contributed by two poles is strictly less than 180°.

Q4.99.1 NAT | 1999 | 5M | Ans: * | ● ○ ○

The characteristic equation is given by $1 + G(s)H(s) = 0$:

$$1 + \frac{K(s + 1)}{s(1 + Ts)(1 + 2s)} = 0$$

$$2Ts^3 + (2 + T)s^2 + (K + 1)s + K = 0$$

Routh-Hurwitz Array:

$$\begin{array}{c|cc} s^3 & 2T & K+1 \\ s^2 & 2+T & K \\ s^1 & \dfrac{(2+T)(K+1)-2TK}{2+T} & 0 \\ s^0 & K & \end{array}$$

For stability, all elements in the first column must be greater than zero ($K > 0, T > 0$):

$$(K+1) - \frac{2KT}{2+T} > 0 \implies K > \frac{2+T}{T-2}$$

$$\text{Alternatively, } T > \frac{2(1+K)}{K-1}$$

For asymptotes and physical boundary limits: As $T \to \infty$, $K \to 1$. As $K \to \infty$, $T \to 2$.
Hence, for absolute stable regions:

$$\boxed{K > 1 \text{ and } T > 2}$$

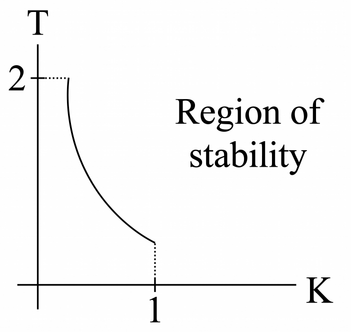


Fig. Solution Q4.99.1

Q4.98.1 MCQ 1998 1M Ans: C ● ○ ○

Given the open-loop transfer function of a unity feedback system ($H(s) = 1$):

$$G(s) = \frac{2s^2 + 6s + 5}{(s+1)^2(s+2)}$$

The characteristic equation of a closed-loop system is defined by:

$$1 + G(s)H(s) = 0$$

$$1 + \frac{2s^2 + 6s + 5}{(s+1)^2(s+2)} = 0$$

$$(s+1)^2(s+2) + 2s^2 + 6s + 5 = 0$$

$$\boxed{(C)\ \ 2s^2 + 6s + 5 + (s+1)^2(s+2) = 0}$$

Q4.98.2 NAT 1998 5M Ans: 0 to 1.49 ● ○ ○

Characteristic equation: $s^4 + 20s^3 + 15s^2 + 2s + K = 0$
Routh-Hurwitz Table:

$$\begin{array}{c|ccc} s^4 & 1 & 15 & K \\ s^3 & 20 & 2 & 0 \\ s^2 & 14.9 & K & \\ s^1 & \dfrac{29.8-20K}{14.9} & 0 & \\ s^0 & K & & \end{array}$$

For the system to be stable, all elements in the first column of the Routh array must be strictly positive (> 0):

$$K > 0$$

$$\frac{29.8 - 20K}{14.9} > 0 \implies K < 1.49$$

Combining the inequalities, the stable range is:

$$\boxed{0 < K < 1.49}$$

Q4.98.3 NAT 1998 5M Ans: 0.316 ● ○ ○

The system becomes marginally stable when the s^1 row becomes completely zero:

$$29.8 - 20K = 0 \implies K = 1.49$$

To find the frequency of sustained oscillation, we form the Auxiliary Equation $A(s)$:

$$A(s) = 14.9s^2 + K = 0$$

Substituting $K = 1.49$:

$$14.9s^2 + 1.49 = 0 \implies s = \pm j\frac{1}{\sqrt{10}}$$

Comparing with $s = \pm j\omega_n$, the frequency of sustained oscillation is:

$$\boxed{\omega_n = \frac{1}{\sqrt{10}} \text{ rad/sec} \approx 0.316 \text{ rad/sec}}$$

Q4.94.1 MCQ 1994 2M Ans: False ● ○ ○

Let the transfer function of $G(s)$ be:

$$G(s) = \frac{(s + Z_1)(s + Z_2)}{(s + P_1)(s + P_2)}$$

Then the inverted transfer function $F(s)$ is:

$$F(s) = \frac{1}{G(s)} = \frac{(s + P_1)(s + P_2)}{(s + Z_1)(s + Z_2)}$$

The condition for the stability of $G(s)$ is that none of its poles (i.e., $-P_1, -P_2$) should lie in the right half of the s-plane. However, a stable transfer function $G(s)$ can still have zeros (i.e., $-Z_1, -Z_2$) in the right half of the s-plane (known as a non-minimum phase system).

When finding $F(s) = \frac{1}{G(s)}$, these zeros of $G(s)$ become the poles of $F(s)$. If any of these zeros lie in the right half of the s-plane, $F(s)$ will possess right-half plane poles, making it unstable. Therefore, $F(s)$ need not always be stable.

$$\boxed{\text{False}}$$

Q4.93.1 MCQ 1993 2M Ans: C ● ○ ○

Given characteristic equation is:

$$s^3 + 3s^2 + 4s + A = 0$$

RH Table :

s^3	1	4
s^2	3	A
s^1	$\frac{12 - A}{3}$	
s^0	A	

For all roots to lie in the left half of the s-plane, elements in the first column must be strictly positive.

- From s^1 row:

$$\frac{12 - A}{3} > 0 \implies 12 - A > 0 \implies A < 12$$

- From s^0 row: A > 0

$$\boxed{(C)\ \ 0 < A < 12}$$

Q4.90.1 MCQ | 1990 | 2M | Ans: B | ● ● ○

Given characteristic equation is:

$$s^3 + 6Ks^2 + (K+2)s + 8 = 0$$

RH Table :

$$\begin{array}{c|cc} s^3 & 1 & K+2 \\ s^2 & 6K & 8 \\ s^1 & \dfrac{6K(K+2)-8}{6K} & 0 \\ s^0 & 8 & 0 \end{array}$$

For system stability,

- From s^2 row: $6K > 0 \implies K > 0$
- From s^1 row: $\dfrac{6K^2 + 12K - 8}{6K} > 0$
  Since $K > 0$, we have:

$$6K^2 + 12K - 8 > 0$$

Solving the quadratic equation $6K^2 + 12K - 8 = 0$ using the quadratic formula:

$$K = \frac{-12 \pm \sqrt{12^2 - 4(6)(-8)}}{2(6)} = \frac{-12 \pm \sqrt{144 + 192}}{12} = \frac{-12 \pm \sqrt{336}}{12}$$

$$K \approx \frac{-12 \pm 18.33}{12} \implies K \approx 0.528 \text{ or } K \approx -2.528$$

Since K must be greater than 0, the valid range for absolute stability is $K > 0.528$.

$$\boxed{(B)\ \ K = 2}$$

Key Points

The value $K = 0.528$ acts as the marginal stability boundary where the roots lie exactly on the $j\omega$-axis.

Q4.89.1 MCQ | 1989 | 2M | Ans: D | ● ○ ○

From the given block diagram, the forward path transfer function $G(s)$ is:

$$G(s) = \left(\frac{1 + sT_i}{s}\right) \cdot \left(\frac{1}{s(1 + sT)}\right) = \frac{1 + sT_i}{s^2(1 + sT)}$$

The characteristic equation of this unity feedback system ($H(s) = 1$) is defined by $1 + G(s)H(s) = 0$:

$$1 + \frac{1 + sT_i}{s^2(1 + sT)} = 0 \implies s^3T + s^2 + sT_i + 1 = 0$$

RH Table :

$$\begin{array}{c|cc} s^3 & T & T_i \\ s^2 & 1 & 1 \\ s^1 & T_i - T & 0 \\ s^0 & 1 & 0 \end{array}$$

For absolute closed-loop stability, all elements in the first column of the Routh array must be strictly positive.
Assuming the physical time constant $T > 0$,
From s^1 row:

$$T_i - T > 0 \implies T_i > T$$

$$\boxed{(D)\ \ T_i > T}$$

Q4.88.1 MCQ 1988 2M Ans: D ● ○ ○

Given characteristic equation is:

$$s^4 + 3s^3 + 5s^2 + 6s + (K + 10) = 0$$

Applying the Routh-Hurwitz stability criterion:

$$\begin{array}{c|ccc} s^4 & 1 & 5 & K+10 \\ s^3 & 3 & 6 & \\ s^2 & 3 & K+10 & \\ s^1 & 6-(K+10) & & \\ s^0 & K+10 & & \end{array}$$

- From s^1 row:

$$6 - (K + 10) > 0$$

$$6 > K + 10 \implies K < -4$$

- From s^0 row:

$$K + 10 > 0 \implies K > -10$$

$$\boxed{(D) \quad -10 < K < -4}$$

DEBUG INFO

Chapter IV

Routh's Stability Criterion

Total Questions : 37

MCQ : 25

MSQ : 0

NAT : 12

SOLUTIONS

V

Root Locus

Topics

- Basics of Root Locus
- Rules for Sketching Root Locus
- Angle of Departure and Arrival
- Complementary Root Locus

Q5.25.1 MSQ 2025 1M Ans: C

From control theory, the root locus represents the trajectory of all possible closed-loop pole locations as the system gain K varies from 0 to ∞.
By inspecting the root locus diagram:

- The complex point $s = -1 + j1$ lies outside the circular and linear paths traced by the root locus branches.
- Since this specific point is not touched or intersected by any branch of the root locus, it cannot be a closed-loop pole for any positive value of K.

Hence, the closed-loop poles never pass through the point $-1 + j1$ for any positive value of K.

$$\boxed{\text{(C) for no positive value of K}}$$

Q5.22.1 MCQ 2022 1M Ans: A

There are 5 root-locus branches from the same point, so there are 5 real multiple poles.
From the root-locus plot, the centroid is located at -1. Since there are 5 branches originating from this single point on the real axis and no finite zeros are visible, it indicates 5 multiple poles at $s = -1$.
The corresponding forward path transfer function is:

$$\boxed{(A)\ \ G(s) = \frac{1}{(s+1)^5}}$$

Key Points

- The number of root-locus branches is equal to the number of open-loop poles (n).
- The angles of asymptotes for n poles and m zeros are given by $\theta = \frac{(2q+1)180^\circ}{n-m}$.

Q5.20.1 MCQ 2020 2M Ans: A

Given: C.E $= s^3 + 3s^2 + (2+K)s + 3K = 0$
Converting to standard form: $1 + G(s)H(s) = 0$

$$s^3 + 3s^2 + 2s + K(s+3) = 0$$

$$1 + \frac{K(s+3)}{s^3 + 3s^2 + 2s} = 0$$

$$1 + \frac{K(s+3)}{s(s+1)(s+2)} = 0$$

Where $G(s)H(s)$ is the open loop transfer function:

$$G(s)H(s) = \frac{K(s+3)}{s(s+1)(s+2)}$$

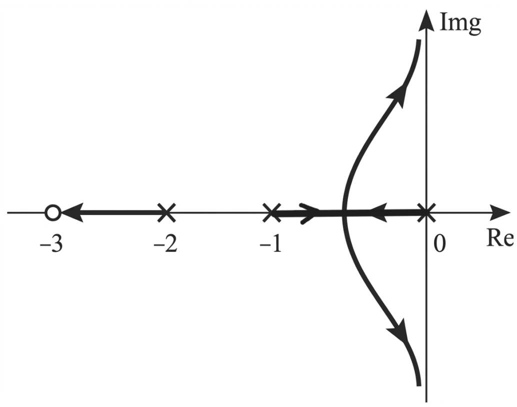


Fig. Solution Q5.20.1

Since there are poles located at -1 and 0, the breakaway point will lie in between -1 and 0.

$$\boxed{(A)\ \ (-1, 0)}$$

Key Points

- A point on the real axis lies on the root locus if the total number of open-loop poles and zeros to its right is odd.
- Breakaway points typically occur between two adjacent poles on the real axis if that segment is part of the root locus. Break-in points typically occur between two adjacent zeros on the real axis if that segment is part of the root locus.

Q5.17.1 MCQ 2017 2M Ans: A ● ○ ○

The open-loop transfer function of the unity feedback configuration is: $G(s) = \dfrac{K(s^2+2s+2)}{s^2-3s+2}$

Method 1

Routh-Hurwitz Stability Criterion:

The characteristic equation is defined by:

$$1 + G(s) = 0$$

$$1 + \frac{K(s^2+2s+2)}{s^2-3s+2} = 0$$

$$(s^2-3s+2) + K(s^2+2s+2) = 0$$

$$s^2(1+K) + s(2K-3) + 2(1+K) = 0$$

For a stable second-order system, all coefficients must have the same sign.

- Given $K > 0$, the first coefficient $(1+K)$ is strictly positive.
- For the system to be absolute stable, the coefficient of s^1 must also be strictly positive:

$$2K - 3 > 0$$

$$\boxed{(A)\ \ K > 1.5}$$

Method 2

Root Locus Analysis:

- Poles at $s = 1, 2$ and zeros at $s = -1 \pm j1$.
- The branches cross the imaginary axis into the left-half of the s-plane precisely at $K = 1.5$.
- Thus, for $K > 1.5$, all closed-loop poles lie entirely in the stable left-half plane.

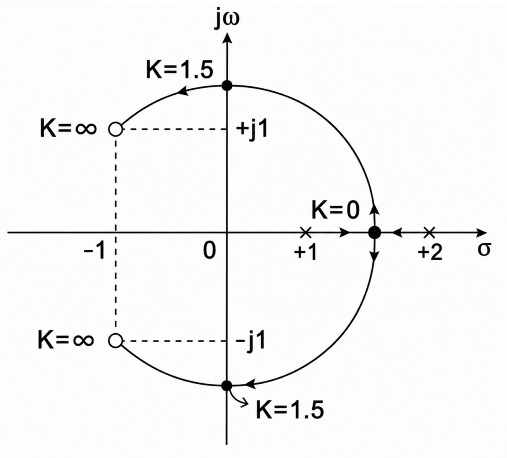


Fig. Solution Q5.17.1

Q5.16.1 NAT 2016 2M Ans: -3.45 to -3.35 ● ○ ○

The characteristic equation is $1 + G(s)H(s) = 0$

$$s^2 + 2s + 2 + K(s+2) = 0$$

$$K = -\frac{s^2 + 2s + 2}{s+2}$$

To find the breakaway or break-in points, set $\frac{\mathrm{d}K}{\mathrm{d}s} = 0$:

$$\frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{s^2 + 2s + 2}{s+2}\right] = 0$$

$$\frac{(2s+2)(s+2) - (s^2 + 2s + 2)(1)}{(s+2)^2} = 0$$

$$s^2 + 4s + 2 = 0$$

$$s = \frac{-4 \pm \sqrt{16-8}}{2} = -2 \pm \sqrt{2}$$

$$s \approx -0.58 \quad \text{or} \quad s \approx -3.41$$

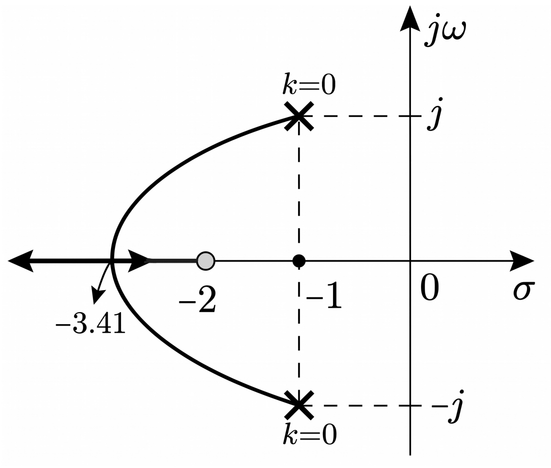


Fig. Solution Q5.16.1

$s = -0.58$ does not lie on the root locus. Therefore, $s = -3.41$ is the valid break-away point.

$$\boxed{-3.41}$$

Key Points

Breakaway/break-in points always satisfy the condition $\frac{\mathrm{d}K}{\mathrm{d}s} = 0$.

Q5.16.2 NAT 2016 2M Ans: 1.2 to 1.3 (or) -1.3 to -1.2 ● ○ ○

The characteristic equation for the unity-feedback system is defined by $1 + G(s)H(s) = 0$:

$$1 + \frac{K}{s^2 + 5s + 5} = 0 \implies K = -s^2 - 5s - 5$$

To find the breakaway points on the root locus, $\frac{\mathrm{d}K}{\mathrm{d}s} = 0$:

$$\frac{\mathrm{d}K}{\mathrm{d}s} = -2s - 5 = 0 \implies s = -2.5$$

Evaluating the value of the forward gain K at this breakaway point using the **magnitude condition**:

$$|G(s)H(s)|_{s=-2.5} = 1$$

$$\frac{|K|}{(-2.5)^2 + 5(-2.5) + 5} = 1$$

$$|K| = 1.25$$

$$\boxed{K = 1.25 \ \ or \ \ -1.25}$$

Key Points

At any valid point on the root locus, the absolute parameter value is determined via the standard magnitude criterion: $|G(s)H(s)| = 1$.

Q5.15.1 NAT 2015 1M Ans: 12 ● ○ ○

The characteristic equation of the unity negative feedback control system is given by $1 + G(s) = 0$:

$$1 + \frac{K}{s(s+1)(s+3)} = 0 \implies s^3 + 4s^2 + 3s + K = 0$$

RH Table:

s^3	1	3
s^2	4	K
s^1	$\frac{12-K}{4}$	0
s^0	K	

For the system branches to cross the imaginary axis, the system must cross the threshold of **marginal stability**. Setting the elements of the s^1 row to zero:

$$\frac{12-K}{4} = 0 \implies \boxed{K = 12}$$

Q5.15.2 NAT 2015 2M Ans: 0.29 to 0.31 ● ● ○

Step 1: Angle Condition Check

The open-loop transfer function is:

$$G(s)H(s) = \frac{10K(s+3)}{s+2}$$

For $s = -2.75$ to lie on the root locus, it must satisfy the angle condition:

$$\angle G(s)H(s) = \frac{\angle K + \angle(s+3)}{\angle(s+2)}$$

$$= \angle K + \angle(s+3) - \angle(s+2)$$

$$= 0^\circ + \angle(-2.75+3) - \angle(-2.75+2)$$

$$= \angle(0.25) - \angle(-0.75)$$

$$= 0^\circ - 180^\circ = -180^\circ$$

Since the angle is -180°, it satisfies the angle condition $\pm 180^\circ(2k+1)$. Thus, the point lies on the root locus.

Step 2: Magnitude Condition

Since the angle condition is verified, we apply the magnitude criteria to find K:

$$|G(s)H(s)|_{s=-2.75} = 1$$

$$\left|\frac{10K(s+3)}{s+2}\right|_{s=-2.75} = 1$$

$$K = \left|\frac{-2.75+2}{10(-2.75+3)}\right|$$

$$K = \frac{0.75}{2.5} = 0.3$$

$$\boxed{0.3}$$

Mistakes to be avoided

The angle condition must be satisfied first to ensure a point truly lies on the root locus; only then should the magnitude condition be applied to calculate the value of K.

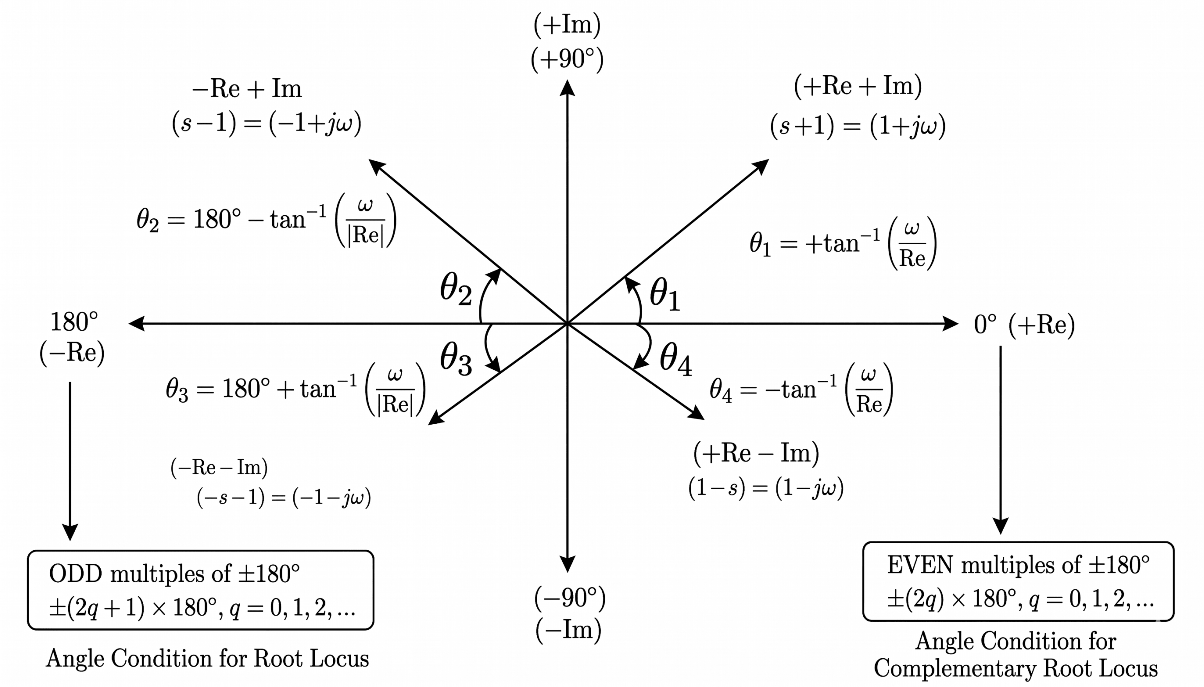


Fig. Solution Q5.15.2

Q5.15.3 NAT 2015 2M Ans: 25 to 26 ● ● ○

Given the open-loop transfer function of a unity feedback system ($H(s) = 1$):

$$G(s) = \frac{K(s+4)}{(s+8)(s^2-9)} = \frac{K(s+4)}{(s+8)(s+3)(s-3)}$$

Step 1: Angle Condition Check

$$\angle G(s)H(s) = \frac{\angle K + \angle(s+4)}{\angle(s+8) + \angle(s+3) + \angle(s-3)}$$

Evaluating at $s = -1 + j2$:

$$= \frac{\angle K + \angle(-1+j2+4)}{\angle(-1+j2+8) + \angle(-1+j2+3) + \angle(-1+j2-3)}$$

$$= \frac{0^\circ + \angle(3+j2)}{\angle(7+j2) + \angle(2+j2) + \angle(-4+j2)}$$

$$= \frac{\tan^{-1}\left(\frac{2}{3}\right)}{\tan^{-1}\left(\frac{2}{7}\right) + \tan^{-1}(1) + \left(180^\circ - \tan^{-1}\left(\frac{1}{2}\right)\right)}$$

$$= 33.69^\circ - 15.945^\circ - 45^\circ - 180^\circ + 26.565^\circ$$

$$= -180.72^\circ \simeq -180^\circ$$

Since it satisfies $\pm 180^\circ(2Q+1)$, the point lies on the root locus.

Step 2: Magnitude Condition

Applying magnitude criteria to calculate system gain K:

$$|G(s)H(s)|_{s=-1+j2} = 1$$

$$\frac{K\sqrt{(-1+4)^2+2^2}}{\sqrt{(-1+8)^2+2^2}\cdot\sqrt{(-1+3)^2+2^2}\cdot\sqrt{(-1-3)^2+2^2}} = 1$$

$$\frac{K\sqrt{9+4}}{\sqrt{49+4}\cdot\sqrt{4+4}\cdot\sqrt{16+4}} = 1$$

$$\frac{K\sqrt{13}}{\sqrt{53}\cdot\sqrt{8}\cdot\sqrt{20}} = 1$$

$$K = \frac{\sqrt{53\times 8\times 20}}{\sqrt{13}} \simeq 25.54$$

$$\boxed{25.54}$$

Q5.14.1 NAT | 2014 | 2M | Ans: 0.32 to 0.41 | ● ● ●

From the given system constraints, the open-loop transfer function is:

$$G(s)H(s) = \frac{K}{s(s+1)^2}$$

Method 1

Analytical Method

Given damping ratio $\xi = 0.5$. The angle of the damping line is:

$$\theta = \cos^{-1}(\xi) = \cos^{-1}(0.5) = 60^\circ$$

Let coordinates of point A be $(-a + jb)$. Given the distance from the origin (O) to point A is $OA = 0.5$.

$$a = OA\cos 60^\circ = 0.5 \times 0.5 = 0.25$$

$$b = OA\sin 60^\circ = 0.5 \times \frac{\sqrt{3}}{2} \simeq 0.433$$

Thus, the position of point A is $s = -0.25 + j0.433$.

Applying magnitude condition $|G(s)H(s)|_{s=A} = 1$:

$$K = |s| \cdot |s+1|^2$$

$$|s| = \sqrt{(-0.25)^2 + (0.433)^2} = 0.5$$

$$|s+1|^2 = (1-0.25)^2 + (0.433)^2 = 0.75$$

$$\therefore K = 0.5 \times 0.75 = 0.375$$

Method 2

Geometric Vector Method

The system value of gain K at any point A on the root locus is defined by:

$$K = \frac{\text{Product of lengths from } A \text{ to poles}}{\text{Product of lengths from } A \text{ to zeros}}$$

The poles are located at $s = 0$ and a double pole at $s = -1$.

- Distance from $s = 0$ to $A = 0.5$ (Given)
- Distance from $s = -1$ to A : (Using the cosine rule)

$$\text{Length}^2 = 1^2 + 0.5^2 - 2(1)(0.5)\cos 60^\circ$$

$$= 1 + 0.25 - 0.5 = 0.75$$

$$\text{Distance} = \sqrt{0.75}$$

Evaluating K:

$$K = 0.5 \times \sqrt{0.75} \times \sqrt{0.75} = 0.375$$

$$\boxed{K = 0.375}$$

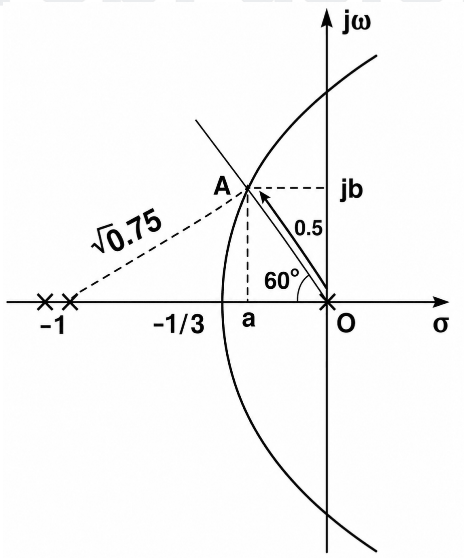


Fig. Solution Q5.14.1

Q5.14.2 MCQ 2014 2M Ans: B ● ● ○

By inspecting the given root locus diagram, a break-away point occurs on the real axis between the two rightmost singularities. This confirms that $s = -1$ and $s = -2$ are open-loop poles, thus **eliminating Option (A) and (D)**

To distinguish between the remaining potential pole-zero placements at $s = -4$ and $s = -7$, we can evaluate the centroid (σ) and angles (θ_l) of the asymptotes.

$$\theta_l = \frac{(2l+1)180°}{P-Z}, \quad \text{where } l = 0, 1, 2, \ldots, (P-Z-1)$$

The intersection point of these asymptotes on the real axis (the centroid) is calculated as:

$$\sigma = \frac{\sum \text{Real parts of open-loop poles} - \sum \text{Real parts of open-loop zeros}}{P-Z}$$

Testing Option (B):
Open-loop poles ($P = 3$) at $s = -1, -2, -7$ and open-loop zero ($Z = 1$) at $s = -4$.

- Asymptote Angles ($P - Z = 2$): $\theta_l = 90°, 270°$
- Centroid $\sigma = \dfrac{(-1-2-7)-(-4)}{3-1} = -3$

The asymptotes intersect at $\sigma = -3$, matching the geometric profile of the given diagram as shown below.

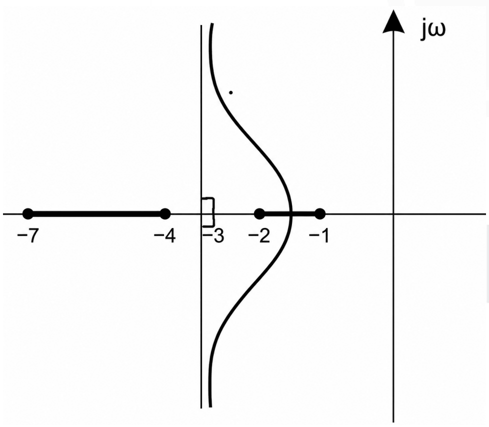


Fig. Solution Q5.14.2

Testing Option (C):
Open-loop poles ($P = 3$) at $s = -1, -2, -4$ and open-loop zero ($Z = 1$) at $s = -7$.

- Asymptote Angles ($P - Z = 2$):

$$\theta_l = 90°, 270°$$

- Centroid $(\sigma) = \dfrac{(-1-2-4)-(-7)}{3-1} = 0$

The asymptotes would intersect precisely at the origin ($s = 0$), which clearly contradicts the provided plot where the complex branches bend well before reaching the imaginary axis.

$$\boxed{(B) \quad \frac{s+4}{(s+1)(s+2)(s+7)}}$$

Q5.11.1 MCQ 2011 1M Ans: B ● ○ ○

- **Zeros:** There is a visible open-loop zero at $s = -1$.
- **Poles on the real axis:** There are poles at $s = 0$ and $s = -2$.
- **Behavior at $s = -3$:** The root locus branch on the real axis originates from $s = -3$ and moves in both directions (to the left towards $-\infty$ and to the right towards the break-away point). For a branch to move out in both directions from a single point on the real axis, there must be a multiple pole at that location. Therefore, there are 2 **poles located at** $s = -3$.

Thus, the overall open-loop transfer function is given by:

$$\boxed{(B) \quad G(s)H(s) = K\frac{(s+1)}{s(s+2)(s+3)^2}}$$

Mistakes to be avoided

Do not overlook the direction arrows near $s = -3$. Mistaking it for a single pole would lead to an incorrect count of branches and asymptotes.

Q5.09.1 MCQ 2009 2M Ans: MTA ● ● ○

The characteristic equation of the closed-loop system is:

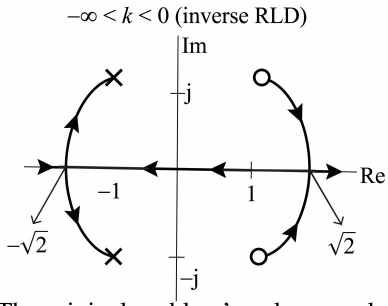
$$1 + KG(s)H(s) = 0 \implies 1 + K\frac{s^2 - 2s + 2}{s^2 + 2s + 2} = 0$$

$$K = -\frac{s^2 + 2s + 2}{s^2 - 2s + 2}$$

Setting $\dfrac{\partial K}{\partial s} = 0$ for breakaway/break-in points:

$$\frac{\partial K}{\partial s} = -\frac{(2s+2)(s^2 - 2s + 2) - (2s-2)(s^2 + 2s + 2)}{(s^2 - 2s + 2)^2} = 0$$

$$(s^3 - s^2 + 2) - (s^3 + s^2 - 2) = 0 \implies \boxed{s = \pm\sqrt{2}}$$



Note: The original problem's pole-zero plot layout is incorrect.

Zeros are at $s = 1 \pm j$ and poles are at $s = -1 \pm j$. The angular contributions to complex pole $P = -1 + j$ are:

- From pole $P^* = -1 - j \implies \phi_{p1} = 90^\circ$
- From zero $Z_1 = 1 + j \implies \phi_{z1} = 180^\circ$
- From zero $Z_2 = 1 - j \implies \phi_{z2} = 180^\circ - \tan^{-1}\left(\frac{2}{2}\right) = 135^\circ$

For a negative root locus ($K < 0$), **the phase criteria equals** 0° **(or even multiples of** 180°**).**

Method 1

Using phasor phase angle method for $K < 0$:

$$\phi_D = 0^\circ + \angle G(s)H(s)\Big|_{s=-1+j}$$

$$\angle G(s)H(s) = (\phi_{z1} + \phi_{z2}) - \phi_{p1}$$

$$\angle G(s)H(s) = (180^\circ + 135^\circ) - 90^\circ = 225^\circ$$

$$\phi_D = 0^\circ + 225^\circ = 225^\circ \equiv -135^\circ$$

Method 2

Using explicit summation method for $K < 0$:

$$\phi_D = 0^\circ - \phi \quad \left[\text{where } \phi = \sum \phi_{p'} - \sum \phi_z\right]$$

$$\phi = \phi_{p1} - (\phi_{z1} + \phi_{z2})$$

$$\phi = 90^\circ - (180^\circ + 135^\circ) = -225^\circ$$

$$\phi_D = 0^\circ - (-225^\circ) = 225^\circ \equiv -135^\circ$$

No option matches $\boxed{\phi_D = -135^\circ \text{ or } 225^\circ}$.

Mistakes to be avoided

For a negative root locus ($K < 0$), the baseline reference angle factor switches from 180° to 0°.

Q5.07.1 MCQ 2007 2M Ans: D ● ○ ○

For a point s to lie on the root locus of a system, it must satisfy the magnitude condition:

$$|G(s)H(s)| = 1$$

$$\left|\frac{K}{s(s+3)(s+4)}\right|_{s=-1+j1} = 1$$

$$\frac{K}{|-1+j1||2+j1||3+j1|}=1$$

Substituting these values back into the magnitude condition equation:

$$\frac{K}{\sqrt{2}\cdot\sqrt{5}\cdot\sqrt{10}}=1 \implies \frac{K}{10}=1 \implies K=10$$

$$\boxed{(D)\ \ 10}$$

Q5.05.1 MCQ 2005 2M Ans: C ● ○ ○

From the characteristic equation $1+G(s)H(s)=0$:

$$1+\frac{K(1-s)}{s(s+3)}=0 \implies s(s+3)+K(1-s)=0$$

$$K=\frac{s(s+3)}{1-s}$$

For determining the breakaway and break-in points, we set $\dfrac{dK}{ds}=0$:

$$\frac{dK}{ds}=\frac{(1-s)(2s+3)-s(s+3)(-1)}{(1-s)^2}=0$$

$$(1-s)(2s+3)+s^2+3s=0$$

$$s^2-2s-3=0$$

$$(s-3)(s+1)=0$$

$$\implies s=-1,3 \text{ are the break points.}$$

$$\boxed{(C)}$$

Q5.04.1 MCQ 2004 1M Ans: C ● ○ ○

The intersection point of the asymptotes on the real axis is known as the centroid.

$$\text{Centroid}=\frac{\sum P-\sum Z}{P-Z}=\frac{-1-3}{3}=-1.33$$

$$\boxed{(C)-1.33}$$

Q5.03.1 MCQ 2003 2M Ans: D ● ○ ○

The characteristic equation is $1+G(s)H(s)=0$

$$1+\frac{K}{s(s+2)(s+3)}=0$$

$$K=-s(s+2)(s+3)=-(s^3+5s^2+6s)$$

To find the breakaway points, set $\frac{dK}{ds}=0$:

$$\frac{dK}{ds}=-(3s^2+10s+6)=0 \implies 3s^2+10s+6=0$$

Solving the quadratic equation gives $s = -0.784$ and $s = -2.55$.
Since $s = -0.784$ lies on the root locus branch, it is a valid breakaway point, corresponding to the coordinates $(-0.784, 0)$.
$s = -2.55$ does not lie on the root locus, making it invalid.

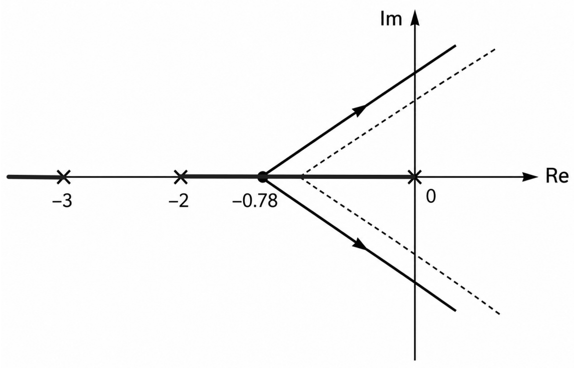


Fig. Solution Q5.03.1

$\boxed{\text{(D) (-0.784, 0)}}$

Key Points

- Breakaway points must lie on the valid sections of the root locus on the real axis to be physically realizable.

Q5.02.1 MCQ 2002 1M Ans: B ● ○ ○

Given the open-loop transfer function: $G(s)H(s) = \dfrac{K}{s(s+1)(s+3)}$
The open-loop poles are located at $s = 0, -1, -3$.

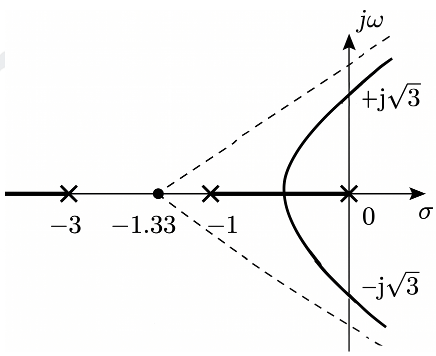


Fig. Solution Q5.02.1

Since $s = -1.5$ lies in the region between $s = -1$ and $s = -3$, it does not lie on the root locus.

$$\boxed{(B) \quad s = -1.5}$$

Q5.01.1 MCQ | 2001 | 1M | Ans: D | ● ○ ○

If the roots of the characteristic equation lie on the negative real axis at different locations (i.e., unequal), then the system response is over-damped.

- For $0 \le K < 1$, the roots are on the real axis.
- For $1 < K < 5$, the roots are complex conjugates.
- For $K > 5$, the roots are on the real axis.

Therefore, for an over-damped system, the range of K values is $0 \le K < 1$ or $K > 5$.

$$\boxed{\text{(D)} \quad \text{if } \; 0 \le K < 1 \text{ or } K > 5}$$

Q5.01.2 NAT | 2001 | 5M | Ans: * | ● ● ○

Given the open-loop transfer function:

$$G(s) = \frac{K(K_D s + K_P)}{s(s+1)}$$

Assuming unity negative feedback ($H(s) = 1$), the closed-loop transfer function is:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)} = \frac{K(K_D s + K_P)}{s^2 + s(KK_D + 1) + KK_P}$$

Given the time constant $\tau = 1 \text{ sec} = \dfrac{1}{\xi\omega_n}$, which implies:

$$\xi\omega_n = 1$$

Given the damping ratio $\xi = 0.707$:

$$\omega_n = \frac{1}{0.707} \approx 1.41 \text{ rad/sec}$$

The characteristic equation poles are given by:

$$\omega_1, \omega_2 = -\xi\omega_n \pm \omega_n\sqrt{1-\xi^2} = \omega_n\left(-\xi \pm \sqrt{1-\xi^2}\right)$$

$$\omega_1, \omega_2 = -1 \pm 1.41\sqrt{1-(0.707)^2} = \boxed{-1 \pm 1j}$$

Q5.01.3 NAT | 2001 | 5M | Ans: * | ● ○ ○

Given the open-loop transfer function:

$$G(s) = \frac{K(s+2)}{s(s+1)}$$

Number of asymptotes $N = 1$; Angle of asymptotes $\theta = 180°$
To find the breakaway and break-in points, set the characteristic equation $1 + G(s) = 0$:

$$-1 = \frac{K(s+2)}{s(s+1)} \implies K = \frac{-s^2 - s}{s+2}$$

$$\frac{dK}{ds} = \frac{(-2s-1)(s+2) + s^2 + s}{(s+2)^2} = 0$$

$$-2s^2 - s - 4s - 2 + s^2 + s = 0$$

$$s^2 + 4s + 2 = 0$$

$$s = \frac{-4 \pm \sqrt{16-8}}{2} = \frac{-4 \pm \sqrt{8}}{2} = \frac{-4 \pm 2\sqrt{2}}{2} = -2 \pm \sqrt{2}$$

Calculating the numerical values:

$$s = -2 + 1.414 = -0.6 \quad \text{(Breakaway point)}$$

$$s = -2 - 1.414 = -3.414 \quad \text{(Break-in point)}$$

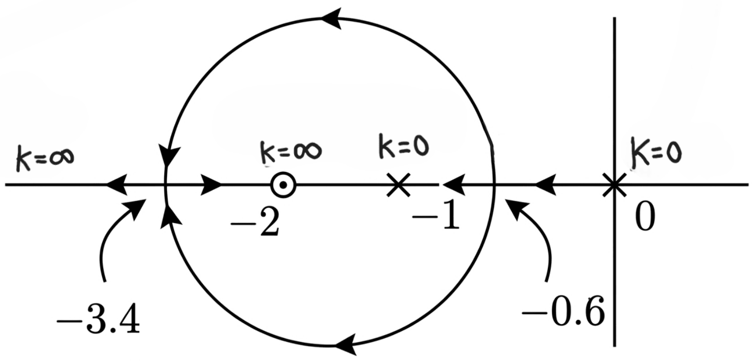


Fig. Solution Q5.01.3

Q5.99.1 | MCQ | 1999 | 2M | Ans: B

The given open-loop transfer function is:

$$GH(s) = \frac{K}{(s+1)^4}$$

To verify if a point lies on the root locus, it must satisfy the angle condition: $\angle GH(s) = \pm(2q+1)180°$.

Case 1: For $s_1 = -3 + j4$

$$GH(s_1) = \frac{K}{(-3+j4+1)^4} = \frac{K}{(-2+j4)^4}$$

$$\begin{aligned}\angle GH(s_1) &= -4(180° - \tan^{-1} 2)\\ &= -4(180° - 63.43°)\\ &= -466.28°\end{aligned}$$

Since $-466.28°$ is not an odd multiple of 180°, s_1 does not lie on the root locus.

Case 2: For $s_2 = -3 - j2$

$$GH(s_2) = \frac{K}{(-3-j2+1)^4} = \frac{K}{(-2-j2)^4}$$

$$\begin{aligned}\angle GH(s_2) &= -4(180° + \tan^{-1} 1)\\ &= -4(180° + 45°)\\ &= -900° = -5 \times 180°\end{aligned}$$

Since $-900°$ is an odd multiple of 180°, s_2 lies on the root locus.

$\boxed{\text{(B) } s_2 \text{ is on the root locus, but not } s_1}$

Q5.94.1 | MCQ | 1994 | 1M | Ans: D

- m = degree of numerator polynomial = number of open-loop zeros
- n = degree of denominator polynomial = number of open-loop poles

he integer $(n - m)$ represents the number of asymptotes.

$\boxed{\text{(D) Asymptotes}}$

Q5.92.1 MCQ 1992 2M Ans: A ● ○ ○

Given the open-loop transfer function:

$$G(s) = \frac{K}{s(s+1)(s+2)}$$

1. Angles of Asymptotes:

$$\theta = \frac{(2Q+1)180^\circ}{P-Z} = 60^\circ, 180^\circ, 300^\circ \implies \pm 60^\circ, 180^\circ$$

2. Centroid (σ):

$$\sigma = \frac{\sum \text{Poles} - \sum \text{Zeros}}{P-Z} = \frac{(-2-1-0)-0}{3} = -1$$

3. Breakaway Points: The characteristic equation is $1 + G(s) = 0$

$$K = -s(s+1)(s+2) = -(s^3 + 3s^2 + 2s)$$

Setting $\frac{dK}{ds} = 0$:

$$-(3s^2 + 6s + 2) = 0 \implies 3s^2 + 6s + 2 = 0$$

$$s = \frac{-6 \pm \sqrt{36-24}}{6} \implies s = -0.42, \ -1.58$$

Since the root locus exists on the real axis between 0 and -1, $s = -0.42$ is the valid breakaway point.

Thus, the root locus features a breakaway point at -0.42 and asymptotically branches out at $\pm 60^\circ$ centered from $\sigma = -1$, matching plot (a).

$$\boxed{\text{(A)}}$$

Q5.91.1 MCQ 1991 2M Ans: B ● ○ ○

Given characteristic equation:

$$s^3 + 5s^2 + (K+6)s + K = 0 \implies s(s^2 + 5s + 6) + K(s+1) = 0$$

Rearranging into the standard form $1 + K \cdot G(s)H(s) = 0$:

$$1 + \frac{K(s+1)}{s(s^2+5s+6)} = 0 \implies 1 + \frac{K(s+1)}{s(s+2)(s+3)} = 0$$

Thus, the open-loop transfer function is $G(s)H(s) = \dfrac{K(s+1)}{s(s+2)(s+3)}$.

Centroid (σ):

$$\sigma = \frac{\sum \text{Real parts of poles} - \sum \text{Real parts of zeros}}{n-m}$$

$$\sigma = \frac{(0-2-3)-(-1)}{3-1} = \frac{-5+1}{2} = -2$$

The asymptotes intersect the real axis at $\sigma = -2$, giving the coordinate point $(-2, 0)$.

$$\boxed{\text{(B)} \ \ (-2, 0)}$$

Q5.90.1 MCQ 1990 2M Ans: B ● ● ○

Given the closed-loop characteristic equation:

$$s^2 + (3-K)s + 1 = 0 \implies s^2 + 3s + 1 - Ks = 0$$

Rearranging into the standard root locus form $1 + K \cdot G(s)H(s) = 0$:

$$1 - \frac{Ks}{s^2+3s+1} = 0 \implies 1 + K\left(\frac{-s}{s^2+3s+1}\right) = 0$$

- **Open-loop Poles ($K = 0$):**

$$s^2 + 3s + 1 = 0$$
$$s = \frac{-3 \pm \sqrt{9-4}}{2} = -0.38,\ -2.61$$

- **Open-loop Zeros ($K = \infty$):** There is one finite zero at $s = 0$.
- **Root Locus Behavior:** As K increases from 0:
  - The two real poles move toward each other, break away into the complex plane, form a circular loop, and re-enter the real axis.
  - One branch terminates at the zero ($s = 0$) and the other goes to $-\infty$.

$$\boxed{\text{(B)}}$$

Q5.89.1 MCQ 1989 2M Ans: A ● ○ ○

Elimination Process:

- Options (b) and (d) are incorrect because the root locus diagram (RLD) must always be symmetrical about the real axis.
- Option (c) is incorrect because root locus branches must always direct from open-loop poles toward open-loop zeros.

$$\boxed{\text{(A)}}$$

Q5.88.1 NAT 1988 2M Ans: D ● ○ ○

From Fig.(a), the closed-loop transfer function is given by:

$$\text{T.F.} = \frac{K \cdot G(s)}{1 + K \cdot G(s)}$$

From the root locus the open-loop transfer function: $G(s) = \dfrac{1}{(s+0.1)(s+2)}$

$$\text{T.F.} = \frac{K \cdot \dfrac{1}{(s+0.1)(s+2)}}{1 + K \cdot \dfrac{1}{(s+0.1)(s+2)}}$$

$$\text{T.F.} = \frac{K}{K + (s+0.1)(s+2)}$$

Rearranging the terms in the denominator to standard time-constant form:

$$\boxed{\text{(D)}\ \ \text{T.F.} = \frac{K}{K + 0.2(10s+1)(0.5s+1)}}.$$

DEBUG INFO

Chapter V

Root Locus

Total Questions : 28

MCQ : 18

MSQ : 1

NAT : 9

SOLUTIONS

VI

Polar Plot and Frequency Response of 2nd Order Systems

Topics

1. Basics of Frequency Response Analysis
2. Frequency Response of 2nd Order Systems
3. Polar Plot
4. Gain and Phase Crossover Frequency
5. Gain Margin and Phase Margin

Q6.23.1 MCQ 2023 1M Ans: A ● ○ ○

Given,

$$G(s) = \frac{k}{s(1 + sT_1)(1 + sT_2)}$$

The phase cross-over frequency is the frequency at which

$$\angle G(j\omega) = -180^\circ \ (-\pi \text{ radians})$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{k}{j\omega(1 + j\omega T_1)(1 + j\omega T_2)}$$

The phase angle is

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2)$$

At phase cross-over frequency,

$$-90^\circ - \tan^{-1}(\omega T_1) - \tan^{-1}(\omega T_2) = -180^\circ$$

$$(\omega T_1)(\omega T_2) = 1$$

$$\tan^{-1}(\omega T_1) + \tan^{-1}(\omega T_2) = 90^\circ$$

Using $\tan^{-1} x + \tan^{-1} y = 90^\circ \Rightarrow xy = 1$, we get

$$(\omega T_1)(\omega T_2) = 1$$

$$\omega^2 T_1 T_2 = 1$$

$$\omega^2 = \frac{1}{T_1 T_2}$$

$$\omega = \frac{1}{\sqrt{T_1 T_2}}$$

$$\boxed{\text{(A)} \ \ \omega = \frac{1}{\sqrt{T_1 T_2}}}$$

Key Points

- Phase cross-over frequency is the frequency at which the phase angle becomes -180° or $-\pi$ radians.
- The phase contribution of $j\omega$ is 90°.
- The phase contribution of $(1 + j\omega T)$ is $\tan^{-1}(\omega T)$.
- For a product of terms in the denominator, phase angles are added with negative sign.

Mistakes to be avoided

- Do not use $+180^\circ$ instead of -180° for phase cross-over frequency.
- Note that $-180^\circ = -\pi$ radians.
- Do not forget the -90° phase contribution due to the pole at origin.
- Do not write $\tan^{-1}(x + y) = \tan^{-1} x + \tan^{-1} y$.

Q6.20.1 NAT 2020 2M Ans: 3.95 to 4.05 ● ○ ○

Given,

$$G(s) = \frac{1}{(s+1)(s+a)}, \qquad a > 0$$

Input:

$$x(t) = 5\cos 3t$$

Steady-state output:

$$y(t) = \frac{1}{\sqrt{10}}\cos(3t - 1.892)$$

For a sinusoidal input, the steady-state output is

$$y(t) = A|G(j\omega_0)|\cos\left(\omega_0 t + \angle G(j\omega_0)\right)$$

Here,

$$\omega_0 = 3\ \text{rad/s}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{1}{(1+j\omega)(a+j\omega)}$$

Magnitude:

$$|G(j\omega)| = \frac{1}{\sqrt{1+\omega^2}\sqrt{a^2+\omega^2}}$$

At $\omega = 3$,

$$|G(j3)| = \frac{1}{\sqrt{1+3^2}\sqrt{a^2+3^2}}$$

$$= \frac{1}{\sqrt{10}\sqrt{a^2+9}}$$

Output amplitude is

$$5 \times |G(j3)| = \frac{5}{\sqrt{10}\sqrt{a^2+9}}$$

Comparing with the given output amplitude,

$$\frac{5}{\sqrt{10}\sqrt{a^2+9}} = \frac{1}{\sqrt{10}}$$

$$\frac{25}{10(a^2+9)} = \frac{1}{10}$$

$$25 = a^2 + 9$$

$$a^2 = 16$$

$$a = \pm 4$$

Since $a > 0$,

$$a = 4$$

$$\boxed{4}$$

Key Points

- For sinusoidal input $A\cos(\omega_0 t)$, the steady-state output is obtained using frequency response $G(j\omega_0)$.
- Output amplitude is multiplied by $|G(j\omega_0)|$.

- Output phase shift is $\angle G(j\omega_0)$.
- The input frequency remains unchanged in steady-state response.

Mistakes to be avoided

- Do not forget to substitute ω_0 equal to the input frequency before calculating magnitude and phase.
- Do not change the input frequency in the steady-state output expression.
- Do not take the negative value of a since $a > 0$ is given.

Q6.18.1 NAT 2018 2M Ans: 14 to 17

Given,

$$G(s) = \frac{K}{s(s+2)}$$

The closed-loop transfer function for unity feedback is

$$T(s) = \frac{G(s)}{1+G(s)}$$

$$T(s) = \frac{\frac{K}{s(s+2)}}{1+\frac{K}{s(s+2)}}$$

$$T(s) = \frac{K}{s^2+2s+K}$$

Comparing with the standard second-order transfer function

$$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$$

we get

$$2\zeta\omega_n = 2 \quad \text{and} \quad \omega_n^2 = K$$

Now,

$$\omega_n = \sqrt{K}$$

Hence,

$$2\zeta\sqrt{K} = 2$$

$$\zeta = \frac{1}{\sqrt{K}}$$

For a second-order system,

$$M_r = \frac{\text{DC Gain}}{2\zeta\sqrt{1-\zeta^2}}$$

Given,

$$M_r = 2$$

$$\frac{1}{2\zeta\sqrt{1-\zeta^2}} = 2$$

$$1 = 4\zeta\sqrt{1-\zeta^2}$$

Squaring both sides,

$$1 = 16\zeta^2(1-\zeta^2)$$

Substituting

$$\zeta = \frac{1}{\sqrt{K}}$$

$$1 = 16\left(\frac{1}{K}\right)\left(1 - \frac{1}{K}\right)$$

$$1 = \frac{16(K-1)}{K^2}$$

$$K^2 = 16K - 16$$

$$K^2 - 16K + 16 = 0$$

The roots for the above equation are:

$$K = 8 \pm 4\sqrt{3}$$

Since resonant peak exists only for $\zeta < \frac{1}{\sqrt{2}}$,

$$K > 2$$

Therefore,

$$K = 8 + 4\sqrt{3} = 14.93$$

$$\boxed{14.93}$$

Key Points

- For unity feedback system,

$$T(s) = \frac{G(s)}{1 + G(s)}$$

- Standard second-order characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

- Resonant peak magnitude is

$$M_r = \frac{Dc\ \ Gain}{2\zeta\sqrt{1-\zeta^2}}$$

- The DC gain of a system is given by

$$K_{dc} = \lim_{s\to 0} (\text{poles at origin}) \times \text{Transfer Function}$$

- Resonant peak exists only if

$$\zeta < \frac{1}{\sqrt{2}}$$

Mistakes to be avoided

- Do not use the resonant peak formula for systems with $\zeta \geq \frac{1}{\sqrt{2}}$.
- Do not forget to check the DC Gain of the system if you are calculating Resonant Peak.

Q6.16.1 NAT 2016 2M Ans: 59.5 to 60.5 ● ● ○

Given,

$$G(s) = \frac{1}{(s+1)(s+2)(s+3)}$$

For the unity feedback system, the characteristic equation is

$$1 + kG(s) = 0$$

$$1 + \frac{k}{(s+1)(s+2)(s+3)} = 0$$

$$(s+1)(s+2)(s+3) + k = 0$$

$$s^3 + 6s^2 + 11s + 6 + k = 0$$

Given that: Gain Margin = 0 dB and Phase Margin = 0° Hence, the system is marginally stable.
Using Routh-Hurwitz criterion,

$$\begin{array}{c|cc} s^3 & 1 & 11 \\ s^2 & 6 & 6+k \\ s^1 & \dfrac{66-(6+k)}{6} & 0 \\ s^0 & 6+k & \end{array}$$

For marginal stability,

$$\frac{66-(6+k)}{6} = 0$$

$$66 - 6 - k = 0$$

$$k = 60$$

$$\boxed{60}$$

Key Points

- Zero Phase margin and zero Gain margin indicate marginal stability.
- In Routh array, marginal stability occurs when an entire row element becomes zero.

Q6.15.1 NAT 2015 1M Ans: 84 to 84.5 ● ○ ○

Given,

$$G(s) = \frac{10}{s(s+10)}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{10}{j\omega(j\omega+10)}$$

Magnitude condition at gain cross-over frequency:

$$|G(j\omega_{gc})| = 1$$

$$\left|\frac{10}{j\omega(j\omega+10)}\right| = 1$$

$$\frac{10}{\omega\sqrt{\omega^2+10^2}} = 1$$

$$100 = \omega^2(\omega^2+100)$$

$$\omega^4 + 100\omega^2 - 100 = 0$$

$$\text{Let } x = \omega^2$$

$$x^2 + 100x - 100 = 0$$

The valid root of the above quadratic expression is:

$$x = 0.99$$

$$\omega_{gc} = \sqrt{0.99} \approx 0.995 \text{ rad/s}$$

Phase angle at gain cross-over frequency:

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{10}\right)$$

At $\omega = \omega_{gc}$,

$$\angle G(j\omega_{gc}) = -90^\circ - \tan^{-1}\left(\frac{0.995}{10}\right) = -95.68^\circ$$

Phase margin is

$$PM = 180^\circ + \angle G(j\omega_{gc}) = 84.32^\circ$$

$$\boxed{84.32^\circ}$$

Key Points

- Gain cross-over frequency is the frequency at which

$$|G(j\omega)| = 1$$

- Phase margin is given by

$$PM = 180^\circ + \angle G(j\omega_{gc})$$

- Pole at origin contributes -90° phase shift.

Q6.15.2

MCQ | 2015 | 1M | Ans: A | ● ○ ○

Given,

$$G(s) = \frac{10(s+1)}{s+10}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{10(1+j\omega)}{10+j\omega}$$

Rationalizing,

$$G(j\omega) = \frac{10(1+j\omega)(10-j\omega)}{(10+j\omega)(10-j\omega)}$$

$$G(j\omega) = \frac{100+10\omega^2}{100+\omega^2} + j\frac{90\omega}{100+\omega^2}$$

Thus, We can see

$$\Re\{G(j\omega)\} = \frac{100+10\omega^2}{100+\omega^2} > 0 \quad and \quad \Im\{G(j\omega)\} = \frac{90\omega}{100+\omega^2} > 0 \quad for \;\; 0 \le \omega < \infty$$

Hence, the polar plot lies entirely in the first quadrant.

$$\boxed{\text{(A) First Quadrant}}$$

Key Points

- If both real and imaginary parts are positive, the plot lies in the first quadrant.

Mistakes to be avoided

- Do not forget to rationalize the denominator before separating real and imaginary parts.
- Be careful during simplification.

Q6.11.1 MCQ 2011 2M Ans: C

Given,

$$G(s) = \frac{100}{s(s+10)^2}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{100}{j\omega(j\omega+10)^2}$$

At phase cross-over frequency,

$$\angle G(j\omega) = -180^\circ$$
$$-90^\circ - 2\tan^{-1}\left(\frac{\omega}{10}\right) = -180^\circ$$
$$\tan^{-1}\left(\frac{\omega}{10}\right) = 45^\circ$$
$$\omega_{pc} = 10\,\text{rad/s}$$

Magnitude at phase cross-over frequency:

$$|G(j\omega)| = \frac{100}{\omega(\omega^2+10^2)}$$

At $\omega = 10$,

$$|G(j10)| = \frac{100}{10(100+100)} = \frac{100}{10(100+100)} = \frac{1}{20}$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|} = 20$$

In dB,

$$GM_{dB} = 20\log_{10}(20) = 26\,\text{dB}$$

$$\boxed{\text{(C) } 26\,\text{dB}}$$

Q6.10.1 MCQ 2010 1M Ans: B

Given,

$$\frac{Y(s)}{X(s)} = \frac{s}{s+p}$$

$$Input: \quad x(t) = p\cos\left(2t - \frac{\pi}{2}\right) \qquad Output: \quad y(t) = \cos\left(2t - \frac{\pi}{3}\right)$$

For sinusoidal input, the steady-state response is

$$y(t) = |G(j\omega)|A\cos\left(\omega t + \phi + \angle G(j\omega)\right)$$

Here,

$$G(j\omega) = \frac{Y(s)}{X(s)} \quad and \quad \omega = 2$$

Substituting $s = j\omega$,

$$G(j2) = \frac{j2}{p + j2}$$

Magnitude:

$$|G(j2)| = \frac{2}{\sqrt{p^2 + 4}}$$

Output amplitude is

$$p \times \frac{2}{\sqrt{p^2 + 4}}$$

Given output amplitude is 1,

$$p \times \frac{2}{\sqrt{p^2 + 4}} = 1$$

$$2p = \sqrt{p^2 + 4}$$

$$p = \frac{2}{\sqrt{3}}$$

$$\boxed{\text{(B) } \frac{2}{\sqrt{3}}}$$

Q6.08.1 MCQ | 2008 | 2M | Ans: A | ● ● ○

For a standard second-order system,

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Given:

$$\text{Magnitude at } \omega = 0 \text{ rad/s} = 5$$

Hence, DC gain is

$$K_{dc} = 5$$

Also,

$$M_r = \frac{10}{\sqrt{3}}$$

We know,

$$M_r = \frac{K_{dc}}{2\zeta\sqrt{1 - \zeta^2}}$$

$$\frac{10}{\sqrt{3}} = \frac{5}{2\zeta\sqrt{1 - \zeta^2}}$$

$$2\zeta\sqrt{1 - \zeta^2} = \frac{\sqrt{3}}{2}$$

$$\zeta = \frac{1}{2}$$

Also, resonant peak occurs at

$$\omega_r = \omega_n\sqrt{1 - 2\zeta^2}$$

Given,

$$\omega_r = 5\sqrt{2} \text{ rad/s}$$

Substituting $\zeta = \frac{1}{2}$,

$$5\sqrt{2} = \omega_n \sqrt{1 - 2\left(\frac{1}{2}\right)^2}$$

$$5\sqrt{2} = \omega_n \sqrt{\frac{1}{2}}$$

$$\omega_n = 10\,\text{rad/s}$$

Now,

$$2\zeta\omega_n = 2 \times \frac{1}{2} \times 10 = 10$$

and

$$\omega_n^2 = 100$$

Hence,

$$T(s) = \frac{K_{dc}\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$T(s) = \frac{5 \times 100}{s^2 + 10s + 100}$$

$$T(s) = \frac{500}{s^2 + 10s + 100}$$

$$\boxed{(\text{A})\ \frac{500}{s^2 + 10s + 100}}$$

Key Points

- Magnitude at $\omega = 0$ gives the DC gain of the system.
- Resonant peak magnitude is

$$M_r = \frac{K_{dc}}{2\zeta\sqrt{1 - \zeta^2}}$$

Mistakes to be avoided

- Do not assume unity DC gain unless explicitly given.
- Do not forget that magnitude at $\omega = 0$ represents DC gain.

Q6.06.1 MCQ 2006 1M Ans: D ● ● ○

Given,

$$G(s) = \frac{K}{(s + 1)(s + 2)}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{K}{(1 + j\omega)(2 + j\omega)}$$

Phase angle is

$$\angle G(j\omega) = -\tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right)$$

For gain margin calculation,

$$\angle G(j\omega) = -180^\circ$$

Thus,

$$\tan^{-1}(\omega) + \tan^{-1}\left(\frac{\omega}{2}\right) = 180^\circ$$

But,

$$\tan^{-1}(\omega) < 90^\circ$$

and

$$\tan^{-1}\left(\frac{\omega}{2}\right) < 90^\circ$$

Hence,

$$\tan^{-1}(\omega) + \tan^{-1}\left(\frac{\omega}{2}\right) < 180^\circ$$

Therefore, phase cross-over frequency does not exist. Hence,

$$GM = \infty$$

Checking stability:
For unity feedback system, characteristic equation is

$$1 + G(s) = 0$$

$$1 + \frac{K}{(s+1)(s+2)} = 0$$

$$(s+1)(s+2) + K = 0$$

$$s^2 + 3s + (2+K) = 0$$

Using Routh-Hurwitz criterion,

$$\begin{array}{c|cc} s^2 & 1 & 2+K \\ s^1 & 3 & 0 \\ s^0 & 2+K & \end{array}$$

For all positive values of K,

$$2 + K > 0$$

Hence, all first-column elements are positive. Therefore, the system remains stable for all positive values of K. Thus, gain can be increased indefinitely before instability occurs. Hence, $GM = \infty$

$$\boxed{\text{(D) } GM = \infty}$$

Key Points

- Gain margin is infinite if phase cross-over frequency does not exist.
- If the system is absolutely stable then the Gain Margin is $+\infty$ and if it is unstable then the Gain Margin is $-\infty$

Mistakes to be avoided

- Do not forget to check whether phase cross-over frequency exists and valid.
- Always check for stability in case of ambiguity.

Q6.06.2 MCQ 2006 1M Ans: A ● ○ ○

Given,

$$G(s) = \frac{1}{s+1}$$

Input:

$$x(t) = \sin t \; u(t)$$

For sinusoidal steady-state response,

$$y(t) = |G(j\omega)| A \sin(\omega t + \angle G(j\omega))$$

Here,

$$\omega = 1 \text{ rad/s}$$

Substituting $s = j\omega$,

$$G(j1) = \frac{1}{1+j}$$

Magnitude:

$$|G(j1)| = \frac{1}{\sqrt{1^2+1^2}} = \frac{1}{\sqrt{2}}$$

Phase angle:

$$\angle G(j1) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

Hence,

$$y(t) = \frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)$$

$$\boxed{\text{(A) } \frac{1}{\sqrt{2}} \sin\left(t - \frac{\pi}{4}\right)}$$

Q6.06.3 MCQ 2006 2M Ans: C ● ● ○

Given:

$$G_1(s) = \frac{1}{s^2 + as + b} \quad = \text{CLTF}$$

$$G_2(s) = \frac{s}{s^2 + as + b} \quad = \text{CLTF}$$

The characteristic equation for $G_1(s)$ is given by,

$$s^2 + as + b = 0$$

Roots of above characteristic equation is given by,

$$s_1, s_2 = \frac{-a \pm \sqrt{a^2 - 4b}}{2}$$

$$G_1(s) = \frac{1}{\left(s + \frac{a}{2} - \frac{\sqrt{a^2-4b}}{2}\right)\left(s + \frac{a}{2} + \frac{\sqrt{a^2-4b}}{2}\right)}$$

$$G_1(s) = \frac{1}{(s - \omega_L)(s - \omega_H)}$$

$G_1(s)$ represents product of transfer functions of two low pass filters with cut-off frequencies of ω_L and ω_H. Thus $G_1(s)$ represents a low pass filter with two poles. When a system has multiple poles, the dominant pole (i.e. nearest $j\omega$-axis) given overall cut-off frequency of transfer functions $G_1(s)$ is

$$\omega_o = \frac{-a + \sqrt{a^2 - 4b}}{2}$$

3-dB bandwidth of system is,

$$BW = \omega_o = \frac{-a + \sqrt{a^2 - 4b}}{2}$$

$$G_2(s) = \frac{s}{\left(s + \frac{a}{2} - \frac{\sqrt{a^2-4b}}{2}\right)\left(s + \frac{a}{2} + \frac{\sqrt{a^2-4b}}{2}\right)}$$

$$G_2(s) = \frac{s}{(s - \omega_L)(s - \omega_H)}$$

$G_2(s)$ represents a bandpass filter with lower and upper cut-off frequencies ω_L and ω_H respectively. Thus $G_2(s)$ represents a bandpass filter with two poles.

$$\omega_L = \frac{a - \sqrt{a^2 - 4b}}{2}$$

$$\omega_H = \frac{a + \sqrt{a^2 - 4b}}{2}$$

3-dB bandwidth is given by,

$$BW = \omega_H - \omega_L$$

$$BW = \left(\frac{a + \sqrt{a^2-4b}}{2}\right) - \left(\frac{a - \sqrt{a^2-4b}}{2}\right)$$

$$BW = \sqrt{a^2 - 4b}$$

Hence, none of the options is correct. Nearest answer is option (C).

$$\boxed{\text{(C) } \sqrt{a^2-4b},\ \sqrt{a^2-4b}}$$

Q6.06.4 MCQ 2006 2M Ans: C ● ● ○

Given,

$$G(s) = \frac{as + 1}{s^2}$$

For unity feedback system,

$$PM = \frac{\pi}{4} = 45^\circ$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{1 + ja\omega}{(j\omega)^2}$$

Phase angle is

$$\angle G(j\omega) = \tan^{-1}(a\omega) - 180^\circ$$

Phase margin is

$$PM = 180^\circ + \angle G(j\omega_{gc})$$

$$45^\circ = 180^\circ + \tan^{-1}(a\omega_{gc}) - 180^\circ$$

$$\tan^{-1}(a\omega_{gc}) = 45^\circ$$

$$a\omega_{gc} = 1$$

$$\omega_{gc} = \frac{1}{a}$$

At gain cross-over frequency,

$$|G(j\omega_{gc})| = 1$$

Magnitude is

$$|G(j\omega)| = \frac{\sqrt{1 + a^2\omega^2}}{\omega^2}$$

Substituting $\omega_{gc} = \frac{1}{a}$,

$$1 = \frac{\sqrt{1 + a^2\left(\frac{1}{a}\right)^2}}{\left(\frac{1}{a}\right)^2}$$

$$a \approx 0.84$$

$$\boxed{\text{(C) } 0.84}$$

Q6.06.5 MCQ | 2006 | 2M | Ans: C | ● ● ○

From Q6.06.4,

$$a = 0.84$$

Given,

$$G(s) = \frac{as + 1}{s^2}$$

Substituting $a = 0.84$,

$$G(s) = \frac{0.84s + 1}{s^2}$$

For unit impulse input,

$$R(s) = 1$$

Hence,

$$C(s) = G(s)R(s)$$

$$C(s) = \frac{0.84s + 1}{s^2}$$

$$C(s) = \frac{0.84}{s} + \frac{1}{s^2}$$

Taking inverse Laplace transform,

$$c(t) = 0.84 + t$$

At

$$t = 1\text{ s}$$

$$c(1) = 0.84 + 1$$

$$c(1) = 1.84$$

$$\boxed{\text{(C) } 1.84}$$

Q6.05.1 MCQ 2005 2M Ans: D ● ● ●

Given,

$$G(s) = \frac{3e^{-2s}}{s(s+2)}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{3e^{-j2\omega}}{j\omega(j\omega+2)}$$

The exponential term contributes only phase and does not affect magnitude. Phase contributed by $e^{-j2\omega}$ is -2ω. Hence,

$$|G(j\omega)| = \frac{3}{\omega\sqrt{\omega^2+4}}$$

Gain crossover frequency is obtained from

$$|G(j\omega_{gc})| = 1$$

Therefore,

$$\frac{3}{\omega_{gc}\sqrt{\omega_{gc}^2+4}} = 1$$

Squaring both sides,

$$9 = \omega_{gc}^2(\omega_{gc}^2+4)$$

$$\omega_{gc}^4 + 4\omega_{gc}^2 - 9 = 0$$

$$\text{Let, } x = \omega_{gc}^2$$

Then,

$$x^2 + 4x - 9 = 0$$

Solving,

$$x = \frac{-4 \pm \sqrt{16+36}}{2}$$

$$x = -2 + \sqrt{13}$$

$$\omega_{gc} = \sqrt{-2+\sqrt{13}}$$

$$\omega_{gc} \approx 1.26 \text{ rad/sec}$$

Now, phase of the transfer function is

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) - 2\omega$$

For phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180^\circ$$

Hence,

$$-90^\circ - \tan^{-1}\left(\frac{\omega_{pc}}{2}\right) - 2\omega_{pc} = -180^\circ$$

$$\tan^{-1}\left(\frac{\omega_{pc}}{2}\right) + 2\omega_{pc} = 90^\circ$$

Method 1

Option Substitution

The above equation is transcendental in nature and ω_{pc} cannot be separated explicitly in closed form. Hence, the phase crossover frequency is obtained by substituting the given options.

$$For \quad \omega_{pc} = 0.632$$

$$\tan^{-1}\left(\frac{0.632}{2}\right) + 2(0.632)$$

$$\approx 1.57 \text{ rad} = 90^\circ$$

Hence,

$$\omega_{pc} \approx 0.632 \text{ rad/sec}$$

Method 2

Approximation Method

Using the identity,

$$\tan^{-1} x = x - \frac{x^3}{3} + \cdots$$

Approximating to the first term,

$$\tan^{-1} x \approx x$$

Therefore,

$$\tan^{-1}\left(\frac{\omega_{pc}}{2}\right) \approx \frac{\omega_{pc}}{2}$$

Substituting,

$$\frac{\omega_{pc}}{2} + 2\omega_{pc} = \frac{\pi}{2}$$

$$\frac{5\omega_{pc}}{2} = \frac{\pi}{2}$$

$$\omega_{pc} = \frac{\pi}{5} \approx 0.628 \text{ rad/sec}$$

Hence,

$$\omega_{pc} \approx 0.632 \text{ rad/sec}$$

(D) 1.26 and 0.632

Key Points

- Time delay term $e^{-j\omega T}$ affects only phase and not magnitude.

Mistakes to be avoided

- While calculating phase, remember that
$$e^{-j\omega T} \Rightarrow -\omega T \text{ radians}$$
- Do not mix degrees and radians in delay phase term.

Q6.05.2 | MCQ | 2005 | 2M | Ans: D | ● ● ○

From Question Q6.05.1,

$$\omega_{gc} = 1.26 \text{ rad/sec} \quad and \quad \omega_{pc} = 0.632 \text{ rad/sec}$$

Gain Margin Gain margin is given by

$$GM = \frac{1}{|G(j\omega_{pc})|}$$

Now,

$$|G(j\omega)| = \frac{3}{\omega\sqrt{\omega^2 + 4}}$$

Substituting $\omega_{pc} = 0.632$,

$$|G(j0.632)| = \frac{3}{0.632\sqrt{0.632^2 + 4}} \approx 2.26$$

Hence,

$$GM = \frac{1}{2.26} \approx 0.442$$

Gain margin in dB is

$$GM_{dB} = 20\log_{10}(0.442) \approx -7.09 \text{ dB}$$

Phase Margin Phase margin is given by

$$PM = 180^\circ + \angle G(j\omega_{gc})$$

Phase of the transfer function is

$$\angle G(j\omega) = -90^\circ - \tan^{-1}\left(\frac{\omega}{2}\right) - 2\omega$$

Substituting $\omega_{gc} = 1.26$,

$$\angle G(j1.26) = -90^\circ - \tan^{-1}(0.63) - 2(1.26)$$

Converting delay term into degrees,

$$2(1.26) = 2.52 \text{ rad}$$

$$2.52 \times \frac{180}{\pi} \approx 144.4^\circ$$

Also,

$$\tan^{-1}(0.63) \approx 32.211^\circ$$

Therefore,

$$\angle G(j1.26) = -90^\circ - 32.211^\circ - 144.4^\circ \approx -266.611^\circ$$

Hence,

$$PM = 180^\circ + (-266.611^\circ) \approx -86.61^\circ$$

Approximating,

$$PM \approx -87.5^\circ$$

$$\boxed{\text{(D)} \ -7.09 \text{ dB and } -87.5^\circ}$$

Q6.04.1 MCQ 2004 1M Ans: B ● ○ ○

Given : The open loop transfer function is,

$$G(s)H(s) = \frac{2(1+s)}{s^2}$$

Put $s = j\omega$ in above equation,

$$G(j\omega)H(j\omega) = \frac{2(1+j\omega)}{(j\omega)^2}$$

Magnitude of open loop transfer function $G(j\omega)H(j\omega)$ is given by,

$$|G(j\omega)H(j\omega)| = \frac{2\sqrt{1+\omega^2}}{\omega^2}$$

Phase angle of open loop transfer function $G(j\omega)H(j\omega)$ is given by,

$$\angle G(j\omega)H(j\omega) = -180^\circ + \tan^{-1}\omega$$

ω	$\lvert G(j\omega)H(j\omega)\rvert$	$\angle G(j\omega)H(j\omega)$
0	∞	-180°
∞	0	-90°

From above table,

$$\omega_{pc} = 0 \text{ rad/sec}$$

and

$$|G(j\omega_{pc})H(j\omega_{pc})| = \infty$$

Gain margin is given by,

$$\text{GM} = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{\infty} = 0$$

$$\boxed{\text{(B) } 0}$$

Q6.04.2 MCQ 2004 2M Ans: A ● ○ ○

Given,

$$\text{Poles at } 0.1 \text{ Hz}, \ 1 \text{ Hz}, \ 80 \text{ Hz} \quad and \quad \text{Zeros at } 5 \text{ Hz}, \ 100 \text{ Hz}, \ 200 \text{ Hz}$$

The phase contribution of a pole or zero depends on the relation between operating frequency and break frequency. At frequency much higher than break frequency,

$$\text{Pole contributes } -90^\circ$$

$$\text{Zero contributes } +90^\circ$$

At frequency much lower than break frequency,

$$\text{Phase contribution} \approx 0^\circ$$

Now, the response is required at

$$f = 20 \text{ Hz}$$

Phase contribution of poles

$$\begin{aligned} 0.1 \text{ Hz} \ll 20 \text{ Hz} &\quad \Rightarrow -90^\circ \\ 1 \text{ Hz} \ll 20 \text{ Hz} &\quad \Rightarrow -90^\circ \\ 80 \text{ Hz} > 20 \text{ Hz} &\quad \Rightarrow 0^\circ \end{aligned}$$

Hence,

$$\phi_p = -180^\circ$$

Phase contribution of zeros

$$\begin{aligned} 5 \text{ Hz} < 20 \text{ Hz} &\quad \Rightarrow +90^\circ \\ 100 \text{ Hz} > 20 \text{ Hz} &\quad \Rightarrow 0^\circ \\ 200 \text{ Hz} > 20 \text{ Hz} &\quad \Rightarrow 0^\circ \end{aligned}$$

Hence,

$$\phi_z = +90^\circ$$

Therefore, total phase is

$$\phi = \phi_z + \phi_p$$

$$\phi = (+90^\circ) + (-180^\circ)$$

$$\phi = -90^\circ$$

$$\boxed{\text{(A) } -90^\circ}$$

Key Points

- Pole contributes approximately -90° at frequencies much higher than its break frequency.
- Zero contributes approximately $+90^\circ$ at frequencies much higher than its break frequency.
- At frequencies much lower than break frequency, phase contribution is approximately 0°.

Q6.03.1 MCQ | 2003 | 2M | Ans: B | ● ● ○

Given,

$$G(s)H(s) = \frac{s}{(s+100)^3}$$

Substituting $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{j\omega}{(100+j\omega)^3}$$

Magnitude is

$$|G(j\omega)H(j\omega)| = \frac{\omega}{(100^2+\omega^2)^{3/2}}$$

Phase of the transfer function is

$$\angle G(j\omega)H(j\omega) = 90^\circ - 3\tan^{-1}\left(\frac{\omega}{100}\right)$$

Gain Crossover Frequency

Gain crossover frequency occurs when

$$|G(j\omega_{gc})H(j\omega_{gc})| = 1$$

Therefore,

$$\frac{\omega_{gc}}{(100^2+\omega_{gc}^2)^{3/2}} = 1$$

$$\omega_{gc} = (100^2+\omega_{gc}^2)^{3/2}$$

Since,

$$(100^2+\omega_{gc}^2)^{3/2} \gg \omega_{gc}$$

the above equation cannot be satisfied for any finite value of ω. Hence, gain crossover frequency does not exist.

$$PM = \infty$$

Phase Crossover Frequency

For phase crossover frequency,

$$\angle G(j\omega_{pc})H(j\omega_{pc}) = -180^\circ$$

Hence,

$$90^\circ - 3\tan^{-1}\left(\frac{\omega_{pc}}{100}\right) = -180^\circ$$

$$3\tan^{-1}\left(\frac{\omega_{pc}}{100}\right) = 270^\circ$$

$$\tan^{-1}\left(\frac{\omega_{pc}}{100}\right) = 90^\circ$$

which is not possible for any finite value of ω. Hence, phase crossover frequency does not exist.

$$GM = \infty$$

To verify the result, let us check the closed-loop stability using Routh-Hurwitz criterion. Characteristic equation is

$$1 + G(s)H(s) = 0$$

$$1 + \frac{s}{(s+100)^3} = 0$$

$$(s+100)^3 + s = 0$$

$$s^3 + 300s^2 + 30001s + 10^6 = 0$$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 30001 \\ s^2 & 300 & 10^6 \\ s^1 & \dfrac{300(30001)-10^6}{300} & 0 \\ s^0 & 10^6 & \end{array}$$

$$\frac{300(30001)-10^6}{300} = 26667.67$$

Hence, first column elements are

$$1, \quad 300, \quad 26667.67, \quad 10^6$$

All are positive. Therefore, the closed-loop system is stable. Hence, the system can tolerate infinite gain increase and phase variation before becoming unstable.

$$\boxed{\text{(B) } \infty,\ \infty}$$

Q6.02.1 MCQ 2002 1M Ans: D ● ○ ○

Given,

$$G(s)H(s) = \frac{1-s}{(1+s)(2+s)}$$

Substituting $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{1-j\omega}{(1+j\omega)(2+j\omega)}$$

Magnitude is

$$|G(j\omega)H(j\omega)| = \frac{\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}\sqrt{4+\omega^2}}$$

$$|G(j\omega)H(j\omega)| = \frac{1}{\sqrt{4+\omega^2}}$$

For gain crossover frequency,

$$|G(j\omega_{gc})H(j\omega_{gc})| = 1$$

Hence,

$$\frac{1}{\sqrt{4+\omega_{gc}^2}} = 1$$

$$4+\omega_{gc}^2 = 1$$

$$\omega_{gc}^2 = -3$$

which is physically impossible. Hence, gain crossover frequency does not exist. Therefore, phase margin becomes infinite.
To verify the result, let us check the closed-loop stability using Routh-Hurwitz criterion.
Characteristic equation is

$$1 + G(s)H(s) = 0$$

$$1 + \frac{1-s}{(1+s)(2+s)} = 0$$

$$(1+s)(2+s) + (1-s) = 0$$

$$2 + 3s + s^2 + 1 - s = 0$$

$$s^2 + 2s + 3 = 0$$

Routh array:

$$\begin{array}{c|cc} s^2 & 1 & 3 \\ s^1 & 2 & 0 \\ s^0 & 3 & \end{array}$$

First column elements are positive. Therefore, the closed-loop system is stable. Hence, the system can tolerate infinite phase variation before becoming unstable. Thus, $PM = \infty$.

$$\boxed{\text{(D) } \infty}$$

Key Points

- For a non minimum phase system with zero in Right Half, the GM - Stability and PM - stability relation remains same.

Q6.02.2 MCQ 2002 2M Ans: B ● ○ ○

Given,

$$G(s)H(s) = \frac{1}{s(s^2+s+1)}$$

Substituting $s = j\omega$,

$$G(j\omega)H(j\omega) = \frac{1}{j\omega\left((j\omega)^2 + j\omega + 1\right)} = \frac{1}{j\omega\left(1-\omega^2+j\omega\right)}$$

Phase Crossover Frequency

Phase crossover frequency occurs when

$$\angle G(j\omega)H(j\omega) = -180^\circ$$

Now,

$$\angle G(j\omega)H(j\omega) = -90^\circ - \angle(1-\omega^2+j\omega)$$

For phase crossover frequency,

$$-90^\circ - \angle(1-\omega_{pc}^2 + j\omega_{pc}) = -180^\circ$$

Hence,

$$\angle(1-\omega_{pc}^2 + j\omega_{pc}) = 90^\circ$$

This occurs when real part becomes zero.
Therefore,

$$1-\omega_{pc}^2 = 0$$

$$\omega_{pc} = 1 \text{ rad/sec}$$

Gain Margin

Gain margin is evaluated at phase crossover frequency. Hence,

$$|G(j\omega_{pc})H(j\omega_{pc})| = |G(j1)H(j1)|$$

Substituting $\omega = 1$,

$$G(j1)H(j1) = \frac{1}{j(1-1+j)} = \frac{1}{j(j)} = -1$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{1} = 1$$

Gain margin in dB is

$$GM_{dB} = 20\log_{10}(1) = 0 \text{ dB}$$

$$\boxed{\text{(B) 0 dB}}$$

Q6.94.1 MCQ 1994 1M Ans: C ● ● ●

Given transfer function of a second-order system is

$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Substituting $s = j\omega$,

$$\frac{C(j\omega)}{R(j\omega)} = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2}$$

Since,

$$(j\omega)^2 = -\omega^2$$

$$\frac{C(j\omega)}{R(j\omega)} = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + j2\zeta\omega_n\omega}$$

Magnitude is

$$\left|\frac{C(j\omega)}{R(j\omega)}\right| = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}}$$

For 3-dB bandwidth,

$$\left|\frac{C(j\omega)}{R(j\omega)}\right| = \frac{1}{\sqrt{2}}$$

Hence,

$$\frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2}} = \frac{1}{\sqrt{2}}$$

Squaring both sides,

$$\frac{\omega_n^4}{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2} = \frac{1}{2}$$

$$2\omega_n^4 = (\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega_n^2\omega^2$$

Expanding,

$$2\omega_n^4 = \omega_n^4 - 2\omega_n^2\omega^2 + \omega^4 + 4\zeta^2\omega_n^2\omega^2$$

$$\omega^4 + (4\zeta^2 - 2)\omega_n^2\omega^2 - \omega_n^4 = 0$$

Dividing by ω_n^4,

$$\left(\frac{\omega}{\omega_n}\right)^4 + (4\zeta^2 - 2)\left(\frac{\omega}{\omega_n}\right)^2 - 1 = 0$$

$$\text{Let,}\quad x = \left(\frac{\omega}{\omega_n}\right)^2$$

Then,

$$x^2 + (4\zeta^2 - 2)x - 1 = 0$$

Using quadratic formula,

$$x = \frac{-(4\zeta^2 - 2) + \sqrt{(4\zeta^2 - 2)^2 + 4}}{2}$$

$$x = -2\zeta^2 + 1 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}$$

Since,

$$x = \left(\frac{\omega}{\omega_n}\right)^2$$

$$\frac{\omega}{\omega_n} = \sqrt{-2\zeta^2 + 1 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

$$\omega = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

$$\boxed{\text{(C) } \omega = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}}$$

Key Points

- The cut-off frequency of a standard second-order system is

$$\omega_c = \omega_n\sqrt{1 - 2\zeta^2 + \sqrt{4\zeta^4 - 4\zeta^2 + 2}}$$

It is better to remember the above expression directly for quick problem solving in competitive exams.

Q6.87.1 MCQ 1987 2M Ans: C ● ● ○

Given,

$$G(s) = \frac{10}{s(s+1)^2}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{10}{j\omega(1+j\omega)^2} = \frac{10}{j\omega(1+2j\omega-\omega^2)} = \frac{10}{-2\omega^2 + j(\omega-\omega^3)}$$

To separate real and imaginary parts, multiply numerator and denominator by complex conjugate.

$$G(j\omega) = \frac{10[-2\omega^2 - j(\omega-\omega^3)]}{4\omega^4 + (\omega-\omega^3)^2}$$

Thus,

$$\Re\{G(j\omega)\} = \frac{-20\omega^2}{4\omega^4 + (\omega-\omega^3)^2}$$

and

$$\Im\{G(j\omega)\} = \frac{-10(\omega-\omega^3)}{4\omega^4 + (\omega-\omega^3)^2}$$

The polar plot intercepts the real axis when imaginary part becomes zero. Hence,

$$\omega - \omega^3 = 0$$

$$\omega(1-\omega^2) = 0$$

$$\omega = 0, \pm 1$$

Ignoring negative frequency and $\omega = 0$. Therefore,

$$\omega_0 = 1$$

Now, substituting $\omega = 1$ in real part,

$$\Re\{G(j1)\} = \frac{-20}{4}$$

$$\Re\{G(j1)\} = -5$$

$$\boxed{\text{(C)} \; -5,\ 1}$$

DEBUG INFO

Chapter VI

Polar Plot and Frequency Response of 2nd Order Systems

Total Questions : 23

MCQ : 19

MSQ : 0

NAT : 4

SOLUTIONS

VII

Nyquist Plot

Topics

1. Nyquist Plot
2. Nyquist Stability Criterion
3. Gain Crossover Frequency (ω_{gc}) from Nyquist Plot
4. Phase Crossover Frequency (ω_{pc}) from Nyquist Plot
5. Gain Margin (GM) using Nyquist Plot
6. Phase Margin (PM) using Nyquist Plot

Q7.25.1 MCQ 2025 1M Ans: D ● ○ ○

Gain crossover frequency occurs when $|G(j\omega)| = 1$
In the Nyquist plot, this corresponds to the point where the plot intersects the unit circle. The dashed circle shown in the figure represents the unit circle.
From the figure, the Nyquist plot intersects the unit circle at point Q. Hence, $\omega_S = \omega_{gc}$
Phase crossover frequency occurs when $\angle G(j\omega) = -180°$
In the Nyquist plot, this corresponds to the point where the plot intersects the negative real axis.
From the figure, point S lies on the negative real axis. Hence, $\omega_Q = \omega_{pc}$

$$\boxed{\text{(D) } \omega_S \text{ is the gain crossover frequency and } \omega_Q \text{ is the phase crossover frequency}}$$

Q7.22.1 MCQ 2022 1M Ans: C ● ○ ○

In the question, the Nyquist plot is given for $G(s)$.
The forward path gain is $K = 2$ with unity negative feedback. Hence, the open-loop transfer function becomes $2G(s)$
Scaling the Nyquist plot of $G(s)$ by a factor of 2 enlarges the plot radially by 2. Therefore, the Nyquist plot for $2G(s)$ is shown below.

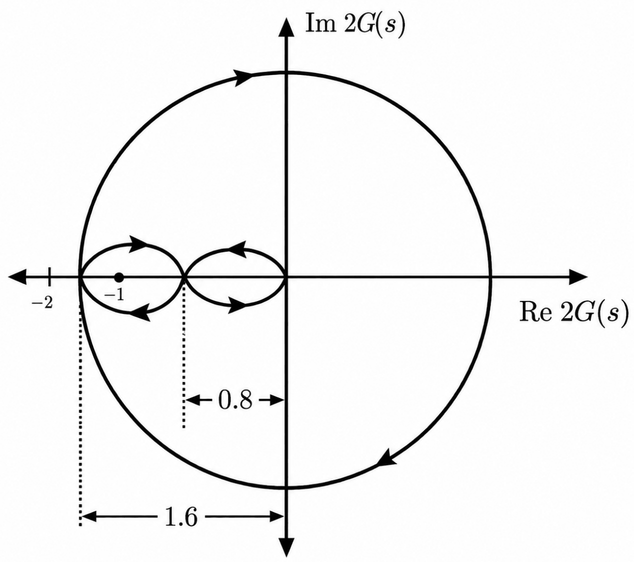


Fig. Solution Q7.22.1

The problem states that $G(s)$ has no poles in the closed right half plane. Scaling by a constant gain does not change the pole locations. Hence, the open-loop transfer function $2G(s)$ also has no poles in the right half plane. Therefore, $P_+ = 0$
For unity negative feedback, the characteristic equation is

$$1 + 2G(s) = 0$$

The poles of the closed-loop transfer function are the zeros of the characteristic equation.
Using Nyquist criterion,

$$N = P_+ - Z_+$$

where,

- N = net number of clockwise encirclements of $(-1, j0)$

- P_+ = number of right half plane poles of characteristic equation, which is same as the poles of open loop system
- Z_+ = number of right half plane zeros of characteristic equation

It is given in the question that the Nyquist contour is traversed in clockwise direction.
From the scaled Nyquist plot, the critical point $(-1, j0)$ is encircled twice in clockwise direction by $2G(s)$ plot. The number of times $2G(s)$ encircles $(-1, j0)$ is same as the number of times $1 + 2G(s) = 0$ encirlces origin.
Hence,

$$N = -2$$

Substituting in Nyquist criterion,

$$-2 = 0 - Z_+$$

$$Z_+ = 2$$

Since the poles of the closed-loop transfer function are the zeros of the characteristic equation. Hence the number of closed-loop poles in RHP = 2.

$\boxed{\text{(C) } 2}$

Key Points

- Scaling the open-loop transfer function by a constant gain scales the Nyquist plot radially.
- Multiplying by gain does not change pole locations.
- Closed-loop poles are the zeros of the characteristic equation. And open loop poles are equal to the poles of characteristic equation.

Mistakes to be avoided

- Do not apply Nyquist criterion directly on the unscaled plot when gain changes.
- Be careful with the sign convention of clockwise encirclements.
- Do not confuse poles of open-loop transfer function with poles of closed-loop transfer function.

Q7.21.1 MCQ 2021 2M Ans: D ● ● ●

It is given that the open-loop transfer function $G(s)H(s)$ has one zero in the right half of the s-plane. Hence, $Z_+ = 1$
Let us first assume that the Nyquist contour is traversed in clockwise direction. Applying Nyquist Stability Criterion (NSC) to the open-loop transfer function,

$$N = P_+ - Z_+$$

From the given Nyquist plot, the $G(s)H(s)$-plane plot encircles the origin twice in clockwise direction. Hence, $N = -2$
Substituting,

$$-2 = P_+ - 1$$

$$P_+ = -1$$

This is physically impossible since number of poles cannot be negative. Therefore, the assumed clockwise traversal is incorrect. Hence, the Nyquist contour must actually be traversed in anticlockwise direction. Thus, $N = +2$
Applying NSC again,

$$2 = P_+ - 1$$

$$P_+ = 3$$

Therefore, the open-loop transfer function has three poles in the right half plane.
The poles of the closed-loop transfer function are the zeros of the characteristic equation.

The characteristic equation is

$$1 + G(s)H(s) = 0$$

Now we will apply the NSC for the characteristic equation. Hence,

$$N = P_+ - Z_+$$

where,

- N = net number of encirclements of $(-1, j0)$
- P_+ = number of right half plane poles of characteristic equation, which is same as the poles of open loop system
- Z_+ = number of right half plane zeros of characteristic equation, which are same as the poles of closed loop system

From the given Nyquist plot, $(-1, j0)$ is enclosed once in anticlockwise direction and once in clockwise direction by the $G(s)H(s) = 0$ plot. As the encirclements of $G(s)H(s) = 0$ around $(-1, j0)$ is same as the encirclements of $1+G(s)H(s) = 0$ around origin.
Hence,

$$N = 0$$

Substituting,

$$0 = 3 - Z_+$$

$$Z_+ = 3$$

Hence, the number of closed-loop transfer function poles in the right half plane are:

$$\boxed{\text{(D) } 3}$$

Key Points

- Physically impossible values such as negative number of poles indicate incorrect contour direction assumption.
- Closed-loop poles are the zeros of the characteristic equation

$$1 + G(s)H(s) = 0$$

- Open loop poles are the poles of characteristic equation.

Mistakes to be avoided

- Do not assume contour direction blindly.
- If P_+ or Z_+ becomes negative, recheck encirclement sign convention.

Q7.20.1 MCQ 2020 1M Ans: B ● ○ ○

From the given pole-zero plot, the number of poles encircled by the closed contour Γ are: $P_+ = 2$
And, the number of zeros encircled by the closed contour Γ are: $Z_+ = 3$
Applying Nyquist Stability Criterion:

$$N = P_+ - Z_+$$

$$N = 2 - 3$$

$$N = -1$$

Since N is negative, the mapping encircles the origin once in the clockwise direction.

$$\boxed{\text{(B) the origin of the } G(s)\text{-plane once in the clockwise direction}}$$

Q7.18.1 MCQ | 2018 | 1M | Ans: D | ● ● ○

Option A
Routh criterion helps in finding the range of gain for which the closed-loop system remains stable.
Nyquist criterion also gives information about stability margins and stable gain range using encirclement concept.
Hence, this statement is true.
Option B
For minimum phase systems, magnitude and phase are related.
Therefore, from the Bode magnitude plot, the general nature of Nyquist plot can be approximately visualized.
Hence, this statement is true.
Option C
Routh criterion is mainly applicable for polynomial characteristic equations.
Transport delay introduces exponential term such as e^{-sT} which makes the characteristic equation transcendental. Hence, Routh criterion cannot be directly applied. Nyquist criterion can easily handle such delay terms using frequency response.
Therefore, this statement is true.
Option D
Nyquist plot contains both magnitude and phase information of the open-loop transfer function.
From Nyquist plot, closed-loop frequency response characteristics can also be determined.
Therefore, the statement "closed-loop frequency response cannot be obtained" is incorrect.
Hence, this statement is false.

$$\boxed{\text{(D) The closed-loop frequency response for a unity feedback system cannot be obtained from the Nyquist plot}}$$

Q7.17.1 MCQ | 2017 | 2M | Ans: C | ● ● ●

Given,

$$G(s) = \frac{10K(s+2)}{s^3 + 3s^2 + 10} \quad \text{and} \quad 0 < K < 1$$

The Nyquist plot of $G(s)$ is already provided in the question.
For unity negative feedback, the characteristic equation is

$$1 + G(s) = 0$$

The poles of the closed-loop transfer function are the zeros of the characteristic equation.
Applying Nyquist criterion,

$$N = P_+ - Z_+$$

where,

- N = net number of encirclements of $(-1, j0)$
- P_+ = number of right half plane poles of characteristic equation, which is same as the poles of open loop system
- Z_+ = number of right half plane zeros of characteristic equation, which are same as the poles of closed loop system

From the given Nyquist path, the contour is traversed in clockwise direction.
Now, for $0 < K < 1$ the point $-K$ lies between 0 and -1 on the real axis. From the Nyquist plot, the critical point $(-1, j0)$ is not encircled. Hence,

$$N = 0$$

Now, let us determine P_+.
Open-loop transfer function denominator is

$$s^3 + 3s^2 + 10 = 0$$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 0 \\ s^2 & 3 & 10 \\ s^1 & -\dfrac{10}{3} & 0 \\ s^0 & 10 & \end{array}$$

First column elements are

$$1, \quad 3, \quad -\frac{10}{3}, \quad 10$$

There are two sign changes. Hence,

$$P_+ = 2$$

Substituting in Nyquist criterion,

$$0 = 2 - Z_+$$

$$Z_+ = 2$$

$$\boxed{\text{(C) } 2}$$

Q7.17.2 MCQ 2017 2M Ans: B ● ● ○

Given,

$$G(s) = \frac{K}{(s^2 + 2s + 2)(s + 2)}$$

For unity negative feedback, the characteristic equation is

$$1 + G(s) = 0$$

Applying Nyquist Stability Criterion (NSC),

$$N = P_+ - Z_+$$

where,

- N = net encirclements of open loop system around $(-1, j0)$ or net encirclements of characteristic equation around origin
- P_+ = number of right half plane poles of open-loop transfer function or characteristic equation
- Z_+ = number of right half plane poles of closed-loop transfer function or zeros of characteristic equation

The open-loop poles are obtained from

$$(s^2 + 2s + 2)(s + 2) = 0$$

$$s = -2, \quad -1 \pm j$$

All poles lie in the left half plane. Hence,

$$P_+ = 0$$

Case 1 : $K = 10$

Assume the Nyquist contour is traversed in clockwise direction.

It is given in the question that the open loop system's Nyquist plot does not encircle the point $(-1, j0)$, it implies that the net encirclements of characteristic equation around origin is 0. Hence,

$$N = 0$$

Applying NSC,

$$0 = 0 - Z_+$$

$$Z_+ = 0$$

Therefore, the closed-loop system has no right half plane poles. Hence, the system is stable for $K = 10$

Case 2 : $K = 100$

It is given that the Nyquist plot of open loop system encircles the point $(-1, j0)$.

Hence,

$$N = \pm a \qquad \text{where} \qquad a \neq 0$$

Applying NSC,

$$\pm a = 0 - Z_+$$

$$Z_+ = \mp a$$

Since $a \neq 0$,

$$Z_+ \neq 0$$

Therefore, the closed-loop system has right half plane poles. Hence, the system is unstable for $K = 100$

$$\boxed{\text{(B) stable for K = 10 and unstable for K = 100}}$$

Q7.16.1 MCQ 2016 1M Ans: A ● ● ○

Given,

$$G(s) = \frac{1-s}{4+2s}$$

The question asks for the number and direction of encirclements of the point $(-1, j0)$ by the Nyquist plot of $G(s)$. Encirclement of the point $(-1, j0)$ by $G(s)$ is equivalent to encirclement of origin by

$$1 + G(s)$$

Hence, Nyquist Stability Criterion can be applied to

$$1 + G(s) = 0$$

Applying NSC,

$$N = P_+ - Z_+$$

where,

- N = encirclements of $(-1, j0)$ by $G(s)$
- P_+ = number of right half plane poles of $G(s)$
- Z_+ = number of right half plane zeros of $1 + G(s)$

Finding P_+,

$$4 + 2s = 0$$

$$s = -2$$

Hence,

$$P_+ = 0$$

Finding Z_+,

$$1 + G(s) = 0$$

$$1 + \frac{1-s}{4+2s} = 0$$

$$4 + 2s + 1 - s = 0$$

$$s = -5$$

Hence,

$$Z_+ = 0$$

Applying NSC,

$$N = 0 - 0$$

$$N = 0$$

Hence, the Nyquist plot of $G(s)$ gives zero encirclements about the point $(-1, j0)$.

(A) zero

Q7.16.2 MCQ 2016 1M Ans: A ● ○ ○

For a closed-loop system,

$$1 + G(s)H(s) = 0$$

Applying Nyquist Stability Criterion,

$$N = P_+ - Z_+$$

where,

- N = net encirclements of the point $(-1, j0)$
- P_+ = number of right half plane poles of open-loop transfer function
- Z_+ = number of right half plane poles of closed-loop transfer function

For closed-loop stability,

$$Z_+ = 0$$

Hence,

$$N = P_+$$

For the standard clockwise Nyquist contour, positive N corresponds to anticlockwise encirclements. Therefore, for stability, the Nyquist plot must encircle the point $(-1, j0)$ in anticlockwise direction as many times as the number of right half plane poles.

(A) encircles the s-plane point (-1+j0) in the counterclockwise direction as many times as the number of right-half s-plane poles

Q7.14.1 MCQ 2014 1M Ans: D ● ● ●

Option A

For an all-pass filter, $|G(j\omega)| = 1$ for all frequencies. Hence, the Nyquist plot of an all-pass filter is symmetric about the origin and lies on a unit circle.

The given Nyquist plot is not symmetric about the origin. Hence, option A is incorrect.

Option B

For a proper transfer function, $P > Z$ where, P is the number of poles and Z is the number of zeros

Assume the Nyquist contour is traversed in clockwise direction.

Applying Nyquist Stability Criterion to determine relation between poles and zeros,

$$N = P - Z$$

From the given Nyquist plot, the origin is encircled once in clockwise direction. Hence, $N = -1$

Therefore,

$$-1 = P - Z$$

$$Z - P = 1$$

Thus,

$$Z > P$$

Hence, the transfer function is not proper. Therefore, option B is incorrect.

Option C

From the previous result, $Z - P = 1$

This implies that there exists at least one right half plane pole or zero.

Hence, the system cannot be minimum phase. Therefore, option C is incorrect.

Option D

For the closed-loop system, $L(s) = kG(s)$

If k is sufficiently large and positive, the Nyquist plot expands radially and encircles the critical point $(-1, j0)$

Assume the Nyquist contour is traversed in clockwise direction.

Applying Nyquist Stability Criterion,

$$N = P - Z$$

Since the critical point is encircled once in clockwise direction, $N = -1$

Therefore,

$$-1 = P - Z$$

$$Z = P + 1$$

Since $P \geq 0$,

$$Z \geq 1$$

Hence, the closed-loop system has at least one right half plane pole. Therefore, the system becomes unstable.

$$\boxed{\text{(D) The closed-loop system is unstable for sufficiently large and positive k}}$$

Q7.11.1 | MCQ | 2011 | 1M | Ans: A | ● ○ ○

Given, $G(j\omega) = 5 + j\omega$

From above equation, $\Re\{G(j\omega)\} = 5$ and $\Im\{G(j\omega)\} = \omega$

For positive frequencies, $0 \leq \omega < \infty$, At $\omega = 0$ rad/sec, $G(j0) = 5$

Hence, the Nyquist plot starts from $(5, 0)$

As $\omega \to \infty$, $\Re\{G(j\omega)\} = 5$ and $\Im\{G(j\omega)\} \to \infty$

Hence, the Nyquist plot moves vertically upward keeping real part constant at 5

Therefore, the Nyquist plot is a vertical straight line passing through $\Re = 5$

$$\boxed{\text{(A)}}$$

Q7.09.1 | MCQ | 2009 | 2M | Ans: B | ● ● ○

Option A For an all-pass filter,

$$|G(j\omega)| = 1 \qquad \text{for all frequencies.}$$

Hence, the Nyquist plot must be symmetric about the origin and lie on a unit circle.

The given Nyquist plot is not symmetric about the origin. Therefore, option A is incorrect.

Option B

Assume the Nyquist contour is traversed in clockwise direction.

Applying Nyquist Stability Criterion to $G(s)$,

$$N = P_+ - Z_+$$

where,

$$P_+ = \text{number of RHP poles}$$
$$Z_+ = \text{number of RHP zeros}$$

From the given Nyquist plot, the origin is encircled once in clockwise direction. Hence,

$$N = -1$$

It is given that $G(s)$ is stable. Therefore,

$$P_+ = 0$$

Substituting,

$$-1 = 0 - Z_+$$

$$Z_+ = 1$$

Hence, $G(s)$ has one zero in the right half plane. Therefore, option B is correct.

Option C

The Nyquist plot of impedance of a passive network generally lies in the right half plane since passive networks do not generate energy.

The given plot extends into the left half plane. Hence, option C is incorrect.

Option D

Marginal stability requires poles on the imaginary axis.

From the given information, only stability of $G(s)$ is mentioned and no imaginary-axis poles are indicated. Hence, option D is incorrect.

$$\boxed{\text{(B) G(s) has a zero in the right-half plane}}$$

Q7.09.2 MCQ | 2009 | 2M | Ans: C | ● ● ○

From the given Nyquist plot, $|G(j\omega)| = 1$ occurs at the point $(0, -j)$

Hence, the gain crossover frequency corresponds to phase $-90°$

Therefore, phase margin is

$$PM = 180° - 90°$$

$$PM = 90°$$

Phase crossover frequency occurs when phase becomes $-180°$

From the Nyquist plot, this occurs at $(-0.5, j0)$

Hence,

$$|G(j\omega_{pc})| = 0.5$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|} = \frac{1}{0.5} = 2$$

Gain margin in dB is

$$GM_{dB} = 20\log_{10}(2) \approx 6 \text{ dB}$$

$$\boxed{\text{(C) 6 dB and } 90°}$$

Key Points

- Gain crossover frequency is the frequency at which $|G(j\omega)| = 1$
- In Nyquist plot, magnitude at any frequency is the distance of the point from the origin.
- Phase crossover frequency is the frequency at which $\angle G(j\omega) = -180°$
- Gain Margin is $GM = \dfrac{1}{|G(j\omega_{pc})|}$
- Phase Margin is $PM = 180° + \angle G(j\omega_{gc})$

Q7.06.1 MCQ 2006 2M Ans: D ● ○ ○

Gain margin is evaluated at phase crossover frequency. At phase crossover frequency, $\angle G(j\omega)H(j\omega) = -180°$
The problem states that the Nyquist plot passes through the point $(-1, j0)$. Hence, at phase crossover frequency,

$$|G(j\omega_{pc})H(j\omega_{pc})| = 1$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{1} = 1$$

Gain margin in dB is

$$GM_{dB} = 20\log_{10}(1) = 0\ dB$$

$\boxed{\text{(D) zero}}$

Q7.05.1 MCQ 2005 2M Ans: B ● ● ●

The given polar plot corresponds to the open loop transfer function for gain $K = 1$.
To analyze the stability of the closed loop system for a general gain K, the complete Nyquist plot can be redrawn by scaling the plot along the real axis by a factor of K.
Hence, the important intercepts on the real axis become $-8K$, $-2K$ and $-0.2K$
The complete Nyquist plot for gain K is shown below.

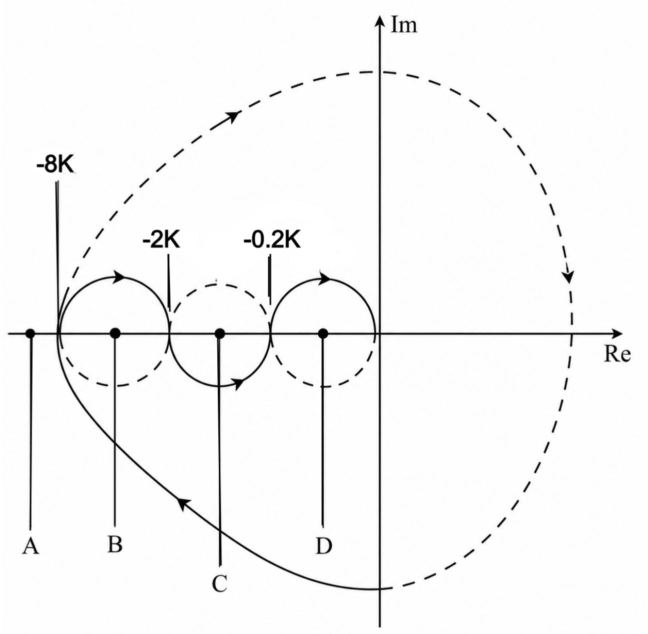


Fig. Solution Q7.05.1

The problem states that the open loop system is stable. Therefore, the open loop transfer function has no poles in the right half of the s-plane. Hence,

$$P_+ = 0$$

Using Nyquist stability criterion,

$$N = P_+ - Z_+$$

Applying the criterion to the characteristic equation,

$$1 + GH = 0$$

Substituting $P_+ = 0$,

$$N = 0 - Z_+$$

$$Z_+ = -N$$

For the closed loop system to be stable, the characteristic equation should not have any roots in the right half plane. Hence,

$$Z_+ = 0$$

Therefore,

$$N = 0$$

Thus, the closed loop system is stable only when the Nyquist plot gives zero net encirclements about the critical point $(-1, j0)$.
The location of the critical point $(-1, j0)$ with respect to the Nyquist plot gives four possible cases.

Case A

$$-1 < -8K$$

$$K < \frac{1}{8}$$

For this case,

$$N = 0$$

Hence,

$$Z_+ = 0$$

Therefore, the closed loop system is stable.

$$\boxed{K < \frac{1}{8} \Rightarrow \text{Stable}}$$

Case B

$$-8K < -1 < -2K$$

$$K > \frac{1}{8} \quad \text{and} \quad K < \frac{1}{2}$$

$$\frac{1}{8} < K < \frac{1}{2}$$

For this case,

$$N = 1$$

Hence,

$$Z_+ \neq 0$$

Therefore, the closed loop system is unstable.

$$\boxed{\frac{1}{8} < K < \frac{1}{2} \Rightarrow \text{Unstable}}$$

Case C

$$-2K < -1 < -0.2K$$

$$K > \frac{1}{2} \quad \text{and} \quad K < 5$$

$$\frac{1}{2} < K < 5$$

For this case,

$$N = 0$$

Hence,

$$Z_+ = 0$$

Therefore, the closed loop system is stable.

$$\boxed{\frac{1}{2} < K < 5 \Rightarrow \text{Stable}}$$

Case D

$$-0.2K < -1$$

$$K > 5$$

For this case,

$$N = -1$$

Hence,

$$Z_+ \neq 0$$

Therefore, the closed loop system is unstable.

$$\boxed{K > 5 \Rightarrow \text{Unstable}}$$

Hence, the closed loop system is stable for $K < \frac{1}{8}$ and $\frac{1}{2} < K < 5$

$$\boxed{\text{(B) } K < \frac{1}{8} \quad \text{and} \quad \frac{1}{2} < K < 5}$$

Key Points

- The poles of the open-loop transfer function and the characteristic equation are the same. Hence,

$$P_+(\text{open loop}) = P_+(\text{characteristic equation})$$

- The zeros of the characteristic equation are the poles of the closed-loop system. Therefore, closed-loop stability depends on the location of zeros of $1 + GH = 0$
- If $GH = -\alpha$ then the encirclement of the point $(-\alpha, 0)$ by GH is equivalent to the encirclement of origin by $\alpha + GH = 0$
- Hence, the number of encirclements of origin by $1 + GH = 0$ is equal to the number of encirclements of the critical point $(-1, j0)$ by GH.

Mistakes to be avoided

- By default, assume the Nyquist contour traversal in clockwise direction unless stated otherwise.
- If the obtained value of Z_+ or P_+ becomes physically impossible or absurd (such as negative number of poles/zeros), then recheck the direction of encirclement and consider anticlockwise traversal.
- Do not check only whether $(-1, j0)$ lies inside the loop; the net number of encirclements must be counted properly.

Q7.03.1 MCQ | 2003 | 1M | Ans: A | ● ○ ○

Given Nyquist plot corresponds to the open-loop transfer function $G(s)H(s)$
It is also given that the open-loop transfer function has one right half plane pole. Hence, $P_+ = 1$
Applying Nyquist Stability Criterion to the characteristic equation, $1 + G(s)H(s) = 0$,

$$N = P_+ - Z_+$$

where,

$$N = \text{number of encirclements of } (-1, j0)$$
$$P_+ = \text{number of RHP poles of } G(s)H(s)$$
$$Z_+ = \text{number of RHP zeros of } 1 + G(s)H(s)$$

Assume the Nyquist contour is traversed in clockwise direction.
From the given Nyquist plot, the point $(-1, j0)$ is encircled once in anti-clockwise direction. Hence, $N = 1$
Substituting,

$$1 = 1 - Z_+$$

$$Z_+ = 0$$

Zeros of $1 + G(s)H(s)$ are the poles of the closed-loop transfer function. Hence, the closed-loop system has zero right half plane poles. Therefore, the closed-loop system is always stable.

$$\boxed{\text{(A) always stable}}.$$

Q7.02.1 NAT 2002 5M Ans: * ● ● ○

Let the transfer function of the all-pole second-order system be

$$G(s) = \frac{K}{s^2 + as + b}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{K}{(b - \omega^2) + ja\omega}$$

From the Nyquist plot,

$$G(j0) = 2$$

Therefore,

$$\frac{K}{b} = 2$$

$$K = 2b$$

Also, the plot passes through the point $(0, -2)$ at $\omega = 2$ rad/sec. Hence,

$$G(j2) = -j2$$

Substituting $\omega = 2$,

$$G(j2) = \frac{K}{(b - 4) + j2a}$$

Rationalizing,

$$G(j2) = \frac{K[(b-4) - j2a]}{(b-4)^2 + 4a^2} = \frac{K(b-4)}{(b-4)^2 + 4a^2} - j\frac{2aK}{(b-4)^2 + 4a^2}$$

Since the Nyquist plot lies on the imaginary axis at $\omega = 2$, the real part must be zero.

$$\frac{K(b-4)}{(b-4)^2 + 4a^2} = 0$$

$$b - 4 = 0$$

$$b = 4$$

From the earlier result,

$$K = 2b$$

$$K = 2(4) = 8$$

Therefore,

$$G(j2) = \frac{8}{j2a}$$

Rationalizing,

$$G(j2) = \frac{8}{j2a} \times \frac{-j}{-j} = \frac{-8j}{2a} = -\frac{4j}{a}$$

But from the Nyquist plot,

$$G(j2) = -j2$$

Comparing imaginary parts,

$$\frac{4}{a} = 2$$
$$a = 2$$

Hence, the transfer function is

$$\boxed{G(s) = \frac{8}{s^2 + 2s + 4}}$$

Q7.01.1 MCQ | 2001 | 1M | Ans: A | ● ○ ○

Given Nyquist plot corresponds to the open-loop transfer function $G(s)$ It is given that the open-loop transfer function has no poles in the right half plane. Hence, $P_+ = 0$
Applying Nyquist Stability Criterion to the characteristic equation, $1 + G(s) = 0$,

$$N = P_+ - Z_+$$

where,

$$N = \text{number of encirclements of } (-1, j0) \text{ by G(s) Plot}$$
$$P_+ = \text{number of RHP poles of } G(s)$$
$$Z_+ = \text{number of RHP zeros of } 1 + G(s)$$

Assume the Nyquist contour is traversed in clockwise direction.
From the given Nyquist plot, the point $(-1, j0)$ is encircled once in clockwise and once in anti-clockwise direction. Hence, net encirclements are $N = 0$
Substituting,

$$0 = 0 - Z_+$$
$$Z_+ = 0$$

Zeros of $1 + G(s)$ are the roots of the characteristic equation. Hence, the system characteristic equation has one root in the right half plane. Therefore, the correct answer is

$$\boxed{(A)\ 0}$$

Q7.94.1 NAT | 1994 | 1M | Ans: Unstable | ● ○ ○

Given open-loop frequency response points are $1.2\angle - 180°$ and $1.0\angle - 190°$
At $\angle G(j\omega) = -180°$, the magnitude is $|G(j\omega)| = 1.2 > 1$
Hence, the Nyquist plot crosses the negative real axis to the left of the critical point $(-1, j0)$
Also, at another frequency, $|G(j\omega)| = 1$ occurs at phase $-190°$
Thus, gain crossover occurs after the phase has already crossed $-180°$. Therefore, $PM < 0$
Also, $GM < 1$ or equivalently, $GM_{dB} < 0$
Negative phase margin and negative gain margin imply that the closed-loop unity feedback system is unstable. Therefore,

$$\boxed{\text{Unstable}}$$

Q7.92.1 NAT 1992 5M Ans: * ● ● ●

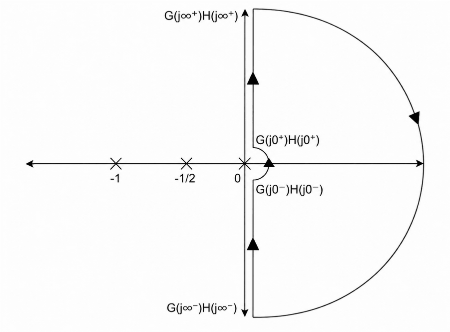
Given open loop transfer function,

$$G(s)H(s) = \frac{1}{s(2s+1)(s+1)}$$

Nyquist contour poles are at: $s = 0,\ s = -\frac{1}{2},\ s = -1$

Hence, the Nyquist contour contains an indentation around the pole at origin.

Segment	Mapping of $G(j\omega)H(j\omega)$
Segment–I	$G(j0^+)H(j0^+) \rightarrow G(j\infty^+)H(j\infty^+)$
Segment–II	$G(j\infty^+)H(j\infty^+) \rightarrow G(j\infty^-)H(j\infty^-)$
Segment–III	$G(j\infty^-)H(j\infty^-) \rightarrow G(j0^-)H(j0^-)$
Segment–IV	$G(j0^-)H(j0^-) \rightarrow G(j0^+)H(j0^+)$



Segment – I :

Put $s = j\omega$ in the given open loop transfer function,

$$G(j\omega)H(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)}$$

$$G(j\omega)H(j\omega) = \frac{1}{\omega\sqrt{1+4\omega^2}\sqrt{1+\omega^2}} \angle\left(-90^\circ - \tan^{-1}(2\omega) - \tan^{-1}(\omega)\right)$$

$$G(j0^+)H(j0^+) = \infty\angle -90^\circ$$

$$G(j\infty)H(j\infty) = 0\angle -270^\circ$$

Again,

$$G(j\omega)H(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)} = \frac{1}{j\omega(1+j3\omega-2\omega^2)} = \frac{1}{-3\omega^2 + j\omega(1-2\omega^2)}$$

Multiplying numerator and denominator by conjugate,

$$= \frac{-3\omega^2 - j\omega(1-2\omega^2)}{9\omega^4 + \omega^2(1-2\omega^2)^2}$$

$$= \left\{\frac{-3\omega^2}{9\omega^4 + \omega^2(1-2\omega^2)^2}\right\} - j\left\{\frac{\omega(1-2\omega^2)}{9\omega^4 + \omega^2(1-2\omega^2)^2}\right\}$$

Real part is negative for all $\omega > 0$.

Imaginary part changes sign when

$$1 - 2\omega^2 = 0$$

$$\omega = \frac{1}{\sqrt{2}}$$

Therefore, the Nyquist plot crosses the real axis at $\omega = \frac{1}{\sqrt{2}}$

Substituting $\omega = \frac{1}{\sqrt{2}}$,

$$G\left(j\frac{1}{\sqrt{2}}\right)H\left(j\frac{1}{\sqrt{2}}\right) = -\frac{2}{3}$$

Therefore, the Nyquist plot cuts the real axis at $\left(-\frac{2}{3},\ 0\right)$

For

$$\omega < \frac{1}{\sqrt{2}}$$

$$1 - 2\omega^2 > 0$$

Hence imaginary part is negative.
Therefore, the Nyquist plot lies in Quadrant–III.
For

$$\omega > \frac{1}{\sqrt{2}}$$

$$1 - 2\omega^2 < 0$$

Hence imaginary part is positive.
Therefore, the Nyquist plot lies in Quadrant–II.

Segment – II :
Put $s = Re^{j\theta}$ where, $R \to \infty$ and $\theta : 90° \to -90°$ in clockwise direction.

$$G(Re^{j\theta})H(Re^{j\theta}) = \lim_{R\to\infty} \frac{1}{Re^{j\theta}(2Re^{j\theta}+1)(Re^{j\theta}+1)} \approx \lim_{R\to\infty} \frac{1}{2R^3 e^{j3\theta}} = 0\, e^{-j3\theta}$$

Hence the mapping is from $-270° \to 270°$ with approximately zero radius in anticlockwise direction.
Segment – III :
Segment–III is the mirror image of Segment–I.
Segment – IV :
Put $s = re^{j\theta}$ where, $r \to 0$ and $\theta : -90° \to 90°$ in anticlockwise direction.

$$G(re^{j\theta})H(re^{j\theta}) = \lim_{r\to 0} \frac{1}{re^{j\theta}(2re^{j\theta}+1)(re^{j\theta}+1)} \approx \lim_{r\to 0} \frac{1}{re^{j\theta}} = \infty\, e^{-j\theta}$$

Hence the mapping is from $90° \to -90°$ with infinite radius in clockwise direction.
Therefore the complete Nyquist plot is,

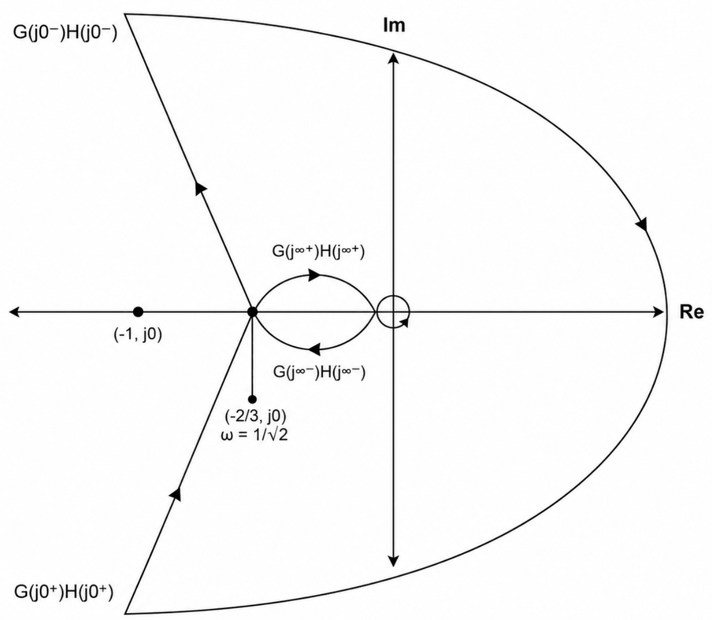


Fig. Solution Q7.92.1

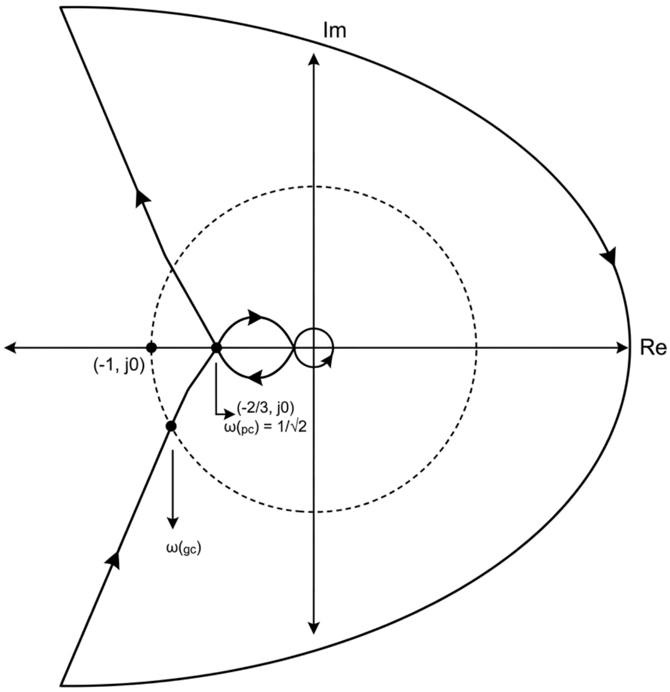
Since the Nyquist plot does not encircle the critical point $(-1, 0)$, $N = 0$
And, the right half poles of open loop transfer function are none. Hence, $P = 0$
Therefore,

$$N = P - Z$$

$$0 = 0 - Z$$

$$Z = 0$$

Therefore the closed loop system is stable.



From the Nyquist plot, phase crossover occurs at $\omega_{pc} = \dfrac{1}{\sqrt{2}}$

At phase crossover frequency,

$$G(j\omega_{pc})H(j\omega_{pc}) = -\frac{2}{3}$$

$$|G(j\omega_{pc})H(j\omega_{pc})| = \frac{2}{3}$$

Hence, Gain Margin is:

$$GM = \frac{1}{|G(j\omega_{pc})H(j\omega_{pc})|} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$$

$$GM_{dB} = 20\log\left(\frac{3}{2}\right) = 3.52\,\text{dB}$$

Gain crossover frequency is obtained from

$$|G(j\omega)H(j\omega)| = 1$$

$$\omega_{gc} \approx 0.57\,\text{rad/sec}$$

At gain crossover frequency,

$$\angle G(j\omega_{gc})H(j\omega_{gc}) = -90^\circ - \tan^{-1}(2\times 0.57) - \tan^{-1}(0.57) = -168.4^\circ$$

Hence, Phase Margin is

$$PM = 180^\circ + \angle G(j\omega_{gc})H(j\omega_{gc}) = 180^\circ - 168.4^\circ = 11.6^\circ$$

$$\boxed{GM = \frac{3}{2} \text{ or } 3.52 \text{ dB} \quad , \quad PM = 11.6^\circ}$$

DEBUG INFO

Chapter VII

Nyquist Plot

Total Questions : 20

MCQ : 17

MSQ : 0

NAT : 3

SOLUTIONS

VIII

Bode Plot

Topics

1. Bode Gain and Phase Plots
2. Minimum and Non-Minimum Phase Systems
3. Bode Plots for Repeated Poles and Zeros
4. Recovery of Transfer Function from Bode Plot
5. Error in Asymptotic Bode Gain Plot
6. Bode Plots of 2nd order systems

Q8.24.1 MCQ | 2024 | 1M | Ans: A | ● ○ ○

It is known that 20 dB/decade = 6 dB/octave. Let it be eq 1.
Given slope is 40 dB/decade

$$\frac{40}{20} = 2$$

Multiplying eq 1 with factor 2,

$$2 \times 20 \text{ dB/decade} = 2 \times 6 \text{ dB/octave}$$

$$40 \text{ dB/decade} = 12 \text{ dB/octave}$$

$$\boxed{\text{(A) 12 dB/octave}}$$

Q8.23.1 NAT | 2023 | 2M | Ans: 4 to 4 | ● ○ ○

Given,

$$G(s) = \frac{k(s+z)^a}{s^b(s+p)^c}$$

A pole contributes a slope of −20 dB/decade and a zero contributes a slope of +20 dB/decade.
From the asymptotic Bode magnitude plot, the initial slope is −20 dB/decade. This indicates the presence of one pole at the origin. Hence, $b = 1$
After the frequency ω_1, the slope becomes 0 dB/decade. Thus, the slope increases by +20 dB/decade, which indicates the presence of one zero at ω_1. Hence, $z = \omega_1$ and $a = 1$
After the frequency ω_2, the slope becomes −40 dB/decade. Since one pole contributes −20 dB/decade, a slope of −40 dB/decade indicates the presence of two poles at ω_2. Hence, $p = \omega_2$ and $c = 2$
Therefore, $a + b + c = 1 + 1 + 2 = 4$

$$\boxed{4}$$

Key Points

- A pole contributes a slope of −20 dB/decade in the asymptotic Bode magnitude plot.
- A zero contributes a slope of +20 dB/decade in the asymptotic Bode magnitude plot.
- Initial slope indicates the number of poles present at the origin.
- Change in slope at a corner frequency indicates the presence of poles or zeros at that frequency.

Q8.19.1 MCQ | 2019 | 1M | Ans: B | ● ○ ○

From the given asymptotic Bode magnitude plot:

Frequency	Slope	Change in Slope	Inference
Initially	0 dB/decade	–	No pole or zero at origin
$f = 10^1$ Hz	$0 \rightarrow -20$	−20 dB/decade	One pole
$f = 10^2$ Hz	$-20 \rightarrow -60$	−40 dB/decade	Two poles
$f = 10^3$ Hz	$-60 \rightarrow -40$	+20 dB/decade	One zero
$f = 10^4$ Hz	$-40 \rightarrow 0$	+40 dB/decade	Two zeros
$f = 10^5$ Hz	$0 \rightarrow -40$	−40 dB/decade	Two poles
$f = 10^6$ Hz	$-40 \rightarrow -60$	−20 dB/decade	One pole

Total number of poles: $N_p = 1 + 2 + 2 + 1 = 6$
Total number of zeros: $N_z = 1 + 2 = 3$

$$\boxed{\text{(B) } N_p = 6,\ N_z = 3}$$

Q8.18.1 NAT | 2018 | 2M | Ans: 0.1 to 0.1 | ● ● ○

It is given that the closed-loop system is stable for $k < k_0$
Therefore, the maximum value of k_0 corresponds to the condition of marginal stability.
For a marginally stable system, Gain Margin = 0 dB
From the given Bode plot, at the phase crossover frequency where $\angle G(j\omega) = -180°$ the magnitude is 20 dB
When gain k is introduced, the magnitude plot shifts vertically by $20\log_{10} k$
For marginal stability, the magnitude at phase crossover frequency must become 0 dB. Hence,

$$20 + 20\log_{10} k_0 = 0$$

$$20\log_{10} k_0 = -20$$

$$k_0 = 0.1$$

Therefore, the maximum value of k_0 is

$$\boxed{0.1}$$

Q8.17.1 MCQ | 2017 | 1M | Ans: A | ● ○ ○

A zero contributes a slope of +20 dB/decade and a pole contributes a slope of −20 dB/decade.
Therefore, the net high-frequency slope is $20(p - q)$ dB/decade
Given high-frequency slope is −60 dB/decade
Hence,

$$20(p - q) = -60$$

$$p - q = -3$$

$$q - p = 3$$

Checking the options:

Option	p	q	$q - p$
A	0	3	3
B	1	7	6
C	2	3	1
D	3	5	2

Only Option A satisfies.

$$\boxed{\text{(A) p = 0 and q = 3}}$$

Q8.16.1 NAT | 2016 | 2M | Ans: 0.95 to 1.05 | ● ○ ○

Given,

$$G(s) = \frac{k}{(s+0.1)(s+10)(s+p_1)}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{k}{(j\omega+0.1)(j\omega+10)(j\omega+p_1)}$$

Magnitude of the transfer function is

$$|G(j\omega)| = \frac{k}{\sqrt{\omega^2+0.1^2}\sqrt{\omega^2+10^2}\sqrt{\omega^2+p_1^2}}$$

Phase of the transfer function is

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{\omega}{0.1}\right) - \tan^{-1}\left(\frac{\omega}{10}\right) - \tan^{-1}\left(\frac{\omega}{p_1}\right)$$

From the given phase plot,

$$\angle G(j1) = -135°$$

Substituting $\omega = 1$,

$$-\tan^{-1}(10) - \tan^{-1}(0.1) - \tan^{-1}\left(\frac{1}{p_1}\right) = -135°$$

Since,

$$\tan^{-1}(10) + \tan^{-1}(0.1) = 90°$$

So,

$$-90° - \tan^{-1}\left(\frac{1}{p_1}\right) = -135°$$

$$\tan^{-1}\left(\frac{1}{p_1}\right) = 45°$$

$$\frac{1}{p_1} = 1$$

$$p_1 = 1$$

$$\boxed{1}$$

Q8.15.1 NAT | 2015 | 1M | Ans: 8970 to 8970 | ● ○ ○

From the given Bode magnitude plot:

Frequency Range	Slope	Magnitude Change	Inference
$f_L \to 300$ Hz	+40 dB/decade	0 dB → 40 dB	Used to find f_L
900 Hz $\to f_H$	−40 dB/decade	40 dB → 0 dB	Used to find f_H

Using,

$$\text{Slope} = \frac{\text{Change in magnitude}}{\log_{10}\left(\frac{f_2}{f_1}\right)}$$

For the frequency range $f_L \rightarrow 300$ Hz

$$40 = \frac{40}{\log_{10}\left(\frac{300}{f_L}\right)}$$

$$\log_{10}\left(\frac{300}{f_L}\right) = 1$$

$$\frac{300}{f_L} = 10$$

$$f_L = 30 \text{ Hz}$$

Similarly, for the frequency range 900 Hz $\rightarrow f_H$

$$-40 = \frac{-40}{\log_{10}\left(\frac{f_H}{900}\right)}$$

$$\log_{10}\left(\frac{f_H}{900}\right) = 1$$

$$\frac{f_H}{900} = 10$$

$$f_H = 9000 \text{ Hz}$$

Therefore,

$$f_H - f_L = 9000 - 30 = 8970$$

$$\boxed{8970}$$

Q8.15.2 | NAT | 2015 | 2M | Ans: 0.10 to 0.13 | ● ○ ○

For a unit step input,

$$R(s) = \frac{1}{s}$$

The steady state value is obtained using Final Value Theorem:

$$c(\infty) = \lim_{s\to 0} sC(s)$$

$$= \lim_{s\to 0} s \times R(s) \times G(s)$$

$$= \lim_{s\to 0} s \times \frac{1}{s} \times G(s)$$

$$= \lim_{s\to 0} G(s)$$

The above expression is nothing but the steady state value which is equal to the DC gain.
From the frequency response table, at very low frequency,

$$|G(j\omega)| = -18.5 \text{ dB}$$

Converting dB into magnitude,

$$20\log_{10}|G(j\omega)| = -18.5$$

$$|G(j\omega)| = 10^{-18.5/20}$$

$$|G(j\omega)| \approx 0.12$$

Therefore, the unit step response approaches the steady state value

$$\boxed{0.12}$$

Q8.14.1 MCQ | 2014 | 1M | Ans: A | ● ○ ○

For an all-pole system, each pole contributes −20 dB/decade to the high-frequency slope of the Bode magnitude plot. Given system is a fourth-order all-pole system.
Hence, total high-frequency slope is

$$4 \times (-20) = -80 \text{ dB/decade}$$

$$\boxed{\text{(A)} \ -80 \text{ dB/decade}}$$

Q8.14.2 NAT | 2014 | 2M | Ans: 0.49 to 0.51 | ● ○ ○

From the given Bode asymptotic magnitude plot,

Frequency Range	Slope Calculation
$0.1 \rightarrow 2$	$\dfrac{26.02 - 0}{\log_{10} 0.1 - \log_{10} 2} = -20\,\text{dB/dec}$
$2 \rightarrow 10$	$\dfrac{0 - 0}{\log_{10} 2 - \log_{10} 10} = 0\,\text{dB/dec}$
$10 \rightarrow 20$	$\dfrac{0 - (-6.02)}{\log_{10} 10 - \log_{10} 20} = -20\,\text{dB/dec}$

From the above table, the initial slope is −20 dB/dec, which indicates one pole at the origin.

$$\therefore \text{ System Type} = 1$$

For a Type-1 system, the steady state error corresponding to a unit ramp input is finite.

$$e_{ss} = \frac{A}{K_v}$$

For finite steady state error, the corresponding error constant is equal to the DC gain.

$$e_{ss} = \frac{A}{K_v} = \frac{1}{K}$$

Method 1

From the slope variations,

- one pole at the origin
- one zero at $\omega = 2\,\text{rad/sec}$
- one pole at $\omega = 10\,\text{rad/sec}$

Hence, the transfer function can be written as,

$$T(s) = \frac{K\left(1 + \frac{s}{2}\right)}{s\left(1 + \frac{s}{10}\right)}$$

For $\omega < 2\,\text{rad/sec}$, only the pole at the origin is effective.

$$T(s) \approx \frac{K(1)}{s(1)} = \frac{K}{s}$$

Substituting $s = j\omega$,

$$T(j\omega) = \frac{K}{j\omega} = \frac{-jK}{\omega}$$

Therefore, magnitude is,

$$|T(j\omega)| = \frac{K}{\omega}$$

In dB,

$$20log(|T(j\omega)|) = 20log(\frac{K}{\omega})$$

At $\omega = 1$ rad/sec, the magnitude from the plot is 6.02 dB. Hence,

$$6.02 = 20\log\left(\frac{K}{1}\right)$$

$$K \simeq 2$$

$$e_{ss} = \frac{1}{K} = \frac{1}{2} = 0.5$$

Method 2

A system having initial slope $-20n$ dB/dec and starts at 0.1 rad/sec, cuts the ω-axis at

$$\omega = K^{1/n}$$

Here, $n = 1$. Hence,

$$\omega = K$$

From the plot, $\omega = 2$

$$\therefore K = 2$$

Therefore,

Method 3

For systems having one pole at origin and if the Bode plot starts from 0.1 rad/sec, initial magnitude is,

$$20 + 20\log K$$

From the figure,

$$20 + 20\log K = 26.02$$

$$20\log K = 6.02$$

$$\therefore K = 2$$

Method 4

For bode plots starting at 0.1 rad/sec,

$$20\log K = |G(j1)|_{\text{dB}}$$

From the plot,

$$20\log K = 6.02$$

$$\therefore K = 2$$

$$e_{ss} = \frac{1}{K} = \frac{1}{2} = 0.5$$

$$\boxed{0.5}$$

Key Points

- If the Bode magnitude plot starts from 0.1 rad/sec and the system has n poles at the origin, then the initial magnitude is given by
$$20n + 20\log K$$
- A slope of $-20n$ dB/dec indicates n poles at the origin.
- For systems having poles only at the origin,
$$G(s) = \frac{K}{s^n}$$

Mistakes to be avoided

- Do not blindly use Method 2, 3 and 4 for every Bode plot. This is valid only when the system has poles at the origin and the plot starts from 0.1 rad/sec.
- In Method 1, While using asymptotic Bode approximation in a frequency range $\omega_1 < \omega < \omega_2$, terms

corresponding to corner frequencies below ω_2 must be approximated as $\frac{s}{\omega_c}$.

- Terms corresponding to corner frequencies above ω_2 must be approximated as 1.
- For example, if

$$(1+\frac{s}{2})(1+\frac{s}{5})$$

is present and $2 < \omega < 5$, then

$$(1+\frac{s}{2})(1+\frac{s}{5}) \approx \left(\frac{s}{2}\right)(1)$$

- Do not mix exact frequency response expressions with asymptotic Bode approximations.

Q8.14.3 NAT | 2014 | 2M | Ans: 42 to 48 | ● ● ○

Given,

$$G(s) = \frac{10}{(s+0.1)(s+1)(s+10)}$$

Writing in time constant form,

$$G(s) = \frac{10}{0.1 \times 1 \times 10} \times \frac{1}{\left(1+\frac{s}{0.1}\right)\left(1+\frac{s}{1}\right)\left(1+\frac{s}{10}\right)}$$

$$\therefore G(s) = \frac{10}{\left(1+\frac{s}{0.1}\right)\left(1+\frac{s}{1}\right)\left(1+\frac{s}{10}\right)}$$

Corner frequencies are, $\omega = 0.1,\ 1,\ 10\,\text{rad/sec}$
Initial magnitude is, $20\log 10 = 20\,\text{dB}$
Hence, the asymptotic magnitude plot starts at 20 dB.
Slopes are,

Frequency Range	Slope
$\omega < 0.1$	0 dB/dec
$0.1 < \omega < 1$	−20 dB/dec
$1 < \omega < 10$	−40 dB/dec
$\omega > 10$	−60 dB/dec

Hence, the asymptotic Bode magnitude plot will be:

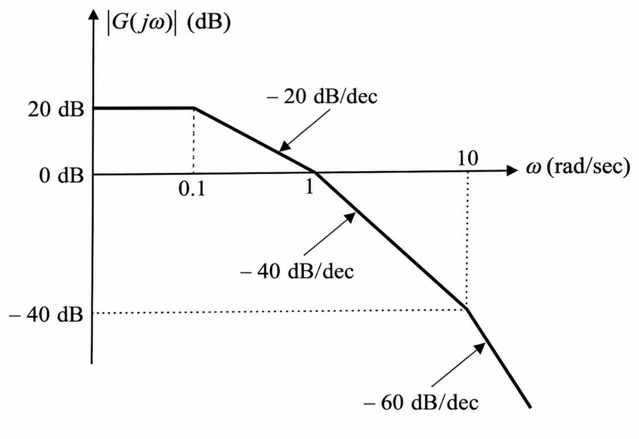


Fig. Solution Q8.14.3

From the asymptotic Bode magnitude plot, gain crossover frequency occurs at $\omega_{gc} = 1$ rad/sec
Phase contribution at $\omega = 1$ rad/sec,

$$\angle G(j\omega) = -\tan^{-1}\left(\frac{1}{0.1}\right) - \tan^{-1}(1) - \tan^{-1}\left(\frac{1}{10}\right) = -84.3^\circ - 45^\circ - 5.7^\circ = -135^\circ$$

Therefore,

$$\text{Phase Margin} = 180^\circ + \angle G(j\omega_{gc}) = 180^\circ - 135^\circ = 45^\circ$$

$$\boxed{45^\circ}$$

Q8.13.1 MCQ 2013 1M Ans: B ● ○ ○

From the given Bode magnitude plot,

$$\text{Slope} = \frac{-8 - 32}{\log_{10} 10 - \log_{10} 1} = -40\,\text{dB/dec}$$

A slope of -40 dB/dec indicates two poles at the origin. Also, the phase is negative for all ω. Hence, the system contains only poles. Therefore, the transfer function can be written as,

$$G(s) = \frac{K}{s^2}$$

For $\omega < 10$ rad/sec, only the two poles at the origin are effective.

$$G(s) \approx \frac{K}{s^2}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{K}{(j\omega)^2}$$

$$G(j\omega) = \frac{-K}{\omega^2}$$

Therefore, magnitude is,

$$|G(j\omega)| = \frac{K}{\omega^2}$$

In dB,

$$20\log(|G(j\omega)|) = 20\log\left(\frac{K}{\omega^2}\right)$$

At $\omega = 1$ rad/sec, the magnitude from the plot is 32 dB. Hence,

$$32 = 20\log\left(\frac{K}{1^2}\right)$$

$$20\log K = 32$$

$$K = 10^{1.6} = 39.8$$

$$\therefore G(s) = \frac{39.8}{s^2}$$

$$\boxed{(\text{B})\ \frac{39.8}{s^2}}$$

Q8.10.1 MCQ | 2010 | 1M | Ans: A | ● ○ ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope Calculation
$0.001 \rightarrow 0.1$	$\frac{0-0}{\log_{10} 0.001 - \log_{10} 0.1} = 0$ dB/dec
$0.1 \rightarrow 10$	$\frac{40-0}{\log_{10} 10 - \log_{10} 0.1} = +20$ dB/dec
$10 \rightarrow 1000$	$\frac{40-40}{\log_{10} 10 - \log_{10} 1000} = 0$ dB/dec

A slope increase of $+20$ dB/dec at $\omega = 0.1$ rad/sec indicates one zero.
A slope decrease of -20 dB/dec at $\omega = 10$ rad/sec indicates one pole.
Therefore, transfer function can be written as,

$$G(s) = K\frac{\left(1 + \frac{s}{0.1}\right)}{\left(1 + \frac{s}{10}\right)}$$

For $\omega < 0.1$ rad/sec, neither zero nor pole is activated.

$$\therefore G(j\omega) \approx \frac{K(1)}{(1)}$$

At $\omega = 0.001$ rad/sec, magnitude from the plot is 0 dB.

$$20\log|G(j0.001)| = 0$$

$$20\log K = 0$$

$$K = 1$$

Hence,

$$G(s) = (1)\frac{\left(1+\frac{s}{0.1}\right)}{\left(1+\frac{s}{10}\right)}$$

$$G(s) = \frac{1+10s}{1+0.1s}$$

$$\boxed{\text{(A) } \frac{10s+1}{0.1s+1}}$$

Q8.07.1 MCQ 2007 2M Ans: D ● ○ ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope
$0.1 \to 1$	-20 dB/dec
$1 \to 20$	-40 dB/dec
$\omega > 20$	-60 dB/dec

Initial slope of -20 dB/dec indicates one pole at the origin.
At $\omega = 1$ rad/sec, slope decreases by another -20 dB/dec. Hence, one more pole is present at $\omega = 1$ rad/sec.
At $\omega = 20$ rad/sec, slope again decreases by -20 dB/dec. Hence, another pole is present at $\omega = 20$ rad/sec.
Therefore, transfer function can be written as,

$$G(s) = \frac{K}{s\left(1+\frac{s}{1}\right)\left(1+\frac{s}{20}\right)}$$

For $1 < \omega < 20$ rad/sec, pole at origin and pole at 1 rad/sec are effective.

$$G(s) \approx \frac{K}{s(\frac{s}{1})(1)}$$

Substituting $s = j\omega$,

$$G(j\omega) \approx \frac{K}{(j\omega)(j\omega)}$$

Magnitude is,

$$|G(j\omega)| = \frac{K}{\omega^2}$$

At $\omega = 1$ rad/sec, magnitude from the plot is 40 dB.

$$20\log|G(j1)| = 40$$

$$20\log\left(\frac{K}{(1)^2}\right) = 40$$

$$K = 100$$

Hence,

$$G(s) = \frac{100}{s(1+s)\left(1+\frac{s}{20}\right)} = \frac{100}{s(1+s)\,(1+0.05s)}$$

$$\boxed{\text{(D)}\ \frac{100}{s(s+1)(1+0.05s)}}$$

Q8.04.1 MCQ | 2004 | 2M | Ans: C | ● ○ ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope Calculation
$\omega < 1$	0 dB/dec
$1 \rightarrow 10$	$\frac{0-(-20)}{\log_{10} 10 - \log_{10} 1} = +20$ dB/dec
$10 \rightarrow 100$	$\frac{0-0}{\log_{10} 100 - \log_{10} 10} = 0$ dB/dec
$\omega > 100$	-20 dB/dec

At $\omega = 1$ rad/sec, slope increases by +20 dB/dec. Hence, one zero is present at $\omega = 1$ rad/sec.
At $\omega = 10$ rad/sec, slope decreases by 20 dB/dec. Hence, one pole is present at $\omega = 10$ rad/sec.
At $\omega = 100$ rad/sec, slope again decreases by 20 dB/dec. Hence, another pole is present at $\omega = 100$ rad/sec.
Therefore, transfer function can be written as,

$$H(s) = K\frac{(1+\frac{s}{1})}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{100}\right)}$$

For $\omega < 1$ rad/sec, no zero or pole is effective.

$$H(j\omega) \approx K\frac{(1)}{(1)\,(1)}$$

From the plot, low frequency magnitude is -20 dB.

$$20\log K = -20$$

$$\log K = -1$$

$$K = 0.1$$

Hence,

$$H(s) = 0.1\frac{(1+\frac{s}{1})}{\left(1+\frac{s}{10}\right)\left(1+\frac{s}{100}\right)} = \frac{100(s+1)}{(s+10)(s+100)}$$

$$\boxed{\text{(C)}\ \frac{10^2(s+1)}{(s+10)(s+100)}}$$

Q8.03.1 MCQ 2003 2M Ans: A ● ● ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope Calculation
$\omega < 0.1$	$0\,\text{dB/dec}$
$0.1 \rightarrow 10$	$\frac{140-20}{\log_{10}10-\log_{10}0.1} = +60\,\text{dB/dec}$
$10 \rightarrow 100$	$\frac{160-140}{\log_{10}100-\log_{10}10} = +20\,\text{dB/dec}$
$\omega > 100$	$0\,\text{dB/dec}$

At $\omega = 0.1$ rad/sec, slope increases by +60 dB/dec. Hence, three zeros are present at $\omega = 0.1$ rad/sec.
At $\omega = 10$ rad/sec, slope decreases by 40 dB/dec. Hence, two poles are present at $\omega = 10$ rad/sec.
At $\omega = 100$ rad/sec, slope decreases by another 20 dB/dec. Hence, one pole is present at $\omega = 100$ rad/sec.
Therefore, transfer function can be written as,

$$G(s) = K\frac{\left(1+\frac{s}{0.1}\right)^3}{\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{100}\right)}$$

For $\omega < 0.1$ rad/sec, no pole or zero is effective.

$$G(j\omega) \approx K\frac{(1)^2}{(1)^2(1)}$$

From the plot, low frequency magnitude is 20 dB.

$$20\log K = 20$$

$$\log K = 1$$

$$K = 10$$

Hence,

$$G(s) = 10\frac{\left(1+\frac{s}{0.1}\right)^3}{\left(1+\frac{s}{10}\right)^2\left(1+\frac{s}{100}\right)} = 10^8\frac{(s+0.1)^3}{(s+10)^2(s+100)}$$

$$\boxed{\text{(A)}\ 10^8\frac{(s+0.1)^3}{(s+10)^2(s+100)}}$$

Q8.99.1 NAT | 1999 | 5M | Ans: * | ● ● ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope Calculation
$\omega < \omega_1$	-40 dB/dec
$\omega_1 \to \omega_2$	-20 dB/dec
$\omega > \omega_2$	-40 dB/dec

At $\omega = \omega_1$, slope changes from

$$-40\,\text{dB/dec} \to -20\,\text{dB/dec}$$

Therefore, slope increases by $+20$ dB/dec.
Hence, one zero is present at $\omega = \omega_1$.

At $\omega = \omega_2$, slope changes from

$$-20\,\text{dB/dec} \to -40\,\text{dB/dec}$$

Therefore, slope decreases by 20 dB/dec.
Hence, one pole is present at $\omega = \omega_2$.

Since the initial slope is -40 dB/dec, two poles are present at the origin.

Therefore, transfer function can be written as,

$$G(s)H(s) = K\frac{\left(1+\frac{s}{\omega_1}\right)}{s^2\left(1+\frac{s}{\omega_2}\right)}$$

From the Bode plot,

$$|G(j\omega)H(j\omega)|_{dB} = 0\,\text{dB} \quad \text{at} \quad \omega = 1\,\text{rad/sec}$$
$$|G(j\omega)H(j\omega)|_{dB} = 20\,\text{dB} \quad \text{at} \quad \omega = \omega_1\,\text{rad/sec}$$
$$|G(j\omega)H(j\omega)|_{dB} = -20\,\text{dB} \quad \text{at} \quad \omega = \omega_2\,\text{rad/sec}$$

From the plot,
For ω_1:

$$\frac{20-0}{\log_{10}\omega_1 - \log_{10}1} = -20\,\text{dB/dec}$$

$$\omega_1 = 0.1\,\text{rad/sec}$$

For ω_2:

$$\frac{0-(-20)}{\log_{10}1 - \log_{10}\omega_2} = -20\,\text{dB/dec}$$

$$\omega_2 = 10 \text{ rad/sec}$$

Therefore,

$$G(s)H(s) = K\frac{\left(1+\frac{s}{0.1}\right)}{s^2\left(1+\frac{s}{10}\right)}$$

For $\omega < 10$ rad/sec,

$$G(s)H(s) = K\frac{\left(\frac{s}{0.1}\right)}{(s^2)\,(1)}$$

Put s=jω,

$$G(j\omega))H(j\omega) = K\frac{\left(\frac{j\omega}{0.1}\right)}{((j\omega)^2)\,(1)} = \frac{-j10K}{\omega}$$

Magnitude,

$$|G(j\omega)H(j\omega)| = \frac{100K^2}{\omega^2}$$

In dB,

$$20\log\left(|G(j\omega)H(j\omega)|\right) = 20\log\left(\frac{100K^2}{\omega^2}\right)$$

At $\omega = 1$ rad/sec, $20\log\left(|G(j1)H(j1)|\right) = 0$ dB

$$20\log\left(\frac{100K^2}{(1)^2}\right) = 0$$

$$\log\left(100K^2\right) = 0$$

$$100K^2 = 1$$

$$K^2 = \frac{1}{100}$$

$$K = \frac{1}{10}$$

$$K = 0.1$$

Hence,

$$G(s)H(s) = 0.1\frac{\left(1+\frac{s}{0.1}\right)}{s^2\left(1+\frac{s}{10}\right)} = \frac{10(s+0.1)}{s^2(s+10)}$$

$$\boxed{G(s)H(s) = \frac{10(s+0.1)}{s^2(s+10)}}$$

Q8.98.1 MCQ | 1998 | 1M | Ans: C | ● ○ ○

For a unity feedback system,

$$\text{Phase Margin} = 180^\circ + \angle G(j\omega_{gc})$$

Given,

$$\angle G(j\omega_{gc}) = -125^\circ$$

Therefore,

$$\text{PM} = 180^\circ - 125^\circ$$

$$\text{PM} = 55^\circ$$

$$\boxed{\text{(C) } 55^\circ}$$

Q8.95.1 MCQ | 1995 | 1M | Ans: A | ● ○ ○

A non-minimum phase transfer function has at least one pole and/or one zero in the right half of the s-plane.
Thus, both right half plane poles and right half plane zeros indicate non-minimum phase behaviour.
Hence, both the given statements are technically correct.
However, in objective type examinations, non-minimum phase systems are generally identified using right half plane zeros.

$$\boxed{\text{(A) which has zeros in the right half } s\text{-plane.}}$$

Q8.92.1 MCQ | 1992 | 2M | Ans: A | ● ○ ○

From the given asymptotic Bode magnitude plot,

Frequency Range	Slope Calculation
$\omega < 1$	0 dB/dec
$\omega > 1$	$\dfrac{0 - 20}{\log_{10} 10 - \log_{10} 1} = -20$ dB/dec

A slope decrease of -20 dB/dec at $\omega = 1$ rad/sec indicates one pole at $\omega = 1$ rad/sec.
Therefore, transfer function can be written as,

$$G(s) = \frac{K}{1 + \frac{s}{1}}$$

For $\omega < 1$ rad/sec, the pole is not effective.

$$G(j\omega) \approx \frac{K}{(1)}$$

From the plot, low frequency magnitude is 20 dB.

$$20 \log K = 20$$

$$\log K = 1$$

$$K = 10$$

Hence,

$$G(s) = \frac{10}{1+s}$$

$$\boxed{\text{(A)}\ \frac{10}{1+s}}$$

Q8.87.1 MCQ 1987 2M Ans: B ● ○ ○

Each pole contributes -20 dB/dec to the high frequency slope.
Each zero contributes $+20$ dB/dec to the high frequency slope.
Given,

$$\text{Number of poles} = 14$$

$$\text{Number of zeros} = 2$$

Therefore, net high frequency slope is,

$$(2 - 14) \times 20 = -12 \times 20 = -240\,\text{dB/dec}$$

$$\boxed{\text{(B)}\ -240\ \text{dB/dec}}$$

DEBUG INFO

Chapter VIII

Bode Plot

Total Questions : 21

MCQ : 13

MSQ : 0

NAT : 8

SOLUTIONS

IX

State Space Analysis

Topics

1. State Variables and State Diagram
2. State Space Representation
3. Eigen Values and Eigen Vectors in State Space
4. Solution of State Space Equations
5. State Transition Matrix and Its Properties
6. Recovery of Transfer Function from State Space
7. Similarity Transforms
8. Controllable and Observable Canonical Forms
9. Parallel and Cascade Decomposition
10. Controllability — Kalman's and Gilbert's Test
11. Observability — Kalman's and Gilbert's Test
12. Observable and Detectable Systems

Q9.26.1 MCQ 2026 2M Ans: C ● ○ ○

Given state space model is,

$$\dot{x} = Ax + Bu$$

where,

$$A = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

and output equation is,

$$y = Cx$$

where,

$$C = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix}$$

Since there is no direct transmission term, we have

$$D = 0$$

Transfer function of the system is given by

$$G(s) = \frac{Y(s)}{U(s)} = C(sI - A)^{-1}B + D$$

First, compute $sI - A$,

$$sI - A = \begin{bmatrix} s+4 & 1.5 \\ -4 & s \end{bmatrix}$$

Determinant is,

$$|sI - A| = \begin{vmatrix} s+4 & 1.5 \\ -4 & s \end{vmatrix}$$

$$= (s+4)(s) - (-4)(1.5)$$

$$= s^2 + 4s + 6$$

Hence,

$$(sI - A)^{-1} = \frac{1}{s^2 + 4s + 6}\begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix}$$

Now multiply $(sI - A)^{-1}$ with B,

$$(sI - A)^{-1}B = \frac{1}{s^2 + 4s + 6}\begin{bmatrix} s & -1.5 \\ 4 & s+4 \end{bmatrix}\begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$= \frac{1}{s^2 + 4s + 6}\begin{bmatrix} 2s \\ 8 \end{bmatrix}$$

Multiplying with matrix C,

$$G(s) = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \frac{1}{s^2 + 4s + 6}\begin{bmatrix} 2s \\ 8 \end{bmatrix}$$

$$= \frac{1}{s^2+4s+6}\,[(1.5)(2s)+(0.625)(8)] = \frac{1}{s^2+4s+6}\,[3s+5] = \frac{3s+5}{s^2+4s+6}$$

$$\boxed{\text{(C) } \frac{3s+5}{s^2+4s+6}}$$

Key Points

- Transfer function from state space model is obtained using

$$G(s) = C(sI - A)^{-1}B + D$$

Q9.25.1 MSQ | 2025 | 2M | Ans: B | ● ● ○ | CLICK FOR VIDEO SOLUTION

Given state equations are,

$$\dot{x}_1(t) = 2x_1(t) + u(t)$$

$$\dot{x}_2(t) = -2x_2(t) + u(t)$$

$$\dot{x}_3(t) = u(t)$$

We will check each option individually using bounded inputs.
Consider a bounded unit step input,

$$u(t) = u^*(t)$$

where $u^*(t)$ denotes the unit step signal.
Laplace transform of unit step signal is,

$$U(s) = \frac{1}{s}$$

Taking Laplace transform of first state equation with zero initial conditions,

$$sX_1(s) = 2X_1(s) + U(s)$$

$$X_1(s) = \frac{U(s)}{s-2}$$

$$X_1(s) = \frac{1}{s(s-2)}$$

Using partial fractions,

$$X_1(s) = -\frac{1}{2s} + \frac{1}{2(s-2)}$$

Taking inverse Laplace transform,

$$x_1(t) = -\frac{1}{2} + \frac{1}{2}e^{2t}$$

Since $e^{2t} \to \infty$ as $t \to \infty$,

$$x_1(t) \text{ is unbounded}$$

Therefore, statement A is false because all states are not bounded for bounded input.
Now examine statement B.
We already found that for bounded input $u(t) = u^*(t)$,

$$x_1(t) = -\frac{1}{2} + \frac{1}{2}e^{2t}$$

is unbounded.
Hence, there exists a bounded input for which at least one state becomes unbounded. Hence it is true.
Now check statement C.
Taking Laplace transform of second state equation,

$$sX_2(s) = -2X_2(s) + U(s)$$

$$X_2(s) = \frac{U(s)}{s+2}$$

$$X_2(s) = \frac{1}{s(s+2)}$$

Using partial fractions,

$$X_2(s) = \frac{1}{2s} - \frac{1}{2(s+2)}$$

Taking inverse Laplace transform,

$$x_2(t) = \frac{1}{2} - \frac{1}{2}e^{-2t}$$

Since $e^{-2t} \to 0$,

$$x_2(t) \text{ is bounded}$$

Now consider third state equation,

$$\dot{x}_3(t) = u(t)$$

$$sX_3(s) = U(s)$$

$$X_3(s) = \frac{1}{s^2}$$

Taking inverse Laplace transform,

$$x_3(t) = t$$

Hence,

$$x_3(t) \text{ is unbounded}$$

Since $x_2(t)$ remains bounded, all three states cannot become unbounded simultaneously.
Therefore, statement C is false.
Statement D says all states are unbounded for all bounded inputs.
This is false because $x_2(t)$ remains bounded for bounded inputs.

$$\boxed{\text{(B) There exists a bounded input such that at least one of the signals } x_1(t),\ x_2(t),\ \text{and } x_3(t) \text{ is unbounded}}$$

Q9.24.1 MSQ 2024 2M Ans: A, C ● ● ○

Given system S is,

$$\dot{x} = Ax + Br$$

$$y = Cx$$

where,

$$A = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -5 \end{bmatrix}$$

Transfer function is given by,

$$G(s) = C(sI - A)^{-1}B$$

First compute $sI - A$,

$$sI - A = \begin{bmatrix} s & 2 \\ -1 & s+3 \end{bmatrix}$$

Determinant is,

$$|sI - A| = s(s+3) - 2(-1)$$

$$= s^2 + 3s + 2$$

$$= (s+1)(s+2)$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix}$$

Now multiply with B,

$$(sI - A)^{-1}B = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & -2 \\ 1 & s \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 \\ 1 \end{bmatrix}$$

Multiplying with C,

$$G(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 2 & -5 \end{bmatrix} \begin{bmatrix} s+3 \\ 1 \end{bmatrix}$$

$$= \frac{2(s+3) - 5}{s^2 + 3s + 2}$$

$$= \frac{2s + 1}{s^2 + 3s + 2}$$

Now we check each option individually.

Option A

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

Compute transfer function.

$$sI - A = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

Determinant is,

$$s(s+3) - (-1)(2) = s^2 + 3s + 2$$

$$(sI - A)^{-1} = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix}$$

$$(sI - A)^{-1} B = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 \\ s \end{bmatrix}$$

$$G_A(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ s \end{bmatrix}$$

$$= \frac{2s + 1}{s^2 + 3s + 2}$$

Hence, option A has same transfer function.

Option B

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad C = \begin{bmatrix} 0 & 2 \end{bmatrix}$$

Using previously calculated inverse matrix,

$$(sI - A)^{-1} B = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} s+3 \\ -2 \end{bmatrix}$$

$$G_B(s) = \frac{1}{s^2 + 3s + 2} \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} s+3 \\ -2 \end{bmatrix}$$

$$= \frac{-4}{s^2 + 3s + 2}$$

This is not equal to original transfer function.
Hence, option B is false.

Option C

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

Since A is diagonal,

$$sI - A = \begin{bmatrix} s+1 & 0 \\ 0 & s+2 \end{bmatrix}$$

$$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

$$(sI - A)^{-1}B = \begin{bmatrix} \frac{-1}{s+1} \\ \frac{3}{s+2} \end{bmatrix}$$

$$G_C(s) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{-1}{s+1} \\ \frac{3}{s+2} \end{bmatrix}$$

$$= \frac{-1}{s+1} + \frac{3}{s+2}$$

$$= \frac{-(s+2) + 3(s+1)}{(s+1)(s+2)}$$

$$= \frac{2s+1}{s^2 + 3s + 2}$$

Hence, option C has same transfer function.

Option D

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$(sI - A)^{-1}B = \begin{bmatrix} \frac{1}{s+1} \\ \frac{1}{s+2} \end{bmatrix}$$

$$G_D(s) = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{s+1} \\ \frac{1}{s+2} \end{bmatrix}$$

$$= \frac{1}{s+1} + \frac{2}{s+2}$$

$$= \frac{(s+2) + 2(s+1)}{(s+1)(s+2)}$$

$$= \frac{3s+4}{s^2 + 3s + 2}$$

This is not equal to original transfer function.
Hence, option D is false.

$$\boxed{\text{(A) } \frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r, \ y = \begin{bmatrix} 1 & 2 \end{bmatrix} x}$$

$$\boxed{\text{(C) } \frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} -1 \\ 3 \end{bmatrix} r, \ y = \begin{bmatrix} 1 & 1 \end{bmatrix} x}$$

Q9.23.1 MCQ 2023 2M Ans: A ● ○ ○

Given state equation is,

$$\dot{x}(t) = Ax(t)$$

with initial condition $x(0)$.
Since there is no input term, the system is a homogeneous state space system.
Let the solution be of the form,

$$x(t) = e^{\lambda t} v$$

where,

λ is an eigenvalue of matrix A and

v is the corresponding eigenvector.

Differentiating $x(t)$,

$$\dot{x}(t) = \lambda e^{\lambda t} v$$

Substituting into the state equation,

$$\lambda e^{\lambda t} v = A e^{\lambda t} v$$

$$e^{\lambda t} \lambda v = e^{\lambda t} A v$$

Cancelling the common term $e^{\lambda t}$,

$$Av = \lambda v$$

which is the eigenvalue equation.
Therefore, for every eigenvalue-eigenvector pair, one solution of the system is,

$$x_i(t) = e^{\lambda_i t} v_i$$

Since the system is second order and eigenvalues are distinct, there exist two independent solutions,

$$x_1(t) = e^{\lambda_1 t} v_1$$

$$x_2(t) = e^{\lambda_2 t} v_2$$

By principle of superposition, the complete solution is obtained as a linear combination of these independent solutions.

$$x(t) = \alpha_1 e^{\lambda_1 t} v_1 + \alpha_2 e^{\lambda_2 t} v_2$$

where,

α_1 and α_2 are constants determined using initial conditions.

Writing the above expression in summation form,

$$x(t) = \sum_{i=1}^{2} \alpha_i e^{\lambda_i t} v_i$$

$$\text{(A)}\ \sum_{i=1}^{2} \alpha_i e^{\lambda_i t} v_i$$

Key Points

For distinct eigenvalues, each eigenvalue-eigenvector pair contributes one exponential mode to the natural response of the system.

Q9.20.1 MCQ 2020 2M Ans: A

Choose the state variables as,

$$v \quad \text{and} \quad i$$

Inputs are,

$$i_1 \quad \text{and} \quad i_2$$

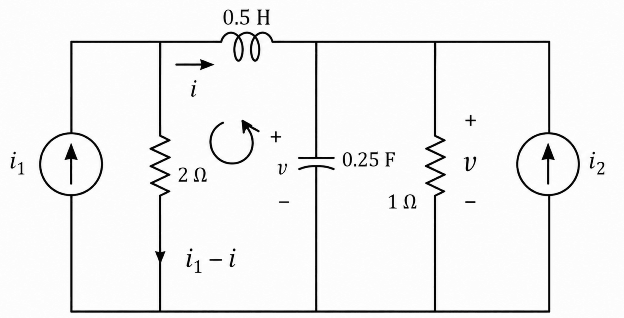


Fig. Solution Q9.20.1

Applying KVL in the left loop,

$$2(i_1 - i) - v - 0.5\frac{di}{dt} = 0$$

$$2i_1 - 2i - v - 0.5\frac{di}{dt} = 0$$

$$0.5\frac{di}{dt} = 2i_1 - 2i - v$$

$$\frac{di}{dt} = 4i_1 - 4i - 2v$$

$$\frac{di}{dt} = -2v - 4i + 4i_1$$

This is equation (1).
Now apply KCL at the capacitor node.

$$i + i_2 = v + 0.25\frac{dv}{dt}$$

$$0.25\frac{dv}{dt} = i + i_2 - v$$

$$\frac{dv}{dt} = 4i + 4i_2 - 4v$$

$$\frac{dv}{dt} = -4v + 4i + 4i_2$$

This is equation (2).
Writing equations (1) and (2) in matrix form,

$$\frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

$$\boxed{\text{(A) } \frac{d}{dt}\begin{bmatrix} v \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} v \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}}$$

Key Points

- For electrical circuits involving inductors and capacitors, choose inductor current and capacitor voltage as state variables. Apply KVL across the inductor loop and KCL at the capacitor node to directly obtain the state equations.

Q9.19.1 MCQ 2019 2M Ans: A ● ● ○

Given transfer function is,

$$H(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^3 + 3s^2 + 2s + 1}$$

Rearranging,

$$(s^3 + 3s^2 + 2s + 1)Y(s) = U(s)$$

Taking inverse Laplace transform,

$$\frac{d^3y(t)}{dt^3} + 3\frac{d^2y(t)}{dt^2} + 2\frac{dy(t)}{dt} + y(t) = u(t)$$

Choose state variables as,

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{x}_1(t) = \frac{dy(t)}{dt}$$

$$x_3(t) = \dot{x}_2(t) = \frac{d^2y(t)}{dt^2}$$

Therefore,

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = x_3(t)$$

Also,

$$\dot{x}_3(t) = \frac{d^3y(t)}{dt^3}$$

From the differential equation,

$$\frac{d^3y(t)}{dt^3} = -y(t) - 2\frac{dy(t)}{dt} - 3\frac{d^2y(t)}{dt^2} + u(t)$$

Substituting state variables,

$$\dot{x}_3(t) = -x_1(t) - 2x_2(t) - 3x_3(t) + u(t)$$

Writing all state equations together,

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = x_3(t)$$

$$\dot{x}_3(t) = -x_1(t) - 2x_2(t) - 3x_3(t) + u(t)$$

Writing in matrix form,

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t)$$

Hence,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix}$$

Comparing with the given options, only options A and C have this matrix A.
Output equation is,

$$y(t) = x_1(t)$$

Therefore,

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

Hence,

$$C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{(A) } A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \text{ and } C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}$$

Key Points

- For transfer function to state space conversion, first write the differential equation in time domain.
- State variables are usually chosen as output and its successive derivatives.

- In controllable canonical form, the last row of matrix A contains the coefficients of the characteristic equation with negative sign.

Q9.17.1 NAT 2017 1M Ans: 4.99 to 5.01 ● ○ ○

Given state space model is,

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -9 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 45 \end{bmatrix}u(t)$$

with initial condition,

$$\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

where $u(t)$ is the unit step input.

Method 1

The given state equations are,

$$\dot{x}_1(t) = 0$$

$$\dot{x}_2(t) = -9x_2(t) + 45u(t)$$

Taking Laplace transform with zero initial conditions,
For first equation,

$$sX_1(s) = 0$$

$$X_1(s) = 0$$

Therefore,

$$x_1(t) = 0$$

For second equation,

$$sX_2(s) = -9X_2(s) + \frac{45}{s}$$

$$(s+9)X_2(s) = \frac{45}{s}$$

$$X_2(s) = \frac{45}{s(s+9)}$$

Using Final Value Theorem,

$$\lim_{t\to\infty} x_2(t) = \lim_{s\to 0} sX_2(s)$$

$$= \lim_{s\to 0} \frac{45}{s+9}$$

$$= 5$$

Therefore,

$$\lim_{t\to\infty} \sqrt{x_1^2(t) + x_2^2(t)} = \sqrt{0^2 + 5^2} = 5$$

Method 2

Using state transition approach,

$$X(s) = [sI - A]^{-1} BU(s)$$

First compute,

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s+9 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{1}{s} & 0 \\ 0 & \frac{1}{s+9} \end{bmatrix}$$

Since,

$$B = \begin{bmatrix} 0 \\ 45 \end{bmatrix}$$

$$U(s) = \frac{1}{s}$$

we get,

$$X(s) = \begin{bmatrix} \frac{1}{s} & 0 \\ 0 & \frac{1}{s+9} \end{bmatrix} \begin{bmatrix} 0 \\ 45 \end{bmatrix} \frac{1}{s}$$

$$X(s) = \begin{bmatrix} 0 \\ \frac{45}{s(s+9)} \end{bmatrix}$$

Therefore,

$$x_1(t) = 0$$

$$x_2(t) = 5 - 5e^{-9t}$$

Hence,

$$\lim_{t\to\infty} \sqrt{x_1^2(t) + x_2^2(t)} = \sqrt{0^2 + 5^2} = 5$$

$$\boxed{5}$$

Q9.17.2 MCQ 2017 2M Ans: D ● ○ ○

Given state equations are,

$$\frac{d}{dt}x_1(t) - x_2(t) = 0$$

$$\frac{d}{dt}x_2(t) + 2x_1(t) + 3x_2(t) = r(t)$$

Rearranging,

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -2x_1(t) - 3x_2(t) + r(t)$$

Output equation is,

$$c(t) = x_1(t)$$

Writing in matrix form,

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r(t)$$

Therefore,

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Characteristic equation is obtained from,

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = 0$$

$$s(s+3) + 2 = 0$$

$$s^2 + 3s + 2 = 0$$

Method 1

Factorizing,

$$(s+1)(s+2) = 0$$

Therefore, poles are,

$$s = -1, \ -2$$

Since poles are real, distinct, and negative, the system is overdamped.

Method 2

Comparing characteristic equation with standard second-order form,

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Comparing coefficients,

$$\omega_n^2 = 2$$

$$\omega_n = \sqrt{2}$$

And,

$$2\zeta\omega_n = 3$$

$$\zeta = \frac{3}{2\sqrt{2}}$$

$$\zeta \approx 1.06$$

Since, $\zeta > 1$ the system is overdamped.

$\boxed{\text{(D) Overdamped}}$

Q9.16.1 MCQ | 2016 | 2M | Ans: D | ● ● ○

Given state equations are,

$$\frac{d}{dt}x_1(t) + 2x_1(t) = 3u(t)$$

$$\frac{d}{dt}x_2(t) + 2x_2(t) = u(t)$$

Rearranging,

$$\dot{x}_1(t) = -2x_1(t) + 3u(t)$$

$$\dot{x}_2(t) = -2x_2(t) + u(t)$$

Output equation is,

$$c(t) = x_1(t)$$

Writing in matrix form,

$$\frac{d}{dt}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u(t)$$

$$c(t) = \begin{bmatrix} 1 & 0 \end{bmatrix}\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -6 \\ -2 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 3 & -6 \\ 1 & -2 \end{bmatrix}$$

Determinant is,

$$|Q_c| = (3)(-2) - (-6)(1) = -6 + 6 = 0$$

Since determinant is zero, rank of controllability matrix is less than 2.
Hence, the system is uncontrollable.
Observability matrix is,

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix}$$

First compute CA,

$$CA = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \end{bmatrix}$$

Therefore,

$$Q_o = \begin{bmatrix} 1 & 0 \\ -2 & 0 \end{bmatrix}$$

Determinant is,

$$|Q_o| = 0$$

Hence, rank of observability matrix is less than 2.
Therefore, the system is unobservable.

$$\boxed{\text{(D) Neither controllable nor observable}}$$

Key Points

- A system is controllable if the controllability matrix has full rank.
- A system is observable if the observability matrix has full rank.
- For a second-order system, determinant of controllability and observability matrices must be non-zero for complete

controllability and observability.

Q9.15.1 MCQ 2015 2M Ans: D ● ○ ○

Given state equation is,

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} x$$

with initial condition,

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Output equation is,

$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

Let,

$$x = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

Expanding the state equation,

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -x_2(t)$$

From output equation,

$$y(t) = x_2(t)$$

Taking Laplace transform of

$$\dot{x}_2(t) = -x_2(t)$$

with initial condition

$$x_2(0) = 0$$

we get,

$$sX_2(s) - x_2(0) = -X_2(s)$$

$$sX_2(s) = -X_2(s)$$

$$(s + 1)X_2(s) = 0$$

$$X_2(s) = 0$$

Taking inverse Laplace transform,

$$x_2(t) = 0$$

Since,

$$y(t) = x_2(t)$$

we get,

$$y(t) = 0$$

$$\boxed{\text{(D) 0}}$$

Q9.14.1 MCQ 2014 2M Ans: D

Given state equation is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

State transition matrix is given by,

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

First compute $sI - A$,

$$sI - A = \begin{bmatrix} s & -1 \\ 0 & s \end{bmatrix}$$

Determinant is,

$$|sI - A| = s^2$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{s^2}\begin{bmatrix} s & 1 \\ 0 & s \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{s} & \frac{1}{s^2} \\ 0 & \frac{1}{s} \end{bmatrix}$$

Taking inverse Laplace transform element-wise,

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$

Hence,

$$\phi(t) = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}$$

$$\boxed{\text{(D)} \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}}$$

Q9.14.2 MCQ 2014 2M Ans: B ● ● ○

Given,

$$\dot{x}(t) = Ax(t)$$

For

$$x_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

the response is,

$$x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

Let,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Since,

$$\dot{x}(t) = Ax(t)$$

we have,

$$\begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

Cancelling common term e^{-t},

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} a - b \\ c - d \end{bmatrix}$$

Therefore,

$$a - b = -1$$

$$c - d = 1$$

Similarly, for

$$x_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

the response is,

$$x(t) = \begin{bmatrix} e^{-t} - e^{-2t} \\ -e^{-t} + 2e^{-2t} \end{bmatrix}$$

Differentiating,

$$\dot{x}(t) = \begin{bmatrix} -e^{-t} + 2e^{-2t} \\ e^{-t} - 4e^{-2t} \end{bmatrix}$$

At $t = 0$,

$$\dot{x}(0) = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Using

$$\dot{x}(0) = Ax(0)$$

we get,

$$\begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix}$$

Therefore,

$$b = 1$$

$$d = -3$$

Substituting in previous equations,

$$a - 1 = -1$$

$$a = 0$$

and,

$$c - (-3) = 1$$

$$c = -2$$

Hence,

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Now for

$$x(0) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

the state equations become,

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -2x_1 - 3x_2$$

Taking Laplace transform of first equation,

$$sX_1(s) - x_1(0) = X_2(s)$$

$$sX_1(s) - 3 = X_2(s)$$

$$X_1(s) = \frac{3}{s} + \frac{X_2(s)}{s}$$

Taking Laplace transform of second equation,

$$sX_2(s) - x_2(0) = -2X_1(s) - 3X_2(s)$$

$$sX_2(s) - 5 = -2X_1(s) - 3X_2(s)$$

Substituting $X_1(s)$,

$$sX_2(s) - 5 = -2\left(\frac{3}{s} + \frac{X_2(s)}{s}\right) - 3X_2(s)$$

$$sX_2(s) + 3X_2(s) + \frac{2X_2(s)}{s} = 5 - \frac{6}{s}$$

$$X_2(s)\left(s^2 + 3s + 2\right) = 5s - 6$$

$$X_2(s) = \frac{5s - 6}{(s+1)(s+2)}$$

Using partial fractions,

$$X_2(s) = \frac{-11}{s+1} + \frac{16}{s+2}$$

Taking inverse Laplace transform,

$$x_2(t) = -11e^{-t} + 16e^{-2t}$$

From

$$X_1(s) = \frac{3}{s} + \frac{X_2(s)}{s}$$

$$X_1(s) = \frac{3}{s} + \frac{1}{s}\left(\frac{5s - 6}{(s+1)(s+2)}\right)$$

$$X_1(s) = \frac{3(s+1)(s+2) + (5s - 6)}{s(s+1)(s+2)}$$

$$= \frac{3s^2 + 14s}{s(s+1)(s+2)}$$

$$= \frac{3s + 14}{(s+1)(s+2)}$$

Using partial fractions,

$$X_1(s) = \frac{11}{s+1} - \frac{8}{s+2}$$

Taking inverse Laplace transform,

$$x_1(t) = 11e^{-t} - 8e^{-2t}$$

Therefore,

$$x(t) = \begin{bmatrix} 11e^{-t} - 8e^{-2t} \\ -11e^{-t} + 16e^{-2t} \end{bmatrix}$$

$$\boxed{(\text{B}) \begin{bmatrix} 11e^{-t} - 8e^{-2t} \\ -11e^{-t} + 16e^{-2t} \end{bmatrix}}$$

Q9.14.3 MCQ 2014 2M Ans: C ● ○ ○

Given state equation is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

with initial conditions, $x_1(0) = 1$ and $x_2(0) = -1$

Method 1

Expanding the state equations,

$$\dot{x}_1 = -x_1$$

$$\dot{x}_2 = -2x_2$$

Taking Laplace transform of first equation,

$$sX_1(s) - x_1(0) = -X_1(s)$$

$$sX_1(s) - 1 = -X_1(s)$$

$$(s+1)X_1(s) = 1$$

$$X_1(s) = \frac{1}{s+1}$$

Taking inverse Laplace transform,

$$x_1(t) = e^{-t}$$

Taking Laplace transform of second equation,

$$sX_2(s) - x_2(0) = -2X_2(s)$$

$$sX_2(s) + 1 = -2X_2(s)$$

$$(s+2)X_2(s) = -1$$

$$X_2(s) = \frac{-1}{s+2}$$

Taking inverse Laplace transform,

$$x_2(t) = -e^{-2t}$$

$$x_1(t) = e^{-t}, \quad x_2(t) = -e^{-2t}$$

Method 2

Using state transition approach,

$$X(s) = [sI - A]^{-1}x(0)$$

First compute,

$$sI - A = \begin{bmatrix} s+1 & 0 \\ 0 & s+2 \end{bmatrix}$$

$$[sI - A]^{-1} = \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

Therefore,

$$X(s) = \begin{bmatrix} \frac{1}{s+1} & 0 \\ 0 & \frac{1}{s+2} \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$X(s) = \begin{bmatrix} \frac{1}{s+1} \\ \frac{-1}{s+2} \end{bmatrix}$$

Taking inverse Laplace transform,

$$x(t) = \begin{bmatrix} e^{-t} \\ -e^{-2t} \end{bmatrix}$$

$$\boxed{(C)\ x_1(t) = e^{-t}, \quad x_2(t) = -e^{-2t}}$$

Q9.14.4 MCQ 2014 2M Ans: A ● ● ○

In the given signal flow diagram, assume the $\dot{x}_3$, $\dot{x}_2$, and $\dot{x}_1$ at:

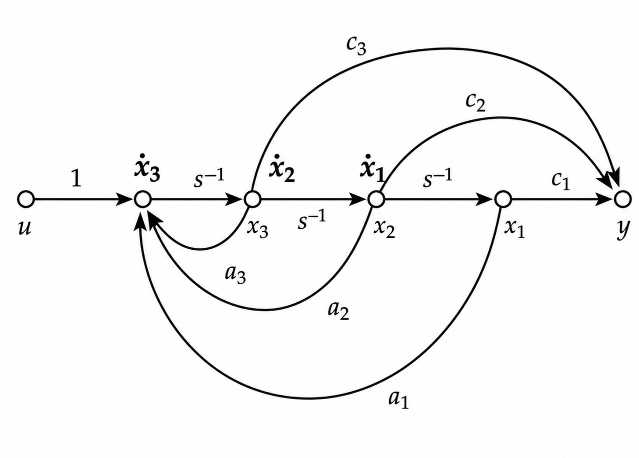


Fig. Solution Q9.14.4

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = a_1x_1 + a_2x_2 + a_3x_3 + u$$

Output equation is,

$$y = c_1x_1 + c_2x_2 + c_3x_3$$

Writing in matrix form,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ a_3 \end{bmatrix}$$

Now compute A^2B,

$$A^2B = A(AB)$$

$$= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ a_3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ a_3 \\ a_2 + a_3^2 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & a_3 \\ 1 & a_3 & a_2 + a_3^2 \end{bmatrix}$$

Determinant is,

$$|Q_c| = -1$$

$$|Q_c| \neq 0$$

Hence, controllability matrix has full rank for all values of a_1, a_2, a_3.
Therefore, the system is always controllable.
Observability matrix is,

$$Q_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Since,

$$C = \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix}$$

observability depends on values of c_1, c_2, c_3.
For example, if

$$c_1 = c_2 = c_3 = 0$$

then,

$$C = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$Q_o = 0$$

which does not have full rank.
Hence, the system is not always observable.
Stability depends on eigenvalues of matrix A.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ a_1 & a_2 & a_3 \end{bmatrix}$$

Characteristic equation is,

$$|sI - A| = s^3 - a_3 s^2 - a_2 s - a_1$$

Since a_1, a_2, a_3 can take arbitrary values, pole locations change accordingly.
For example, if $a_1 = -1, \quad a_2 = -2, \quad a_3 = -3$, the system is stable. But if $a_1 = 1, \quad a_2 = 2, \quad a_3 = 3$, the system becomes unstable.
Hence, the system is neither always stable nor always unstable.

$$\boxed{\text{(A) Always controllable}}$$

Key Points

- A system is controllable if the controllability matrix has full rank.
- Companion canonical form systems are always controllable.
- Stability depends on eigenvalues of matrix A, not on controllability.

Q9.14.5 MCQ 2014 2M Ans: B ● ● ○

Given state space model is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}\begin{bmatrix} 0 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ -4 \\ 0 \end{bmatrix}$$

Now compute A^2B,

$$A^2B = A(AB)$$

$$= \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}\begin{bmatrix} 4 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} -8 \\ 4 \\ 0 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 0 & 4 & -8 \\ 4 & -4 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

Since third row is completely zero,

$$|Q_c| = 0$$

Hence, rank of controllability matrix is less than 3.
Therefore, the system is uncontrollable.
Observability matrix is,

$$Q_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

First compute CA,

$$CA = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & -2 \end{bmatrix}$$

Now compute CA^2,

$$CA^2 = (CA)A$$

$$= \begin{bmatrix} -1 & 0 & -2 \end{bmatrix}\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 4 \end{bmatrix}$$

Therefore,

$$Q_o = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 0 & -2 \\ 1 & -1 & 4 \end{bmatrix}$$

Determinant is,

$$|Q_o| = -6$$

$$|Q_o| \neq 0$$

Hence, observability matrix has full rank.
Therefore, the system is observable.

$$\boxed{\text{(B) Uncontrollable and observable}}$$

Q9.13.1

MCQ | 2013 | 2M | Ans: A | ● ○ ○

From the given signal flow diagram, assume $\dot{\boldsymbol{x}}_1$, $\dot{\boldsymbol{x}}_2$, x_1 and x_2 as following:

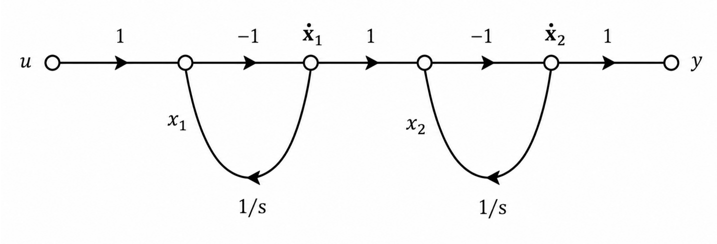


Fig. Solution Q9.13.1

From the signal flow graph,

$$\dot{x}_1 = -x_1 + u$$

$$\dot{x}_2 = x_1 - x_2 + u$$

$$y = x_1 - x_2 + u$$

Therefore, state equations are,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u$$

$$\boxed{\text{(A) } \dot{\mathbf{X}} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}\mathbf{X} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}u, \qquad y = \begin{bmatrix} 1 & -1 \end{bmatrix}\mathbf{X} + u}$$

Q9.13.2

MCQ | 2013 | 2M | Ans: A | ● ○ ○

From Q9.13.1, the state matrix is,

$$A = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$$

State transition matrix is given by,

$$\phi(t) = e^{At} = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

First compute $sI - A$,

$$sI - A = \begin{bmatrix} s+1 & 0 \\ -1 & s+1 \end{bmatrix}$$

Determinant is,

$$|sI - A| = (s+1)^2$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{(s+1)^2}\begin{bmatrix} s+1 & 0 \\ 1 & s+1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{s+1} & 0 \\ \frac{1}{(s+1)^2} & \frac{1}{s+1} \end{bmatrix}$$

Taking inverse Laplace transform element-wise,

$$\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = e^{-t}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\} = te^{-t}$$

Hence,

$$e^{At} = \begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}$$

$$\boxed{\text{(A)} \begin{bmatrix} e^{-t} & 0 \\ te^{-t} & e^{-t} \end{bmatrix}}$$

Q9.12.1 MCQ 2012 2M Ans: D ● ○ ○

Given state space model is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix}$$

Now compute A^2B,

$$A^2B = A(AB)$$

$$= \begin{bmatrix} 0 & a_1 & 0 \\ 0 & 0 & a_2 \\ a_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} a_1a_2 \\ 0 \\ 0 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 0 & 0 & a_1a_2 \\ 0 & a_2 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Determinant is,

$$|Q_c| = -a_1a_2^2$$

For controllability,

$$|Q_c| \neq 0$$

Hence, $a_1 \neq 0$ and $a_2 \neq 0$
Observe that a_3 does not appear in controllability determinant.
Therefore, controllability is independent of a_3.
Hence, the system is controllable for

$$a_1 \neq 0, \qquad a_2 \neq 0$$

$$\boxed{\text{(D) } a_1 \neq 0,\ a_2 \neq 0,\ a_3 = 0}$$

Q9.11.1 MCQ 2011 2M Ans: B ● ● ○

Given system is,

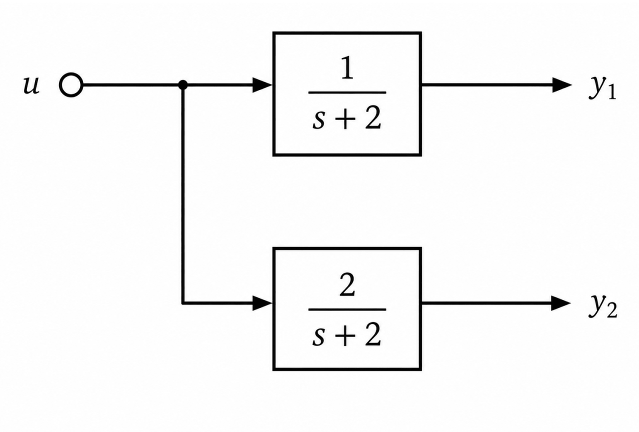


Fig. Solution Q9.11.1

The block diagram can be redrawn as,

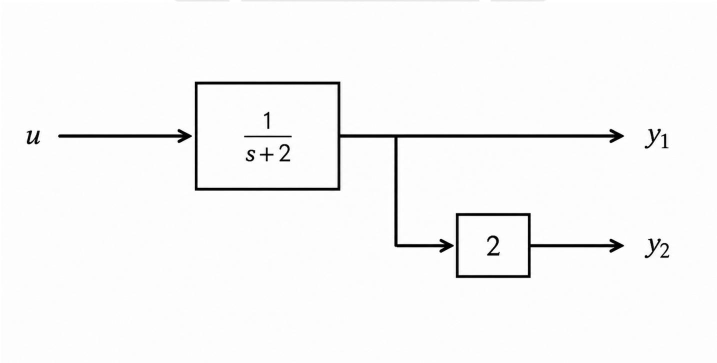


Fig. Solution Q9.11.1

From the above block diagram,

$$y_1 = \frac{1}{s+2}u$$

$$y_2 = \frac{2}{s+2}u$$

Rewrite the transfer function as,

$$\frac{1}{s+2} = \frac{\frac{1}{s}}{1+\frac{2}{s}}$$

The above transfer function can be realized using feedback form shown below.

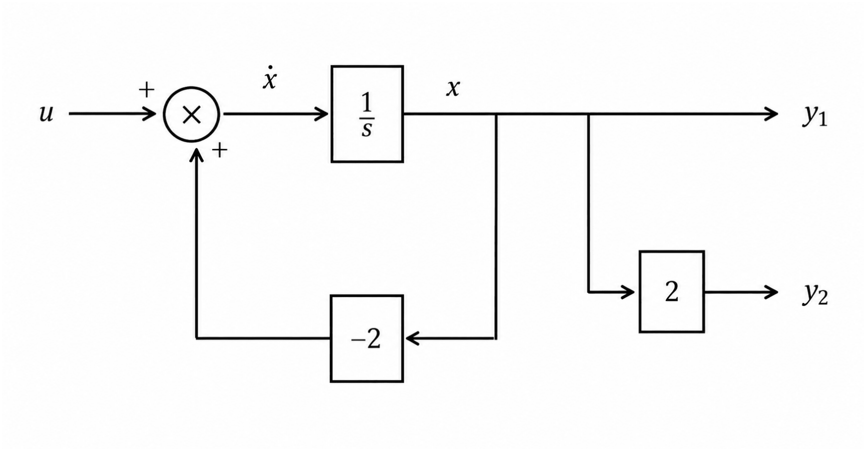


Fig. Solution Q9.11.1

Assume output of the integrator as state variable x.
From the summing junction,

$$\dot{x} = -2x + u$$

From the output equations,

$$y_1 = x$$

$$y_2 = 2x$$

Hence, state space model is,

$$\frac{d}{dt}\begin{bmatrix} x \end{bmatrix} = \begin{bmatrix} -2 \end{bmatrix}\begin{bmatrix} x \end{bmatrix} + \begin{bmatrix} 1 \end{bmatrix} u$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}\begin{bmatrix} x \end{bmatrix}$$

$$\boxed{\text{(B) } \dot{x} = [-2]x + [1]u, \qquad y = \begin{bmatrix} 1 \\ 2 \end{bmatrix} x}$$

Q9.10.1 MCQ 2010 2M Ans: B ● ● ○

From the given signal flow graph, assume the $\dot{x}_1$, $\dot{x}_2$, x_1 and x_2 as:

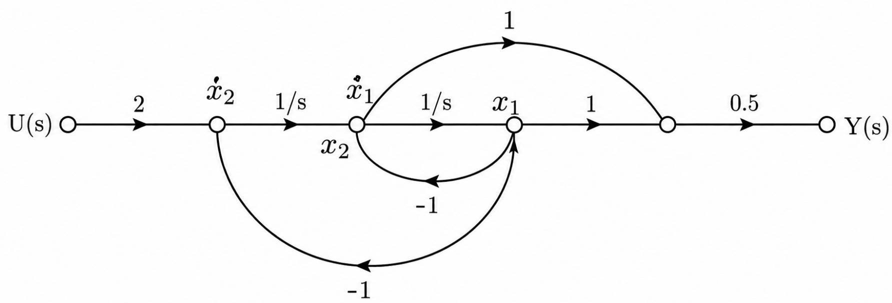


Fig. Solution Q9.10.1

From the signal flow graph,

$$\dot{x}_1 = -x_1 + x_2$$

$$\dot{x}_2 = -x_1 + 2u$$

$$y = 0.5(x_1 + x_2 - x_1) = 0.5x_2$$

Therefore, state equations are,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0.5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\boxed{\text{(B) } \dot{x} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u, \qquad y = \begin{bmatrix} 0 & 0.5 \end{bmatrix} x}$$

Q9.10.2 MCQ 2010 2M Ans: C ● ● ○

From Q9.10.1, the state space model is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & 0.5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Therefore,

$$A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 0.5 \end{bmatrix}$$

Method 1

Transfer function is,

$$G(s) = C(sI - A)^{-1}B$$

First compute,

$$sI - A = \begin{bmatrix} s+1 & -1 \\ 1 & s \end{bmatrix}$$

Determinant is,

$$|sI - A| = s(s+1) + 1 = s^2 + s + 1$$

$$(sI - A)^{-1} = \frac{1}{s^2 + s + 1} \begin{bmatrix} s & 1 \\ -1 & s+1 \end{bmatrix}$$

$$(sI - A)^{-1}B = \frac{1}{s^2 + s + 1} \begin{bmatrix} s & 1 \\ -1 & s+1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$= \frac{1}{s^2 + s + 1} \begin{bmatrix} 2 \\ 2s+2 \end{bmatrix}$$

$$G(s) = \begin{bmatrix} 0 & 0.5 \end{bmatrix} \frac{1}{s^2 + s + 1} \begin{bmatrix} 2 \\ 2s+2 \end{bmatrix}$$

$$= \frac{s+1}{s^2 + s + 1}$$

Method 2

Using Mason gain formula,

$$T = \frac{\sum P_k \Delta_k}{\Delta}$$

Forward paths are,

$$P_1 = \frac{1}{s^2} \qquad P_2 = \frac{1}{s}$$

$$\Delta_1 = 1 \qquad \Delta_2 = 1$$

Loop gains are,

$$L_1 = -\frac{1}{s}$$

$$L_2 = -\frac{1}{s^2}$$

Hence,

$$\Delta = 1 - (L_1 + L_2)$$

$$= 1 - \left(-\frac{1}{s} - \frac{1}{s^2}\right)$$

$$= 1 + \frac{1}{s} + \frac{1}{s^2}$$

$$\frac{Y(s)}{U(s)} = \frac{\frac{1}{s^2} + \frac{1}{s}}{1 + \frac{1}{s} + \frac{1}{s^2}}$$

$$= \frac{s+1}{s^2 + s + 1}$$

$$\boxed{\text{(C) } \frac{s+1}{s^2 + s + 1}}$$

Q9.09.1 | MCQ | 2009 | 2M | Ans: C | ● ○ ○

Given,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} p \\ q \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} p & p \\ q & q \end{bmatrix}$$

Determinant is,

$$|Q_c| = pq - pq = 0$$

Hence, controllability matrix never has full rank.
Therefore, the system is uncontrollable for all values of p and q.

$$\boxed{\text{(C) The system is uncontrollable for all values of } p \text{ and } q}$$

Q9.08.1 | MCQ | 2008 | 2M | Ans: C | ● ● ○

From the given signal flow graph, assume the $\dot{x}_1$, $\dot{x}_2$, and $\dot{x}_3$ as.

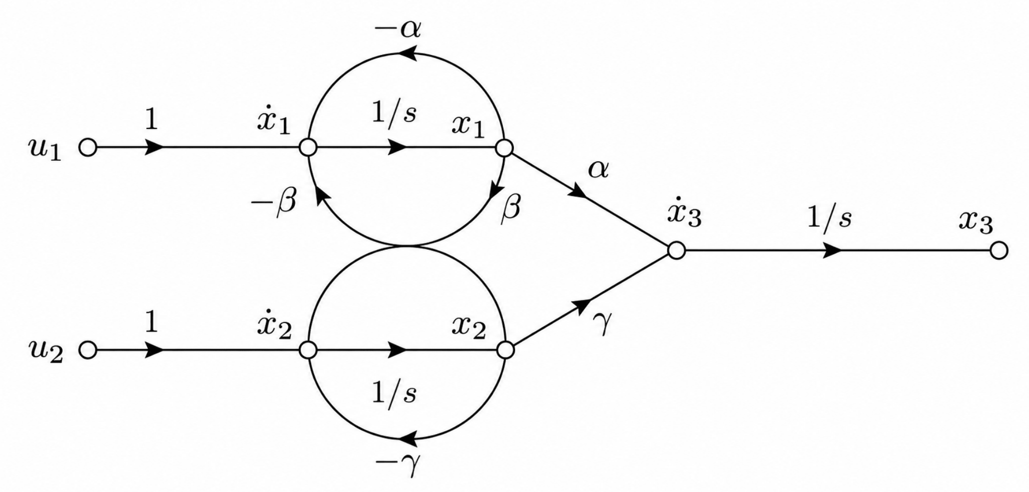


Fig. Solution Q9.08.1

From the signal flow graph,

$$\dot{x}_1 = -\alpha x_1 - \beta x_2 + u_1$$

$$\dot{x}_2 = \beta x_1 - \gamma x_2 + u_2$$

$$\dot{x}_3 = \alpha x_1 + \gamma x_2$$

Therefore, state equations are,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\alpha & -\beta & 0 \\ \beta & -\gamma & 0 \\ \alpha & \gamma & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\boxed{(C)\ \frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -\alpha & -\beta & 0 \\ \beta & -\gamma & 0 \\ \alpha & \gamma & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix}}$$

Q9.07.1 MCQ 2007 2M Ans: A ● ○ ○

Given state equations are,

$$\dot{\omega} = -\omega + i_a$$

$$\dot{i}_a = -\omega - 10 i_a + 10u$$

Taking Laplace transform of first equation with zero initial conditions,

$$s\omega(s) = -\omega(s) + I_a(s)$$

$$I_a(s) = (s+1)\omega(s)$$

Taking Laplace transform of second equation,

$$sI_a(s) = -\omega(s) - 10I_a(s) + 10U(s)$$

$$\omega(s) + (s+10)I_a(s) = 10U(s)$$

Substituting $I_a(s) = (s+1)\omega(s)$ in the above equation. We get,

$$\omega(s) + (s+10)(s+1)\omega(s) = 10U(s)$$

$$\omega(s)\left[1 + s^2 + 11s + 10\right] = 10U(s)$$

$$\omega(s)\left[s^2 + 11s + 11\right] = 10U(s)$$

Therefore,

$$\frac{\omega(s)}{U(s)} = \frac{10}{s^2 + 11s + 11}$$

$$\boxed{(A)\ \frac{10}{s^2 + 11s + 11}}$$

Q9.07.2 MCQ 2007 2M Ans: A ● ● ●

Given,

$$x(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \text{and corresponding response is,} \quad x(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$$

Also,

$$x(0) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and corresponding response is,} \quad x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

Since,

$$\dot{x}(t) = Ax(t)$$

we have,

$$\dot{x}(0) = Ax(0)$$

For the first response,

$$x(t) = \begin{bmatrix} e^{-2t} \\ -2e^{-2t} \end{bmatrix}$$

$$\dot{x}(t) = \begin{bmatrix} -2e^{-2t} \\ 4e^{-2t} \end{bmatrix}$$

At $t = 0$,

$$\dot{x}(0) = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

Let,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Then,

$$\begin{bmatrix} -2 \\ 4 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$a - 2b = -2$$

$$c - 2d = 4$$

For the second response,

$$x(t) = \begin{bmatrix} e^{-t} \\ -e^{-t} \end{bmatrix}$$

$$\dot{x}(t) = \begin{bmatrix} -e^{-t} \\ e^{-t} \end{bmatrix}$$

At $t = 0$,

$$\dot{x}(0) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$a - b = -1$$

$$c - d = 1$$

Solving,

$$a = 0, \qquad b = 1$$

$$c = -2, \qquad d = -3$$

Therefore,

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

Characteristic equation is,

$$|sI - A| = 0$$

$$\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = 0$$

$$s(s+3) + 2 = 0$$

$$s^2 + 3s + 2 = 0$$

$$(s+1)(s+2) = 0$$

Hence, eigenvalues are,

$$\lambda_1 = -1, \qquad \lambda_2 = -2$$

For $\lambda_1 = -1$,

$$(A + I)v_1 = 0$$

$$\begin{bmatrix} 1 & 1 \\ -2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + x_2 = 0$$

Taking $x_1 = 1$, then $x_2 = -1$

$$v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

For $\lambda_2 = -2$,

$$(A + 2I)v_2 = 0$$

$$\begin{bmatrix} 2 & 1 \\ -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$2x_1 + x_2 = 0$$

Taking $x_1 = 1$, then $x_2 = -2$,

$$v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Hence, eigenvalue-eigenvector pairs are,

$$\left(-1, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right)$$

and

$$\left(-2, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$$

$$\boxed{\text{(A)} \left(-1, \begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) \text{ and } \left(-2, \begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)}$$

Q9.07.3 MCQ | 2007 | 2M | Ans: D | ● ○ ○

From Q9.07.2, the system matrix obtained is,

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$\boxed{\text{(D)} \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}}$$

Q9.06.1 MCQ | 2006 | 2M | Ans: A | ● ● ○

Given,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

State transition matrix is given by,

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

First compute,

$$sI - A = \begin{bmatrix} s & -1 \\ 1 & s \end{bmatrix}$$

Determinant is,

$$|sI - A| = s^2 + 1$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{s^2 + 1} \begin{bmatrix} s & 1 \\ -1 & s \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{s}{s^2+1} & \dfrac{1}{s^2+1} \\ \dfrac{-1}{s^2+1} & \dfrac{s}{s^2+1} \end{bmatrix}$$

Taking inverse Laplace transform,

$$\phi(t) = \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}$$

$$\boxed{(A) \begin{bmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{bmatrix}}$$

Q9.05.1 | MCQ | 2005 | 2M | Ans: A or B | ● ● ○

The eigenvalues of a system are nothing but the poles of its transfer function. A system may be represented in different forms such as, Transfer Function Representation or State Space Representation but the poles of the system always remain unchanged. Here, the same linear system is represented using two different state space models,

$$\dot{X} = AX + BU$$

and

$$\dot{W} = CW + DU$$

Since both represent the same system, their poles must be identical.
Therefore, the eigenvalues are equal.

$$[\lambda] = [\mu]$$

Now, the state variables X and W need not be the same. One realization may be, Controllable Canonical Form while another may be, Observable Canonical Form or diagonal form, etc.
Hence, the state vectors are generally different.

$$X \neq W$$

(A) $[\lambda] = [\mu]$ and $X = W$
or
(B) $[\lambda] = [\mu]$ and $X \neq W$

Q9.04.1 | MCQ | 2004 | 2M | Ans: D | ● ● ○

Given state equations are,

$$\dot{x}_1 = -3x_1 - x_2 + u$$

$$\dot{x}_2 = 2x_1$$

$$y = x_1 + u$$

Comparing with standard state space form,

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

we get,

$$A = \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Controllability matrix is,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 1 & -3 \\ 0 & 2 \end{bmatrix}$$

$$|Q_c| = 2 \neq 0$$

Hence, the system is controllable.
Observability matrix is,

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix}$$

First compute CA,

$$CA = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -1 \end{bmatrix}$$

Therefore,

$$Q_o = \begin{bmatrix} 1 & 0 \\ -3 & -1 \end{bmatrix}$$

$$|Q_o| = -1 \neq 0$$

Hence, the system is observable.

$$\boxed{\text{(D) controllable and observable}}$$

Q9.04.2 MCQ 2004 2M Ans: B ● ○ ○

Given,

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

State transition matrix is defined as,

$$\phi(t) = e^{At}$$

Since,

$$A = I$$

we have,

$$e^{At} = e^{It}$$

$$= e^{t} I$$

Therefore,

$$e^{At} = e^{t} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$e^{At} = \begin{bmatrix} e^{t} & 0 \\ 0 & e^{t} \end{bmatrix}$$

$$\boxed{(\text{B}) \begin{bmatrix} e^{t} & 0 \\ 0 & e^{t} \end{bmatrix}}$$

Key Points

- Matrix exponential is defined as,

$$e^{A} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

- For identity matrix I,

$$I^2 = I, \qquad I^3 = I, \qquad \cdots$$

Hence,

$$e^{tI} = \left(1 + t + \frac{t^2}{2!} + \cdots\right) I = e^{t} I$$

Q9.03.1 MCQ 2003 2M Ans: C ● ● ○

Given state equation is,

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

with initial condition,

$$x(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Zero input response is given by,

$$X(s) = [sI - A]^{-1}x(0)$$

First compute,

$$sI - A = \begin{bmatrix} s-1 & 0 \\ -1 & s-1 \end{bmatrix}$$

Determinant is,

$$|sI - A| = (s-1)^2$$

Therefore,

$$[sI - A]^{-1} = \frac{1}{(s-1)^2}\begin{bmatrix} s-1 & 0 \\ 1 & s-1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{s-1} & 0 \\ \frac{1}{(s-1)^2} & \frac{1}{s-1} \end{bmatrix}$$

Hence,

$$X(s) = \begin{bmatrix} \frac{1}{s-1} & 0 \\ \frac{1}{(s-1)^2} & \frac{1}{s-1} \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$X(s) = \begin{bmatrix} \frac{1}{s-1} \\ \frac{1}{(s-1)^2} \end{bmatrix}$$

Taking inverse Laplace transform,

$$x(t) = \begin{bmatrix} e^t \\ te^t \end{bmatrix}$$

$$\boxed{\text{(C)} \begin{bmatrix} e^t \\ te^t \end{bmatrix}}$$

Q9.02.1 MCQ | 2002 | 2M | Ans: D | ● ○ ○

Given state equation is,

$$\dot{x}(t) = -2x(t) + 2u(t)$$

and output equation is,

$$y(t) = 0.5x(t)$$

Taking Laplace transform with zero initial conditions,

$$sX(s) = -2X(s) + 2U(s)$$

$$(s+2)X(s) = 2U(s)$$

$$X(s) = \frac{2}{s+2}U(s)$$

Since,

$$Y(s) = 0.5X(s)$$

substituting $X(s)$,

$$Y(s) = 0.5\left(\frac{2}{s+2}\right)U(s)$$

$$Y(s) = \frac{1}{s+2}U(s)$$

Therefore,

$$\frac{Y(s)}{U(s)} = \frac{1}{s+2}$$

$$\boxed{\text{(D) } \frac{1}{(s+2)}}$$

Q9.02.2 NAT 2002 Ans: * ● ● ○

From the given block diagram, $x_1(t)$ and $x_2(t)$ are chosen as the state variables.

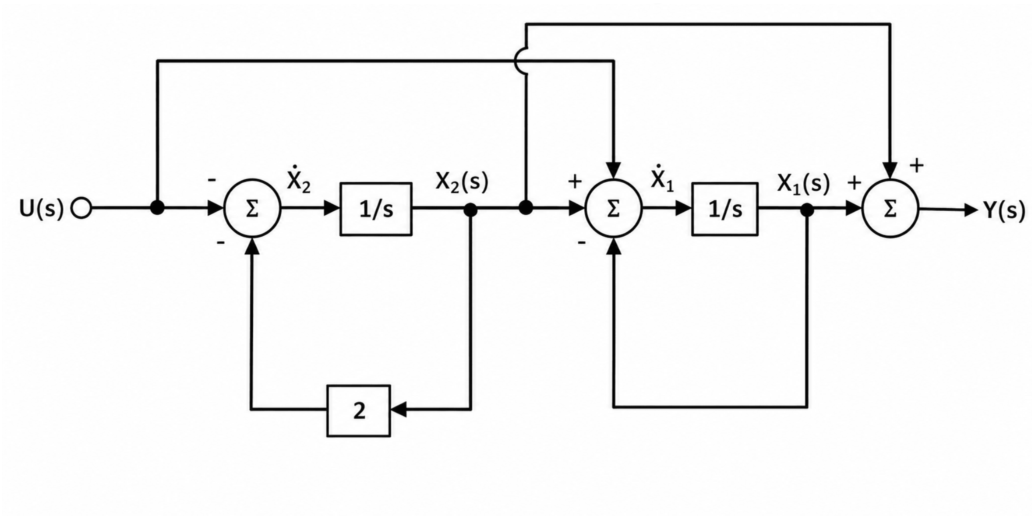


Fig. Solution Q9.02.2

From the block diagram, the input state equations are:

$$\dot{x}_1(t) = -x_1(t) + x_2(t) + u(t)$$

$$\dot{x}_2(t) = -2x_2(t) - u(t)$$

The output state equation is,

$$y(t) = x_1(t) + x_2(t)$$

Arranging the equations in matrix form,

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u(t)$$

Q9.02.3 NAT 2002 Ans: * ● ● ○

From the state model,

$$\dot{x}(t) = Ax(t) + Bu(t)$$

Comparing with the equations obtained in Q9.02.2,

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -2 \end{bmatrix}$$

State transition matrix is given by,

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

First compute,

$$sI - A = \begin{bmatrix} s+1 & -1 \\ 0 & s+2 \end{bmatrix}$$

Determinant is,

$$|sI - A| = (s+1)(s+2)$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s+1 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{1}{s+1} & \dfrac{1}{(s+1)(s+2)} \\ 0 & \dfrac{1}{s+2} \end{bmatrix}$$

Using partial fractions,

$$\frac{1}{(s+1)(s+2)} = \frac{1}{s+1} - \frac{1}{s+2}$$

Taking inverse Laplace transform element wise,

$$\phi(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$$

Q9.02.4 NAT 2002 Ans: * ● ● ○

Given initial conditions are,

$$x_1(0) = 1, \qquad x_2(0) = 1$$

Therefore,

$$x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Assuming zero input,

$$u(t) = 0$$

Hence, the state equation becomes

$$\dot{x}(t) = Ax(t)$$

The solution of the homogeneous state equation is,

$$x(t) = \phi(t)x(0)$$

From Q9.02.3,

$$\phi(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix}$$

Substituting the initial condition vector,

$$x(t) = \begin{bmatrix} e^{-t} & e^{-t} - e^{-2t} \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x(t) = \begin{bmatrix} e^{-t}(1) + \left(e^{-t} - e^{-2t}\right)(1) \\ 0(1) + e^{-2t}(1) \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t} + e^{-t} - e^{-2t} \\ e^{-2t} \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{-t} - e^{-2t} \\ e^{-2t} \end{bmatrix}$$

Therefore,

$$x_1(t) = 2e^{-t} - e^{-2t}$$

$$x_2(t) = e^{-2t}$$

From the output equation,

$$y(t) = x_1(t) + x_2(t)$$

Substituting the values of the state variables,

$$y(t) = \left(2e^{-t} - e^{-2t}\right) + e^{-2t}$$

$$y(t) = 2e^{-t}$$

$$\boxed{y(t) = 2e^{-t}}$$

Q9.01.1 MCQ 2001 2M Ans: A ● ● ○

Given state equation is,

$$\dot{x} = Ax + Bu$$

where,

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0.5 & 1 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Control law is,

$$u = \begin{bmatrix} -0.5 & -3 & -5 \end{bmatrix} x + v$$

Let,

$$K = \begin{bmatrix} -0.5 & -3 & -5 \end{bmatrix}$$

Closed loop state equation becomes,

$$\dot{x} = Ax + B(Kx + v)$$

$$\dot{x} = (A + BK)x + Bv$$

First compute BK,

$$BK = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} -0.5 & -3 & -5 \end{bmatrix}$$

$$BK = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -0.5 & -3 & -5 \end{bmatrix}$$

Therefore,

$$A_{cl} = A + BK$$

$$A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0.5 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -0.5 & -3 & -5 \end{bmatrix}$$

$$A_{cl} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix}$$

Characteristic equation is obtained from,

$$|sI - A_{cl}| = 0$$

First compute $sI - A_{cl}$,

$$sI - A_{cl} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 2 & s+3 \end{bmatrix}$$

Taking determinant,

$$\begin{vmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 0 & 2 & s+3 \end{vmatrix} = 0$$

Expanding along first column,

$$s\begin{vmatrix} s & -1 \\ 2 & s+3 \end{vmatrix} = 0$$

$$s[s(s+3)+2] = 0$$

$$s(s^2+3s+2) = 0$$

$$s(s+1)(s+2) = 0$$

$$s = 0, -1, -2$$

$$\boxed{\text{(A) } 0, -1, -2}$$

Q9.00.1 NAT 2000 Ans: * ● ● ○

Given,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}u$$

$$y = \begin{bmatrix} 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The system matrix is,

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}$$

Eigenvalues are obtained from,

$$|sI - A| = 0$$

Therefore,

$$\begin{vmatrix} s+2 & -1 \\ 0 & s+3 \end{vmatrix} = 0$$

$$(s+2)(s+3) = 0$$

$$s = -2, -3$$

Hence,

$$\boxed{\lambda_1 = -2, \qquad \lambda_2 = -3}$$

Q9.00.2 NAT 2000 Ans: * ● ● ○

Given,

$$u(t) = \delta(t)$$

$$x_1(0^+) = 0, \qquad x_2(0^+) = 0$$

State transition matrix is given by,

$$\phi(t) = \mathcal{L}^{-1}\left\{(sI - A)^{-1}\right\}$$

First compute,

$$sI - A = \begin{bmatrix} s+2 & -1 \\ 0 & s+3 \end{bmatrix}$$

Determinant is,

$$|sI - A| = (s+2)(s+3)$$

Therefore,

$$(sI - A)^{-1} = \frac{1}{(s+2)(s+3)} \begin{bmatrix} s+3 & 1 \\ 0 & s+2 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{1}{s+2} & \dfrac{1}{(s+2)(s+3)} \\ 0 & \dfrac{1}{s+3} \end{bmatrix}$$

Using partial fractions,

$$\frac{1}{(s+2)(s+3)} = \frac{1}{s+2} - \frac{1}{s+3}$$

Taking inverse Laplace transform element wise,

$$\phi(t) = \begin{bmatrix} e^{-2t} & e^{-2t} - e^{-3t} \\ 0 & e^{-3t} \end{bmatrix}$$

Since, $u(t) = \delta(t)$ and $x(0^+) = 0$, the state response is,

$$x(t) = \phi(t)B$$

Here,

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Therefore,

$$x(t) = \begin{bmatrix} e^{-2t} & e^{-2t} - e^{-3t} \\ 0 & e^{-3t} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} e^{-2t} \\ 0 \end{bmatrix}$$

Hence,

$$\boxed{x_1(t) = e^{-2t}}$$

$$\boxed{x_2(t) = 0}$$

Output equation is,

$$y(t) = x_1(t) + x_2(t)$$

Therefore,

$$y(t) = e^{-2t} + 0$$

$$\boxed{y(t) = e^{-2t}}$$

Q9.00.3 NAT 2000 Ans: * ● ● ○

Given,

$$A = \begin{bmatrix} -2 & 1 \\ 0 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

The desired output is

$$y(t) = Ae^{-2t}.$$

Since the output contains only the mode e^{-2t}, the initial state must lie along the eigenvector corresponding to the eigenvalue $\lambda = -2$.

$$(A + 2I)v = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$v_2 = 0.$$

There is no restriction on v_1, so we choose $v_1 = 1$ for simplicity.

$$v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Therefore, let

$$x(0^+) = k \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The corresponding state response is

$$x(t) = ke^{-2t}\begin{bmatrix}1\\0\end{bmatrix}.$$

The output is

$$y(t) = \begin{bmatrix}1 & 1\end{bmatrix}x(t) = \begin{bmatrix}1 & 1\end{bmatrix}ke^{-2t}\begin{bmatrix}1\\0\end{bmatrix} = ke^{-2t}.$$

Comparing with the desired output

$$y(t) = Ae^{-2t},$$

gives

$$k = A.$$

Hence,

$$x(0^+) = A\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}A\\0\end{bmatrix}.$$

Therefore, the required initial conditions are

$$\boxed{x_1(0^+) = A, \qquad x_2(0^+) = 0}$$

Q9.99.1 MCQ 1999 2M Ans: A ● ○ ○

Given,

$$A = \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

To check controllability, form the controllability matrix,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 0 & 1 \\ 1 & -3 \end{bmatrix}$$

$$|Q_c| = -1 \neq 0$$

Hence, the system is controllable.
To check observability, form the observability matrix,

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix}$$

First compute CA,

$$CA = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 \end{bmatrix}$$

Therefore,

$$Q_o = \begin{bmatrix} 1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$|Q_o| = -4 \neq 0$$

Hence, the system is observable.

$$\boxed{\text{(A) Controllable and Observable.}}$$

Q9.97.1 MCQ | 1997 | 2M | Ans: A | ● ● ○

Given output equation is,

$$y(t) = x_1(t) + x_2(t)$$

Differentiating,

$$\frac{dy}{dt} = \dot{x}_1(t) + \dot{x}_2(t)$$

From the given state equations,

$$\dot{x}_1(t) = x_1(t) - x_2(t)$$

$$\dot{x}_2(t) = x_2(t) + u(t)$$

Given,

$$x_1(0) = 1, \quad x_2(0) = -1, \quad u(0) = 0$$

Therefore,

$$\dot{x}_1(0) = x_1(0) - x_2(0) = 1 - (-1) = 2$$

And,

$$\dot{x}_2(0) = x_2(0) + u(0) = -1 + 0 = -1$$

Hence,

$$\left.\frac{dy}{dt}\right|_{t=0} = \dot{x}_1(0) + \dot{x}_2(0) = 2 + (-1) = 1$$

$$\boxed{\text{(A) 1}}$$

Q9.97.2 NAT 1997 Ans: * ● ● ○

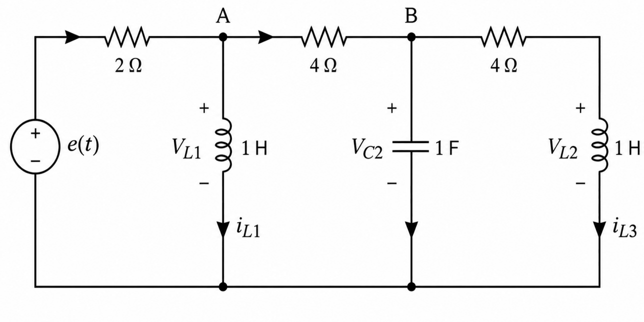


Fig. Solution Q9.97.2

Given,

$$x_1 = i_{L1}(t), \qquad x_2 = v_{C2}(t), \qquad x_3 = i_{L3}(t)$$

Input,

$$u(t) = e(t)$$

Let the voltage across the first inductor be V_{L1}.

$$V_{L1} = L\frac{di_{L1}}{dt}$$

Since $L = 1H$,

$$V_{L1} = \dot{x}_1$$

Now, KCL at node A,

$$\frac{e(t) - V_{L1}}{2} = i_{L1} + \frac{V_{L1} - V_{C2}}{4}$$

$$\frac{e(t) - \dot{x}_1}{2} = x_1 + \frac{\dot{x}_1 - x_2}{4}$$

$$2e(t) - 2\dot{x}_1 = 4x_1 + \dot{x}_1 - x_2$$

$$2e(t) + x_2 - 4x_1 = 3\dot{x}_1$$

$$\dot{x}_1 = \frac{2}{3}e(t) - \frac{4}{3}x_1 + \frac{1}{3}x_2$$

$$\dot{x}_1 = -\frac{4}{3}x_1 + \frac{1}{3}x_2 + \frac{2}{3}e(t)$$

KCL at node B,

$$\frac{V_{L1} - V_{C2}}{4} = C\frac{dV_{C2}}{dt} + i_{L3}$$

$$\frac{\dot{x}_1 - x_2}{4} = \dot{x}_2 + x_3$$

$$\dot{x}_1 - x_2 = 4\dot{x}_2 + x_3$$

$$\left\{-\frac{4}{3}x_1 + \frac{1}{3}x_2 + \frac{2}{3}e(t)\right\} - x_2 = 4\dot{x}_2 + x_3$$

$$\dot{x}_2 = -\frac{1}{3}x_1 - \frac{1}{6}x_2 - \frac{1}{4}x_3 + \frac{1}{6}e(t)$$

KVL in the right side loop,

$$V_{L2} - V_{C2} + 4i_{L3} = 0$$

$$\dot{x}_3 - x_2 + 4x_3 = 0$$

$$\dot{x}_3 = x_2 - 4x_3$$

Hence, the state equations are

$$\dot{x}_1 = -\frac{4}{3}x_1 + \frac{1}{3}x_2 + \frac{2}{3}e(t)$$
$$\dot{x}_2 = -\frac{1}{3}x_1 - \frac{1}{6}x_2 - \frac{1}{4}x_3 + \frac{1}{6}e(t)$$
$$\dot{x}_3 = x_2 - 4x_3$$

Matrix form,

$$\boxed{\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-\frac{4}{3} & \frac{1}{3} & 0\\ -\frac{1}{3} & -\frac{1}{6} & -\frac{1}{4}\\ 0 & 1 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}\frac{2}{3}\\ \frac{1}{6}\\ 0\end{bmatrix}e(t)}$$

Q9.97.3 NAT 1997 Ans: * ● ● ○

For $t \geq 0$,

$$e(t) = 0$$

Initial conditions,

$$i_{L1}(0) = 0$$

$$V_{C2}(0) = 0$$

$$i_{L3}(0) = 1A$$

Energy stored in an inductor,

$$W_L = \frac{1}{2}Li^2$$

Energy stored in a capacitor,

$$W_C = \frac{1}{2}CV^2$$

Since both inductors are $1H$ and capacitor is $1F$,

$$W_{L1}(0) = \frac{1}{2}(1)(0)^2 = 0$$

$$W_C(0) = \frac{1}{2}(1)(0)^2 = 0$$

$$W_{L3}(0) = \frac{1}{2}(1)(1)^2 = \frac{1}{2}$$

Total initial stored energy,

$$W(0) = 0 + 0 + \frac{1}{2}$$

$$W(0) = \frac{1}{2}J$$

Since $e(t) = 0$, there is no external source supplying energy.
As $t \to \infty$, all currents and voltages become zero due to dissipation in resistors.

$$W(\infty) = 0$$

Therefore, total energy dissipated in the resistors over $(0, \infty)$ is

$$W_{\text{dissipated}} = W(0) - W(\infty)$$

$$W_{\text{dissipated}} = \frac{1}{2} - 0$$

$$\boxed{W_{\text{dissipated}} = \frac{1}{2}\text{ J}}$$

Q9.92.1 MSQ | 1992 | 2M | Ans: B, C | ● ○ ○

Given,

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 \end{bmatrix}$$

To check controllability, form the controllability matrix,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

Therefore,

$$Q_c = \begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}$$

$$|Q_c| = 0$$

Hence, controllability matrix does not have full rank.
Therefore, the system is not completely controllable.
Hence,Statement B is true
Now check observability.
Observability matrix is,

$$Q_o = \begin{bmatrix} C \\ CA \end{bmatrix}$$

First compute CA,

$$CA = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} -1 & -4 \end{bmatrix}$$

Therefore,

$$Q_o = \begin{bmatrix} 1 & 2 \\ -1 & -4 \end{bmatrix}$$

$$|Q_o| = -4 + 2 = -2 \neq 0$$

Hence, observability matrix has full rank.
Therefore, the system is completely observable.
Hence, Statement C is true
Therefore, correct options are B and C.

$$\boxed{\text{(B) The system is not completely controllable}}$$

$$\boxed{\text{(C) The system is completely observable}}$$

Q9.91.1 MCQ 1991 2M Ans: B ● ● ○

Given state equations are,

$$\dot{x}_1 = -2x_1 + 4x_2$$

$$\dot{x}_2 = 2x_1 - x_2 + u$$

Writing in state-space form,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Therefore,

$$A = \begin{bmatrix} -2 & 4 \\ 2 & -1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

To check controllability,

$$Q_c = \begin{bmatrix} B & AB \end{bmatrix}$$

First compute AB,

$$AB = \begin{bmatrix} -2 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

Hence,

$$Q_c = \begin{bmatrix} 0 & 4 \\ 1 & -1 \end{bmatrix}$$

$$|Q_c| = -4 \neq 0$$

Therefore, the system is controllable.
Now check stability.

$$sI - A = \begin{bmatrix} s+2 & -4 \\ -2 & s+1 \end{bmatrix}$$

Characteristic equation is,

$$|sI - A| = 0$$

$$(s+2)(s+1) - 8 = 0$$

$$s^2 + 3s - 6 = 0$$

Since constant term is negative, one pole lies in the right half plane.
Hence, the system is unstable.
Therefore, the system is controllable but unstable.

$$\boxed{\text{(B) controllable but unstable}}$$

DEBUG INFO

Chapter IX

State Space Analysis

Total Questions : 45

MCQ : 33

MSQ : 3

NAT : 9

SOLUTIONS

X

Controllers and Compensators

Topics

1. Basics and Types of Controllers — P, PI, PD, PID
2. Lag Compensator
3. Lead Compensator
4. Lag-Lead Compensator
5. Design Problems on Controllers and Compensators

Q10.26.1 MCQ 2026 2M Ans: B ● ○ ○

Given,

$$G(s) = \frac{1}{(s+1)(s+2)}$$

The compensator is

$$C(s) = K(s+\alpha)$$

For a unity feedback system,

$$1 + C(s)G(s) = 0$$

Substituting $G(s)$ and $C(s)$,

$$1 + \frac{K(s+\alpha)}{(s+1)(s+2)} = 0$$

Therefore,

$$(s+1)(s+2) + K(s+\alpha) = 0$$

Expanding,

$$s^2 + 3s + 2 + Ks + K\alpha = 0$$

$$s^2 + (3+K)s + (2+K\alpha) = 0$$

The closed loop poles are given as

$$-3 \pm j\sqrt{5}$$

Hence the characteristic equation becomes

$$(s+3-j\sqrt{5})(s+3+j\sqrt{5}) = 0$$

Using

$$(a-b)(a+b) = a^2 - b^2$$

$$(s+3)^2 + 5 = 0$$

$$s^2 + 6s + 14 = 0$$

Comparing coefficients,

$$3 + K = 6$$

$$K = 3$$

Also,

$$2 + K\alpha = 14$$

$$K\alpha = 12$$

Substituting $K = 3$,

$$3\alpha = 12$$

$$\alpha = 4$$

$$\boxed{(B)\ 3, 4}$$

Q10.25.1 MCQ 2025 2M Ans: D ● ● ○

Given,

$$G(s) = \frac{1}{10s^2}$$

The system is connected in unity negative feedback configuration.
We analyze both statements separately.

Statement (i):
The controller is

$$M(s) = \frac{K_I}{s}$$

Therefore, the open loop transfer function becomes

$$G(s)M(s) = \frac{K_I}{10s^3}$$

For unity feedback,

$$1 + G(s)M(s) = 0$$

$$1 + \frac{K_I}{10s^3} = 0$$

$$10s^3 + K_I = 0$$

Since, the terms s^2, s^1 are absent, that implies that the system is unstable for any value of K_I.
Hence, statement (i) is TRUE.

Statement (ii):
The controller is

$$M(s) = K_P + sK_D$$

Therefore,

$$G(s)M(s) = \frac{K_P + sK_D}{10s^2}$$

Using the characteristic equation,

$$1 + \frac{K_p + sK_D}{10s^2} = 0$$

$$10s^2 + sK_D + K_p = 0$$

For a second order system, $as^2 + bs + c$ is stable if $a > 0$, $b > 0$ and $c > 0$.
Since $K_p > 0$ and $K_D > 0$,
Therefore, the closed loop system is stable.
Hence, statement (ii) is TRUE.

$$\boxed{\text{(D) Both (i) and (ii) are TRUE}}$$

Key Points

- In the characteristic equation, if any s term is missing, then the system is unstable
- A second-order system $as^2 + as + b$ is stable if all of it's coefficients are positive.

Q10.21.1 NAT 2021 2M Ans: 3.125 to 3.125 ● ● ●

Given,

$$G(s) = \frac{2}{s^3 + 4s^2 + 5s + 2}$$

The PI controller is

$$C(s) = K_p + \frac{K_I}{s}$$

Rewriting,

$$C(s) = \frac{sK_P + K_I}{s}$$

Therefore, the open loop transfer function becomes

$$G(s)C(s) = \frac{2(sK_P + K_I)}{s(s^3 + 4s^2 + 5s + 2)}$$

For unity feedback systems,

$$1 + G(s)C(s) = 0$$

Hence,

$$1 + \frac{2(sK_P + K_I)}{s(s^3 + 4s^2 + 5s + 2)} = 0$$

$$s(s^3 + 4s^2 + 5s + 2) + 2(sK_P + K_I) = 0$$

$$s^4 + 4s^3 + 5s^2 + 2s + 2K_p s + 2K_I = 0$$

$$s^4 + 4s^3 + 5s^2 + (2 + 2K_p)s + 2K_I = 0$$

Constructing the Routh array,

$$
\begin{array}{c|ccc}
s^4 & 1 & 5 & 2K_I \\
s^3 & 4 & 2+2K_P & 0 \\
s^2 & \dfrac{18-2K_P}{4} & 2K_I & 0 \\
s^1 & \dfrac{\left(\dfrac{18-2K_P}{4}\right)(2+2K_P)-8K_I}{\left(\dfrac{18-2K_P}{4}\right)} & 0 & \\
s^0 & 2K_I & &
\end{array}
$$

For stability, all elements in the first column must be positive.

From the s^2 row,

$$\frac{18-2K_P}{4} > 0$$

$$18-2K_P > 0$$

$$K_P < 9$$

Now consider the s^1 row.
Since the denominator is already positive from the previous condition, only the numerator needs to be positive.

$$\left(\frac{18-2K_P}{4}\right)(2+2K_P)-8K_I > 0$$

$$(9-K_P)(1+K_P) > 8K_I$$

$$K_I < \frac{(9-K_P)(1+K_P)}{8}$$

To find the maximum possible value of K_I, define

$$f(K_P) = \frac{(9-K_P)(1+K_P)}{8}$$

$$f(K_P) = \frac{9+8K_P-K_P^2}{8}$$

Maximum value occurs when

$$\frac{df}{dK_P} = 0$$

$$\frac{1}{8}(8-2K_P) = 0$$

$$8-2K_P = 0$$

$$K_P = 4$$

Substituting $K_P = 4$,

$$K_I < \frac{(9-4)(1+4)}{8}$$

$$K_I < \frac{25}{8}$$

$$K_I < 3.125$$

Since the question allows marginal stability also,

$$K_I = 3.125$$

$$\boxed{3.125}$$

Q10.19.1 NAT | 2019 | 2M | Ans: 5.99 to 6.01 | ● ○ ○

Given,

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

The compensator is

$$\frac{K}{s}$$

Therefore, the open loop transfer function becomes

$$G_{OL}(s) = \frac{K}{s(s^2 + 3s + 2)}$$

For unity feedback systems,

$$1 + G_{OL}(s) = 0$$

$$1 + \frac{K}{s(s^2 + 3s + 2)} = 0$$

$$s(s^2 + 3s + 2) + K = 0$$

$$s^3 + 3s^2 + 2s + K = 0$$

The question asks for the value of K for which there are exactly two poles on the $j\omega$-axis.
This corresponds to the condition of marginal stability.
Hence, we use the Routh criterion.

Constructing the Routh array,

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 3 & K \\ s^1 & \dfrac{6-K}{3} & 0 \\ s^0 & K & \end{array}$$

For marginal stability, one complete row must become zero.
Therefore,

$$\frac{6-K}{3} = 0$$

$$6 - K = 0$$

$$K = 6$$

To verify that the roots actually lie on the $j\omega$-axis, form the auxiliary equation using the row just above the zero row.
The auxiliary equation is obtained from the s^2 row:

$$3s^2 + 6 = 0$$

$$s^2 + 2 = 0$$

$$s = \pm j\sqrt{2}$$

Hence, two poles lie on the imaginary axis.
Therefore, the value of K is:

$$\boxed{6}$$

Q10.17.1 MCQ 2017 1M Ans: D ● ○ ○

We analyze each statement individually.

Option A:
Lead compensator is mainly used to improve transient response.
A lead compensator introduces a zero closer to the origin and a pole farther away from the origin.

$$G_c(s) = K\frac{s+z}{s+p} \qquad \text{where } z < p$$

Due to this arrangement, the compensator adds positive phase to the system.
Positive phase increases the phase margin and improves damping.
Better damping makes the response faster and reduces oscillations.
Hence, settling time decreases.
Therefore, option A is correct.

Option B:
Lag compensator is mainly used to improve steady state performance.
A lag compensator introduces a pole closer to the origin and a zero farther away from the origin.

$$G_c(s) = K\frac{s+z}{s+p} \qquad \text{where } p < z$$

Because the pole is nearer to the origin, the low-frequency gain increases.
Steady state error mainly depends on low-frequency gain.
Higher low-frequency gain results in smaller steady state error.
Therefore, option B is correct.

Option C:
A lead compensator introduces one additional pole and one additional zero.

Hence, the order of the system may increase.
Therefore, option C is correct.

Option D:
A lag compensator is mainly designed for improving steady state accuracy.
It does not guarantee stabilization of every unstable system.
Stability must always be checked separately using methods such as Routh criterion, Root Locus or Nyquist criterion.
Therefore, this statement is incorrect.

$$\boxed{\text{(D) Lag compensator always stabilizes an unstable system}}$$

Q10.17.2 MCQ 2017 1M Ans: A ● ○ ○

A lag compensator is also called a phase-lag compensator.
Its standard form is

$$G_c(s) = \frac{s+z}{s+p}$$

For a lag compensator,

$$p < z$$

This means:

- the pole lies closer to the origin,
- the zero lies farther from the origin.

The reason behind this arrangement is that the pole increases the low-frequency effect first, thereby increasing low-frequency gain. Increased low-frequency gain helps in reducing steady state error.

Now checking the options:

Option A:
Pole is closer to the origin than the Zero.
This corresponds to a lag compensator.

Option B:
Zero is closer to the origin than the Zero.
Hence, this represents a lead compensator.

Option C:
Two Zeros are present between two Poles. The pole-zero pair nearer to origin acts as pole dominant, hence lag compensator. The pole-zero pair farther to the origin acts as zero dominant, hence lead compensator. Therefore, this entire arrangement is called as lag-lead compensator.

Option D:
Two Poles are present between two Zeros. The pole-zero pair nearer to origin acts as zero dominant, hence lead compensator. The pole-zero pair farther to the origin acts as pole dominant, hence lag compensator. Therefore, this entire arrangement is called as lead-lag compensator.

Therefore, the correct option is,

$$\boxed{\text{(A)}}$$

Key Points

- Pole closer to origin $\Rightarrow$ lag compensator.
- Zero closer to origin $\Rightarrow$ lead compensator.

Q10.15.1 MCQ 2015 1M Ans: A

Given,

$$G_c(s) = \frac{K(s+a)}{s+b}$$

where K, a and b are positive real numbers.

Comparing with the standard compensator form,

$$G_c(s) = K\frac{s+z}{s+p}$$

we get,

$$z = a \qquad \text{and} \qquad p = b$$

For a phase lead compensator,

$$z < p$$

because the zero must lie closer to the origin than the pole.

Therefore, $a < b$

$$\boxed{\text{(A) a<b}}$$

Mistakes to be avoided

- Do not compare the numerical values directly without understanding pole-zero locations on the negative real axis.
- Remember that the locations are
$$s = -a \quad \text{and} \quad s = -b$$
not $+a$ and $+b$.

Q10.15.2 NAT 2015 2M Ans: 0.5 to 0.5

For a lead compensator,

$$G_c(s) = \frac{1+\tau s}{1+\alpha\tau s}$$

Given,

$$G_c(s) = \frac{s+2}{s+4}$$

Rewriting,

$$G_c(s) = \frac{1+\frac{s}{2}}{1+\frac{s}{4}}$$

Comparing with the standard form,

$$\tau = \frac{1}{2}$$

Since

$$\tau = RC$$

$$RC = \frac{1}{2}$$

$$RC = 0.5$$

$$\boxed{0.5}$$

Q10.14.3 MCQ 2014 2M Ans: C ● ● ○

Given,

$$G(s) = \frac{1}{(s+1)(s+2)}$$

For the uncompensated unity feedback system,

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{1}{s^2+3s+3}$$

Comparing the denominator with $s^2 + 2\zeta\omega_n s + \omega_n^2$,

$$2\zeta\omega_n = 3$$

$$\zeta\omega_n = \frac{3}{2}$$

Time constant is

$$\tau = \frac{1}{\zeta\omega_n} = \frac{2}{3}$$

Settling time for 2% criterion is

$$T_s \approx 4\tau$$

$$T_s \approx 4\left(\frac{2}{3}\right) = 2.67\,\text{s}$$

But the required settling time is less than 2 s.
Hence, the compensated system should reduce the settling time compared to the uncompensated system.
To reduce settling time, we require a PD controller or a lead compensator since they improve transient response and shift poles towards the left-half plane.

Checking the options:

Option A:

$$C(s) = 3\left(\frac{1}{s+5}\right)$$

This introduces an additional pole.
Hence, response becomes slower.

Option B:

$$C(s) = 5\left(\frac{0.03}{s} + 1\right)$$

This contains a pole at origin.
Hence, it also slows down the system.

Option C:

$$C(s) = 2(s+4)$$

This introduces a zero.
Hence, it behaves like a PD controller and improves transient response.

Option D:

$$C(s) = 4\left(\frac{s+8}{s+3}\right)$$

Since the pole is dominant compared to the zero, it behaves like a lag compensator.
Lag compensators mainly improve steady-state accuracy and do not reduce settling time significantly.

Therefore, the correct compensator is option C.

$$\boxed{\text{(C) } 2(s+4)}$$

Key Points

- PD and lead compensators improve transient response and reduce settling time.
- Additional poles generally slow down the response.
- Lag compensators mainly improve steady-state performance.

Q10.12.1 MCQ 2012 2M Ans: A ● ○ ○

Given,

$$G_c(s) = \frac{s+a}{s+b}$$

For a lead compensator, the zero must be dominant.
That means the zero should be nearer to the origin than the pole.
Zero location:

$$s = -a$$

Pole location:

$$s = -b$$

For lead compensation,

$$-a > -b$$

$$a < b$$

Checking the options:

Option A:

$$a = 1, \quad b = 2$$

$$1 < 2$$

Satisfies the lead compensator condition.

$$\boxed{\text{(A) a=1, b=2}}$$

Q10.12.2 MCQ 2012 2M Ans: A ● ○ ○

From Q10.12.1, the lead compensator is

$$G_c(s) = \frac{s+1}{s+2}$$

Maximum phase occurs at the geometric mean of pole and zero locations.

$$\omega_m = \sqrt{z \times p}$$

Here,

$$z = -1$$

$$p = -2$$

Therefore,

$$\omega_m = \sqrt{1 \times 2}$$

$$\omega_m = \sqrt{2}\,\text{rad/s}$$

$$\boxed{(A)\ \sqrt{2}\,\text{rad/s}}$$

Q10.10.1 MCQ 2010 2M Ans: D ● ○ ○

Given,

$$G(s) = \frac{1}{s^2 + 2s + 2}$$

The controller $G_c(s)$ is connected in cascade with the plant.

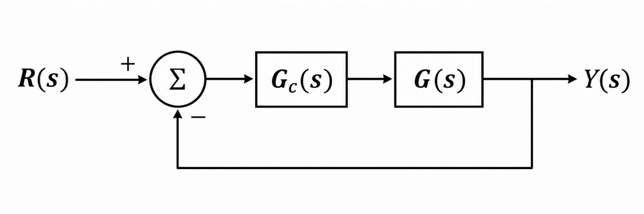


Fig. Solution Q10.10.1

First, assume the system contains only $G(s)$.
Since there is no pole at origin, the system is a Type-0 system.
For a unit step input, a Type-0 system gives finite steady-state error.
Now, we need a controller $G_c(s)$ such that the steady-state error becomes smaller than before.
To reduce steady-state error, we generally use:

- Lag compensator
- Integral controller

Checking the options:

Option A:

$$G_c(s) = \frac{s+1}{s+2}$$

Zero is nearer to origin than pole.
Hence, it is a lead compensator.

Option B:

$$G_c(s) = \frac{s+2}{s+1}$$

Pole is nearer to origin than zero.
Hence, it is a lag compensator. And it has no pole at orign hence it is Type-0 system

Option C:

$$G_c(s) = \frac{(s+1)(s+4)}{(s+2)(s+3)}$$

Dominant effect is lead compensation.

Option D:

$$G_c(s) = 1 + \frac{2}{s} + 3s$$

This contains proportional and integral terms.

Hence, it behaves like a PI controller.

Therefore, among all the options, Options B and D help in reducing steady-state error.

- Option B $\Rightarrow$ Lag compensator
- Option D $\Rightarrow$ PI controller

To break this tie, we check the system type.
Originally, the system is Type-0.
Option B does not introduce any pole at origin.
Hence, the system remains Type-0.
For a unit step input, a Type-0 system gives finite steady-state error.

Option D contains the term

$$\frac{2}{s}$$

Hence, it introduces a pole at origin.
Therefore, system type increases from Type-0 to Type-1.
For a unit step input, a Type-1 system gives zero steady-state error.

Hence, Option D gives minimum steady-state error.

$$\boxed{(D)\ G_c(s) = 1 + \frac{2}{s} + 3s}$$

Key Points

- Integral control increases system type.
- Increasing system type improves steady-state accuracy.
- Type-1 systems give zero steady-state error for unit step input.

Q10.08.1 MCQ 2008 2M Ans: B ● ● ○

The given circuit is an inverting op-amp configuration.
The input impedance is the parallel combination of R_1 and C_1.

$$Z_{in} = R_1 \parallel \frac{1}{sC_1}$$

$$Z_{in} = \frac{R_1\left(\frac{1}{sC_1}\right)}{R_1 + \frac{1}{sC_1}}$$

$$Z_{in} = \frac{R_1}{1 + sR_1C_1}$$

Therefore, transfer function is

$$\frac{V_o}{V_i} = -\frac{\text{Feedback path impedance}}{\text{Input impedance}} = -\frac{Z}{Z_{in}}$$

$$\frac{V_o}{V_i} = -\frac{Z(1 + sR_1C_1)}{R_1}$$

Now analyze both cases.
Case 1: For Q,

$$Z = R_2 + \frac{1}{sC_2}$$

Substituting into transfer function,

$$\frac{V_o}{V_i} = -\frac{\left(R_2 + \frac{1}{sC_2}\right)(1 + sR_1C_1)}{R_1}$$

$$\frac{V_o}{V_i} = -\frac{(1 + sR_1C_1)(1 + sR_2C_2)}{sR_1C_2}$$

$$\frac{V_o}{V_i} = -\frac{1 + sR_1C_1 + sR_2C_2 + s^2R_1R_2C_1C_2}{sR_1C_2}$$

$$\frac{V_o}{V_i} = -\left[\frac{1}{sR_1C_2} + \frac{sR_1C_1}{sR_1C_2} + \frac{sR_2C_2}{sR_1C_2} + \frac{s^2R_1R_2C_1C_2}{sR_1C_2}\right]$$

$$\frac{V_o}{V_i} = -\left[\frac{1}{sR_1C_2} + \frac{C_1}{C_2} + \frac{R_2}{R_1} + sR_2C_1\right]$$

$$\frac{V_o}{V_i} = -\left[\left(\frac{R_1C_1 \;+\; R_2C_2}{R_1C_2}\right) + \frac{1}{sR_1C_2} + sR_2C_1\right]$$

Therefore,

$$\frac{V_o}{V_i} = -\,[\text{Proportional term} + \text{Integral term} + \text{Derivative term}]$$

Hence, the controller is a PID controller.

$$Q \rightarrow 1$$

Case 2: For R,

$$Z = R_2 \parallel \frac{1}{sC_2}$$

$$Z = \frac{R_2}{1 + sR_2C_2}$$

Substituting into transfer function,

$$\frac{V_o}{V_i} = -\frac{R_2(1 + sR_1C_1)}{R_1(1 + sR_2C_2)}$$

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1} \cdot \frac{1 + sR_1C_1}{1 + sR_2C_2}$$

Given,

$$R_2C_2 > R_1C_1$$

Therefore,

$$\frac{1}{R_2C_2} < \frac{1}{R_1C_1}$$

Pole frequency is less than zero frequency.
Hence, it is a lag compensator.

$$R \rightarrow 3$$

$$\boxed{\text{(B) Q-1, R-3}}$$

Key Points

- Integral action appears when transfer function contains a pole at origin.
- For lag compensator, pole frequency is smaller than zero frequency.

Q10.09.1 MCQ 2009 1M Ans: *

From the magnitude plot, gain is high at low frequencies. As frequency increases, the magnitude decreases. Hence, the network initially behaves like a lag compensator.
In the mid-frequency region, magnitude remains almost constant at a low value.
As frequency increases further, magnitude starts increasing. Hence, the network now behaves like a lead compensator.
Therefore, the overall characteristic is:

$$\text{Lag action} \rightarrow \text{Lead action}$$

Thus, the compensator is a lag-lead compensator.
However, this option is not explicitly present in the given choices.

$$\boxed{\text{Lag-lead compensator}}$$

Key Points

- Lag compensator attenuates gain with increasing frequency.
- Lead compensator increases gain with increasing frequency.
- Sequence of magnitude variation determines whether the network is lead-lag or lag-lead.

Q10.07.1 MCQ 2007 2M Ans: B

Open loop transfer function is

$$G(s) = \frac{100(K_P + K_D s)}{s(s+10)}$$

Velocity error constant is given by

$$K_v = \lim_{s\to 0} sG(s)$$

$$1000 = \lim_{s\to 0} \frac{100(K_P + K_D s)}{s+10}$$

$$1000 = \frac{100K_P}{10}$$

$$K_P = 100$$

Characteristic equation is

$$1 + G(s) = 0$$

$$1 + \frac{100(K_P + K_D s)}{s(s+10)} = 0$$

$$s(s+10) + 100(K_P + K_D s) = 0$$

$$s^2 + 10s + 100K_D s + 100K_P = 0$$

$$s^2 + (10 + 100K_D)s + 100K_P = 0$$

Substitute $K_P = 100$,

$$s^2 + (10 + 100K_D)s + 10000 = 0$$

Compare with standard second order characteristic equation,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Comparing constant terms,

$$\omega_n^2 = 10000$$

$$\omega_n = 100$$

Given,

$$\zeta = 0.5$$

Therefore,

$$2\zeta\omega_n = 10 + 100K_D$$

$$2(0.5)(100) = 10 + 100K_D$$

$$100 = 10 + 100K_D$$

$$100K_D = 90$$

$$K_D = 0.9$$

$$\boxed{\text{(B) } K_P = 100, \quad K_D = 0.9}$$

Q10.07.2 MCQ | 2007 | 2M | Ans: C | ● ● ○

Given plant transfer function is

$$G(s) = \frac{1}{s^2 - 1}$$

For unity feedback system,

$$1 + G(s) = 0$$

$$1 + \frac{1}{s^2 - 1} = 0$$

$$s^2 = 0$$

$$s = 0, 0$$

Hence, closed loop poles lie at origin.
Therefore, the system is unstable.
Now as per the question, we need to cascade a lead compensator so that the closed loop system becomes stable.

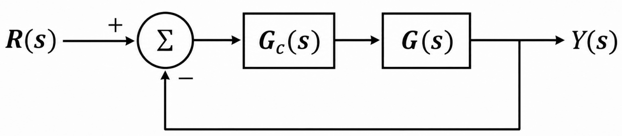


Fig. Solution Q10.07.2

Option A:

$$\frac{10(s-1)}{s+2}$$

Zero lies in right half plane.
Hence, it is a non-minimum phase compensator.
Therefore, it is not suitable.

Option B:

$$\frac{10(s+4)}{s+2}$$

Pole is closer to origin than zero.
Hence, it behaves as a lag compensator.

Options C and D are lead compensators.
We now check stability using Routh criterion.

For **Option C**,

$$G_c(s) = \frac{10(s+2)}{s+10}$$

Open loop transfer function becomes

$$G(s)H(s) = \frac{10(s+2)}{(s+10)(s^2-1)}$$

Characteristic equation:

$$1 + G(s)H(s) = 0$$

$$(s+10)(s^2-1) + 10(s+2) = 0$$

$$s^3 + 10s^2 + 9s + 10 = 0$$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 9 \\ s^2 & 10 & 10 \\ s^1 & 8 & 0 \\ s^0 & 10 & \end{array}$$

All elements of first column are positive.
Hence, system is stable.

For **Option D**,

$$G_c(s) = \frac{2(s+2)}{s+10}$$

Open loop transfer function becomes

$$G(s)H(s) = \frac{2(s+2)}{(s+10)(s^2-1)}$$

Characteristic equation:

$$1 + G(s)H(s) = 0$$

$$(s+10)(s^2-1) + 2(s+2) = 0$$

$$s^3 + 10s^2 + s - 6 = 0$$

Routh array:

$$\begin{array}{c|cc} s^3 & 1 & 1 \\ s^2 & 10 & -6 \\ s^1 & 1.6 & 0 \\ s^0 & -6 & \end{array}$$

Sign change occurs in first column.
Hence, system is unstable.

$$\boxed{\text{(C)}\ \frac{10(s+2)}{s+10}}$$

Q10.06.1 MCQ | 2006 | 2M | Ans: D | ● ○ ○

Given phase-lead compensator is

$$G_c(s) = \frac{1+3Ts}{1+Ts}$$

Standard form of lead compensator is

$$G_c(s) = \frac{1+\tau s}{1+\alpha\tau s}$$

Comparing with given transfer function,

$$\tau = 3T$$

$$\alpha\tau = T$$

Therefore,

$$\alpha(3T) = T$$

$$\alpha = \frac{1}{3}$$

Maximum phase shift for lead compensator is

$$\phi_m = \sin^{-1}\left(\frac{1-\alpha}{1+\alpha}\right) = \sin^{-1}\left(\frac{1-\frac{1}{3}}{1+\frac{1}{3}}\right) = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$$

$$\boxed{\text{(D) } \frac{\pi}{6}}$$

Q10.05.1 MCQ 2005 1M Ans: D ● ● ○

For a lag network, phase angle must be negative.
Transfer function of a lag network can be written as

$$T(s) = \frac{1+\tau s}{1+\beta\tau s} \qquad \text{where } \beta > 1$$

Putting $s = j\omega$,

$$T(j\omega) = \frac{1+j\omega\tau}{1+j\omega\beta\tau}$$

Since lag network produces negative phase,

$$\angle T(j\omega) < 0$$

In polar plot,

$$T(j\omega) = \text{Re} + j\text{Im}$$

Phase angle is

$$\angle T(j\omega) = \tan^{-1}\left(\frac{\text{Im}}{\text{Re}}\right)$$

In all options, real part is positive.
Therefore, for negative phase angle, imaginary part must be negative.
Hence, only Options B and D are possible because their plots lie below the real axis.

Now check starting and ending points.
At $\omega = 0$,

$$T(j0) = 1\angle 0^\circ$$

At $\omega \to \infty$,

$$T(j\infty) = \frac{1}{\beta}\angle 0^\circ$$

Since

$$\beta > 1$$

$$\frac{1}{\beta} < 1$$

Therefore, magnitude decreases as frequency increases.
Hence, final point must move closer to origin.

$$\boxed{\text{(D)}}$$

Q10.05.2 MCQ 2005 2M Ans: A ● ○ ○

Given plant is

$$G(s) = \frac{K}{s^2}$$

$$H(s) = 1$$

Closed loop transfer function of uncompensated system is

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)}$$

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + K}$$

Characteristic equation is

$$s^2 + K = 0$$

Compare with standard second order equation,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Comparing coefficients,

$$2\zeta\omega_n = 0$$

$$\zeta = 0$$

Also,

$$s = \pm j\sqrt{K}$$

Therefore, uncompensated system is undamped.
Given desired damping ratio is

$$\zeta = 0.5$$

Hence, damping ratio must be increased.
Increase in damping ratio is achieved using a phase-lead compensator.

Option A:

$$\frac{s+3}{s+9.9}$$

Zero is closer to origin than pole. Hence, zero dominant network. Therefore, it is a phase-lead compensator.

Option B:

$$\frac{s+9.9}{s+3}$$

Pole is closer to origin than zero. Hence, it is a phase-lag compensator.

Option C:

$$\frac{s-6}{s+8.33}$$

Zero lies in right half plane. Hence, it is a non-minimum phase system.

Option D:

$$\frac{s+6}{s}$$

Contains pole at origin. Hence, it behaves as an integrator.

$$\boxed{\text{(A) } \frac{s+3}{s+9.9}}$$

Q10.03.1 MCQ 2003 1M Ans: C ● ○ ○

A PD controller has the form

$$G_c(s) = K_P + K_D s$$

Option A:
Type of system increases only when a pole at origin is introduced. PD controller does not contain any pole at origin. Hence, type number does not increase. Therefore, Option A is incorrect.

Option B:
PD controller improves transient response. It increases damping ratio ζ. Hence, damping increases and not decreases. Therefore, Option B is incorrect.

Option C:
Derivative action behaves like a differentiator (High pass filter). Differentiators amplify high-frequency noise. Hence, PD controller causes higher noise amplification.
Therefore, Option C is correct.

Option D:
As damping ratio increases, overshoot decreases. Therefore, larger transient overshoot is not obtained. Hence, Option D is incorrect.

$$\boxed{\text{(C) Higher noise amplification}}$$

Q10.02.1 NAT 2002 Ans: * ● ● ○

Given,

$$G_p(s) = \frac{1}{(s+1)(2s+1)}$$

Substituting $s = j\omega$,

$$G_p(j\omega) = \frac{1}{(1+j\omega)(1+2j\omega)}$$

The phase angle of the plant transfer function is

$$\angle G_p(j\omega) = -\tan^{-1}(\omega) - \tan^{-1}(2\omega)$$

For a phase lag of 90°,

$$-\tan^{-1}(\omega) - \tan^{-1}(2\omega) = -90^\circ$$

$$\tan^{-1}(\omega) + \tan^{-1}(2\omega) = 90^\circ$$

Using the identity,

$$\tan^{-1}(A) + \tan^{-1}(B) = 90^\circ \quad \Rightarrow \quad AB = 1$$

Therefore,

$$\omega(2\omega) = 1$$

$$2\omega^2 = 1$$

$$\omega^2 = \frac{1}{2}$$

$$\omega = \frac{1}{\sqrt{2}}$$

$$\boxed{\omega = \frac{1}{\sqrt{2}} \text{ rad/s}}$$

Q10.02.2 NAT | 2002 | Ans: * | ● ● ○

The controller is

$$G_c(s) = \frac{k}{s}$$

The open-loop transfer function of the compensated system is

$$G(s) = G_c(s)G_p(s) = \frac{k}{s(s+1)(2s+1)}$$

Substituting $s = j\omega$,

$$G(j\omega) = \frac{k}{j\omega(1+j\omega)(1+2j\omega)}$$

The phase angle is

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(2\omega)$$

At phase crossover frequency,

$$\angle G(j\omega_{pc}) = -180^\circ$$

Hence,

$$-90^\circ - \tan^{-1}(\omega_{pc}) - \tan^{-1}(2\omega_{pc}) = 180^\circ$$

$$\tan^{-1}(\omega_{pc}) + \tan^{-1}(2\omega_{pc}) = 90^\circ$$

Using the identity,

$$\tan^{-1}(A) + \tan^{-1}(B) = 90^\circ \quad \Rightarrow \quad AB = 1$$

Therefore,

$$\omega_{pc}(2\omega_{pc}) = 1$$

$$2\omega_{pc}^2 = 1$$

$$\omega_{pc}^2 = \frac{1}{2}$$

$$\omega_{pc} = \frac{1}{\sqrt{2}}$$

Now,

$$|G(j\omega)| = \frac{k}{\omega\sqrt{1+\omega^2}\sqrt{1+4\omega^2}}$$

Substituting

$$\omega = \frac{1}{\sqrt{2}}$$

$$|G(j\omega_{pc})| = \frac{k}{\left(\frac{1}{\sqrt{2}}\right)\sqrt{1+\frac{1}{2}}\sqrt{1+2}} = \frac{k}{\left(\frac{1}{\sqrt{2}}\right)\sqrt{\frac{3}{2}}\sqrt{3}} = \frac{k}{3/2}$$

$$|G(j\omega_{pc})| = \frac{2k}{3}$$

Gain margin is

$$GM = \frac{1}{|G(j\omega_{pc})|}$$

Given,

$$GM = 2.5$$

Therefore,

$$2.5 = \frac{1}{2k/3}$$

$$k = \frac{3}{5} = 0.6$$

$$\boxed{k = 0.6}$$

Q10.02.3 NAT 2002 Ans: * ● ● ○

The compensated open-loop transfer function is

$$G(s) = \frac{0.6}{s(s+1)(2s+1)}$$

Since there is one pole at origin, the system is a Type-1 system.

Steady state error for unit-step input

For a Type-1 system,

$$e_{ss,\text{step}} = 0$$

$$\boxed{e_{ss,\text{step}} = 0}$$

Steady state error for unit-ramp input

Velocity error constant,

$$K_v = \lim_{s\to 0} sG(s)$$

$$K_v = \lim_{s\to 0} \frac{0.6}{(s+1)(2s+1)}$$

$$K_v = 0.6$$

Hence,

$$e_{ss,\text{ramp}} = \frac{1}{K_v} = \frac{1}{0.6} = \frac{5}{3}$$

$$\boxed{e_{ss,\text{ramp}} = \frac{5}{3}}$$

Key Points

- A Type-1 system has zero steady state error for unit-step input and finite steady state error for unit-ramp input.

Q10.01.2 NAT 2001 Ans: * ● ● ○

The process transfer function is

$$G_p(s) = \frac{1}{s(s+1)}$$

Given,

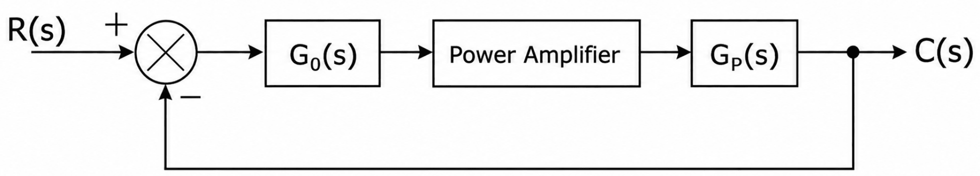


Fig. Solution Q10.01.2

The forward path transfer function is

$$G(s) = G_0(s) \times K \times G_p(s)$$

$$G(s) = KG_0(s)G_p(s)$$

Given,

$$G_p(s) = \frac{1}{s(s+1)}$$

Therefore,

$$G(s) = \frac{KG_0(s)}{s(s+1)}$$

For a unity feedback system, the characteristic equation is

$$1 + G(s) = 0$$

Substituting $G(s)$,

$$1 + \frac{KG_0(s)}{s(s+1)} = 0$$

$$s(s+1) + KG_0(s) = 0$$

$$s^2 + s + KG_0(s) = 0$$

Let the PD controller be

$$G_0(s) = K_P + K_D s$$

then the characteristic equation becomes

$$s^2 + s + K(K_D s + K_P) = 0$$

$$s^2 + s + KK_D s + KK_P = 0$$

$$s^2 + (1 + KK_D)s + KK_P = 0$$

Therefore, the characteristic equation is,

$$\boxed{s^2 + (1 + KK_D)s + KK_p = 0}$$

The amplification factor of the power amplifier is given as $K = 1$. Hence the above characteristic equation becomes,

$$s^2 + (1 + K_D)s + K_P = 0$$

The standard second-order characteristic equation for a unity feedback system is,

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Given damping ratio, $\zeta = 0.707$ and time constant, $\tau = 1$ sec
For a second-order system,

$$\tau = \frac{1}{\zeta\omega_n}$$

$$1 = \frac{1}{0.707\omega_n}$$

$$0.707\omega_n = 1$$

$$\omega_n = \frac{1}{0.707} \approx 1.414$$

Therefore,

$$\omega_n^2 = (1.414)^2 = 2$$

Substituting into the standard characteristic equation,

$$s^2 + 2(0.707)(1.414)s + 2 = 0$$

$$s^2 + 2s + 2 = 0$$

Now, comparing the above equation with $s^2 + (1 + K_D)s + K_P = 0$,

$$1 + K_D = 2 \qquad \text{and} \qquad K_P = 2$$

$$K_D = 1$$

Therefore, the PD controller transfer function is

$$G_0(s) = K_P + K_D s$$

$$G_0(s) = 2 + s$$

$$\boxed{G_0(s) = s + 2}$$

Q10.92.1 MSQ 1992 2M Ans: B, C

In a PID controller:

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s$$

Option A:
Integral mode mainly improves steady state accuracy. It generally slows down transient response. Therefore, Option A is incorrect.

Option B:
Integral action reduces steady state error. Hence, steady state performance improves. Therefore, Option B is correct.

Option C:
Derivative mode improves transient response. It increases damping and reduces overshoot. Therefore, Option C is correct.

Option D:
Derivative action has very little effect on steady state performance. Therefore, Option D is incorrect.

$$\boxed{\text{(B) The system is not completely controllable}}$$

$$\boxed{\text{(C) The system is completely observable}}$$

Q10.90.1 MCQ 1990 2M Ans: D

Given controller transfer function is

$$G_c(s) = \frac{s + Z_1}{s + P_1}$$

From the given transfer function,

$$s = -Z_1 \quad \text{is the zero and}$$

$$s = -P_1 \quad \text{is the pole.}$$

For a phase lead compensator, zero must be dominant. That means, zero should be nearer to origin than the pole. Therefore,

$$-Z_1 > -P_1$$

$$Z_1 < P_1$$

$$P_1 > Z_1$$

$$\boxed{\text{(D) } P_1 > Z_1}$$

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Chapter X

Controllers and Compensators

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